

REAL ANALYSIS

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This is a set of course notes for Math 131A: Real Analysis, taught in the Spring 2022 term at UCLA. There is a hyperlinked table of contents below that indicates the content of each lecture and the overall structure of the course. Throughout the notes there is a combination of casual narration and what I consider *well-written, professional mathematical prose*. Any writing that I consider professional quality will be written in *purple*. This writing is indicative of how I would solve a homework or exam problem. Thus, when you read anything written in *purple*, take note of not only the content, but also how I say and format whatever is written. This includes the use of language, the level of justification, and the actual typesetting and formatting. The *purple writing* is there as a helpful proof writing reference.

Acknowledgments. Thanks to Shayan Saadat, Rupin Mittal, and Allen Tong for pointing out some typos.

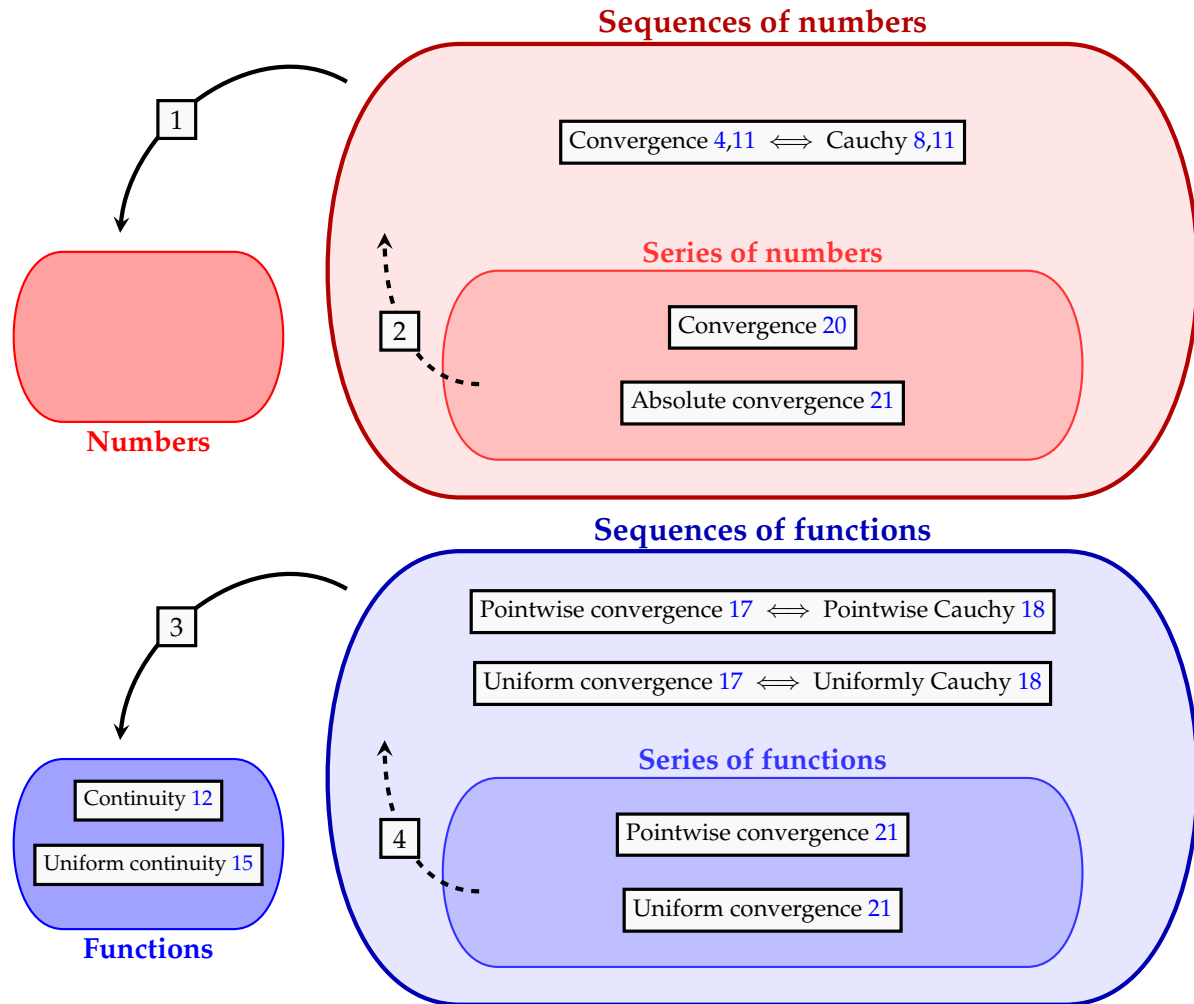
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MAP OF REAL ANALYSIS CONCEPTS

There are many concepts and terms in analysis, and it can be difficult to keep all of them straight. This is especially the case with concepts that sound similar, like uniform continuity and uniform convergence. This map is intended to help organize the main terms that are defined and studied in these notes. Each concept node is hyperlinked with the corresponding lecture where the term is introduced and studied.



Legend. This legend describes the meaning of the arrows in the map.

- 1** One passes from a *sequence of numbers* to a *single number* by taking a *limit*. See Lecture 4.
- 2** One passes from a *series of numbers* to a *sequence of numbers* by considering the *partial sums*. In particular, $\sum_{n=0}^{\infty} a_n := \lim_{N \rightarrow \infty} S_N$. See Lecture 19.
- 3** One passes from a *sequence of functions* to a *single function* by taking a *limit*. See Lecture 17.
- 4** One passes from a *series of functions* to a *sequence of functions* by considering the *partial sums*. In particular, $\sum_{n=0}^{\infty} f_n(x) := \lim_{N \rightarrow \infty} S_N(x)$. See Lecture 21.

Part 1

The Real Numbers

1. LECTURE 1: WHAT IS A REAL NUMBER?

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 1, some of Section 1 and Section 2, but not everything from those sections.
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 1, “Introduction”.
- ◇ Abbott, *Understanding Analysis*: Chapter 1, Section 1.

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Be able to prove that there is no rational number $q \in \mathbb{Q}$ such that $q^2 = 2$, and to prove other similar statements.
- ◇ Think about why defining the concept of a real number is more delicate than that of integers or rational numbers.
- ◇ Be prepared to learn about the two concepts that are needed to discuss the real numbers carefully: ordered fields, and supremums and infimums.

To get into the right mindset for this course, I want you to forget almost everything you know about math. Not *really*, but formally. We want to wipe the slate clean and develop the theory of calculus from nearly the ground up, and the first step of this is to go back to the beginning and think carefully about the concept of a number. Here’s what I will assume you know:

- ◇ You know what the **natural numbers** are:¹

$$\mathbb{N} := \{0, 1, 2, \dots\}.$$

Note that some people like to think of the natural numbers as starting at 1. Some people also like to make a big deal of this, as if it matters at all. To me, it doesn’t matter, and if we’re ever in a situation where it is important to distinguish between starting at 0 or 1, then it will be made clear in context.

- ◇ You know what the **integers** are:

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}.$$

- ◇ You know what the **rational numbers** are:

$$\mathbb{Q} := \left\{ \frac{m}{n} \mid m, n \in \mathbb{Z}, n \neq 0 \right\}.$$

¹Here, the notation “ $:=$ ” means *is defined as*. In other words, whenever I write $x := \text{blah}$, I am defining the symbol x to mean blah. This is in contrast to the statement $x = \text{blah}$, which means that two previously defined quantities, x and blah, are equal.

- ◊ You know what an abstract field is from Math 115A. If you've forgotten what a field is, you can check out Appendix B, but we won't deal with fields much in 131A. I only want you to briefly remember what a field is for the purpose of the first couple lectures. If you'd like, anytime you read the word "field" in the first few lectures you can think " $\mathbb{Q}$."

If you're asking *what about the real numbers*, then you've touched one of the central nerves of real analysis. Do you actually know what a real number is?

Remark 1.1. Although I'm assuming you know about $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ already, these sets can actually be constructed from very first principles using equivalence relations. I won't bother doing this, but I will add some challenge problems to the homework for those of you that are interested. You can also Google the "Peano axioms."

Remark 1.2. Here is a simple fact that isn't worth stating as a proposition. Given any two rational numbers $q, r \in \mathbb{Q}$, there is always a third rational number between them. *Indeed, assume without loss of generality that $q < r$ and consider $\frac{q+r}{2}$, the mean of the numbers q and r . Note that $\frac{q+r}{2} \in \mathbb{Q}$. Also, note that*

$$\frac{q+r}{2} > \frac{q+q}{2} = q$$

and

$$\frac{q+r}{2} < \frac{r+r}{2} = r$$

so that

$$q < \frac{q+r}{2} < r.$$

Here is a (well-known) proposition that demonstrates one of the important drawbacks of the rational numbers.

Proposition 1.3. *There is no rational number $q \in \mathbb{Q}$ such that $q^2 = 2$.*

Proof. Suppose for the sake of contradiction that $q \in \mathbb{Q}$ satisfies $q^2 = 2$. Because q is rational, we can write

$$q = \frac{m}{n}$$

for some $m, n \in \mathbb{Z}$ with $n \neq 0$. Furthermore, we can assume that at least one of m or n is odd. Indeed, if both m and n were even, we could factor out a (power of) 2 from each and simplify the fraction.

Since $q^2 = 2$, we have $\frac{m^2}{n^2} = 2$. This means that

$$m^2 = 2n^2.$$

Thus, m^2 is divisible by 2, and so m^2 is even. We can then write $m = 2k$ for some $k \in \mathbb{Z}$. The above equation then becomes

$$(2k)^2 = 2n^2 \quad \Rightarrow \quad 4k^2 = 2n^2 \quad \Rightarrow \quad 2k^2 = n^2.$$

By a similar reasoning, this implies that n^2 , and hence n , is even. Therefore, both m and n are even. This is a contradiction, because we assumed that at least one of m and n is odd. $\square$

Remark 1.4. Note that I did *not* say "the square root of 2 is irrational." We don't know what irrational numbers are just yet, and so that sentence doesn't make sense. Instead, the proposition just says that there is no rational number that squares to 2, which is different!

Remark 1.5. The proof above is not what I would consider an “analysis” proof, for what it’s worth. I wouldn’t be too concerned with the details of the argument.

Remark 1.6. Of course, the fact that there is no rational that squares to 2 is just one example of a gap in the rational numbers. There are many, many more. See if you kind find some!

It is hard to actually construct $\mathbb{R}$, and we will not do this in 131A. However, in Lecture 3 we will state (without proof) an existence theorem that correctly describes the important properties of $\mathbb{R}$. To state the theorem we need to talk about (1) ordered fields, and (2) supremums and infimums. These are the main concepts for the next couple of lectures.

2. LECTURE 2: ORDERED FIELDS, ABSOLUTE VALUES, AND THE TRIANGLE INEQUALITY

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◊ Ross, *Elementary Analysis*: Chapter 1, the first part of Section 3.
- ◊ Rudin, *Principles of Mathematical Analysis*: Chapter 1, some of “Ordered Sets” and some of “Fields”.
- ◊ Abbott, *Understanding Analysis*: Chapter 1, some of Section 2.

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◊ Understand the definition of an ordered field and be able to prove basic properties about one from the definition.
- ◊ Understand the definition of the absolute value function on an ordered field and be able to prove basic properties about it from the definition.
- ◊ Be able to use the triangle inequality in simple proofs, and be prepared to use it throughout the rest of the course.

Recall that a *field* is an abstract set with two algebraic operations (addition and multiplication) that obey most of the usual algebraic rules you are secretly familiar with. If you are unfamiliar with fields or need a refresher, you can check out Appendix B. For now, the main example of a field that you know is $\mathbb{Q}$, the set of rational numbers.

In an abstract field, there is no notion of *order* or *inequality*. That is, if $\mathbb{F}$ is a field and you pick two elements $a, b \in \mathbb{F}$, you cannot ask the question *is a bigger than b* , because “bigger” doesn’t mean anything in a field. Thus, I want you to forget that you know what the symbols $<$ and $\leq$ mean. However, in a field like $\mathbb{Q}$ we do have a (predisposed, secret) understanding of order. We want to generalize this in the setting of abstract fields.

Definition 2.1. An **ordered field** is a field $\mathbb{F}$ equipped with a symbol^a $<$, that satisfies the following properties.

- (i) Given $x, y \in \mathbb{F}$, it is either the case that $x < y$, $y < x$, or $x = y$.
- (ii) If $x < y$ and $y < z$, then $x < z$.
- (iii) If $y < z$, then $x + y < x + z$ for all $x \in \mathbb{F}$.
- (iv) If $0 < x$ and $0 < y$, then $0 < xy$.

^aFormally, a *relation*.

Remark 2.2. In an ordered field, we define “ $x > y$ ” to mean $y < x$. Similarly, we define “ $x \leq y$ ” to mean $x < y$ or $x = y$, and likewise with “ $x \geq y$ ”.

Example 2.3. The rational numbers $\mathbb{Q}$ form an ordered field with the usual notion of $<$.

Proposition 2.4 (Properties of ordered fields). *Let $\mathbb{F}$ be an ordered field.*

- (i) *If $x > 0$, then $-x < 0$. Likewise, if $x < 0$, then $-x > 0$.*
- (ii) *If $x > 0$, and if $y < z$, then $xy < yz$. Likewise, if $x < 0$, and if $y < z$, then $xy > yz$.*
- (iii) *If $x \neq 0$, then $x^2 > 0$. In particular, $1 > 0$.*
- (iv) *If $0 < x < y$, then $0 < \frac{1}{y} < \frac{1}{x}$.*

Proof. We will prove two of the properties, and leave the rest as exercises.

- (i) **Let $x > 0$. Then**

$$x + -x > 0 + -x$$

by (iii) in the definition of an ordered field. Thus, $0 > -x$ as desired. Likewise, if $x < 0$, then

$$x + -x < 0 + -x$$

and so $0 < -x$.

- (ii) **Suppose that $x > 0$ and $y < z$. Then since $y < z$, by (iii) in the definition of an ordered field we have**

$$z + -y > y + -y = 0$$

so that $z - y > 0$. By (iv) in the definition of an ordered field, it follows that $x(z - y) > 0$. Thus, $xz - xy > 0$, and so

$$(xz - xy) + xy > 0 + xy.$$

Therefore, $xy < xz$, as desired. The “likewise” statement is similar.

- (iii) Exercise.

- (iv) Exercise.

□

In an ordered field, we can define the concept of *absolute value*, which generalizes the notion of absolute value that you are (secretly) familiar with in $\mathbb{Q}$.

Definition 2.5. Let $\mathbb{F}$ be an ordered field. For all $x \in \mathbb{F}$, define the **absolute value**

$$|x| := \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}.$$

Here are basic properties of the absolute value function.

Proposition 2.6. *Let $\mathbb{F}$ be an ordered field. Then*

- (i) *For all $x \in \mathbb{F}$ we have $|x| \geq 0$.*
- (ii) *For all $x, y \in \mathbb{F}$ we have $|xy| = |x||y|$.*

Proof.

- (i) **There are two cases to consider. If $x \geq 0$, then $|x| = x \geq 0$. If $x < 0$, then $|x| = -x > 0$.**

- (ii) **There are four cases to consider.**

First, if $x \geq 0$ and $y \geq 0$, then $|x| = x$ and $|y| = y$. By properties of ordered fields, $xy \geq 0$. Thus,

$$|xy| = xy = |x||y|.$$

Second, if $x < 0$ and $y < 0$, then $|x| = -x$ and $|y| = -y$. By properties of ordered fields, $xy \geq 0$. Thus,

$$|xy| = xy = (-x)(-y) = |x||y|.$$

Third, if $x \geq 0$ and $y < 0$, then $|x| = x$ and $|y| = -y$. By properties of ordered fields, $xy \leq 0$. Thus,

$$|xy| = -xy = x(-y) = |x||y|.$$

The last case where $x < 0$ and $y \geq 0$ is identical to the third case.

□

Here is a more significant property of ordered fields. This is one of the most widely used inequalities in all of math, and you would do well to keep it at the forefront of your tool kit.

Theorem 2.7 (The triangle inequality). *Let $\mathbb{F}$ be an ordered field. Then for all $x, y \in \mathbb{F}$ we have*

$$|x + y| \leq |x| + |y|.$$

Proof. As $|x + y| = x + y$ or $|x + y| = -(x + y)$, to prove the theorem it suffices to separately prove

$$(2.1) \quad x + y \leq |x| + |y|$$

and

$$(2.2) \quad -(x + y) \leq |x| + |y|$$

Note that $-|x| \leq |x|$. Since $|x| = x$ or $|x| = -x$, it then follows that $-|x| \leq x \leq |x|$. Similarly, $-|y| \leq y \leq |y|$. Therefore, by two applications of property (iii) in the definition of an ordered field, we have

$$x + y \leq |x| + |y| \leq |x| + |y|.$$

This gives (2.1). Similarly,

$$-|x| + -|y| \leq x + -|y| \leq x + y.$$

Multiplying both sides of the resulting inequality by -1 gives $|x| + |y| \geq -(x + y)$, which is (2.2), as desired. □

Example 2.8. Here is a simple example of how the triangle inequality is frequently used in analysis. Suppose that we have two numbers $x, y \in \mathbb{Q}$, and we are told that $|x - y| < \frac{1}{4}$ and $|y| < \frac{1}{4}$. In words, the distance between x and y is less than $\frac{1}{4}$, and the distance from y to 0 is less than $\frac{1}{4}$. Using the triangle inequality we can conclude that $|x| < \frac{1}{2}$. **Indeed, note that**

$$x = x - y + y.$$

Thus, by the triangle inequality,

$$\begin{aligned} |x| &= |x - y + y| \\ &\leq |x - y| + |y| \\ &< \frac{1}{4} + \frac{1}{4} \\ &= \frac{1}{2}. \end{aligned}$$

3. LECTURE 3: SUPREMUMS, INFIMUMS, AND THE REAL NUMBERS

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 1, Section 4.
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 1, some of “Ordered Sets” and some of “The Real Field”.
- ◇ Abbott, *Understanding Analysis*: Chapter 1, Section 3.

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Understand the definition of the supremum and infimum of a set, and how these concepts compare and contrast to the notion of maximum, minimum, and upper and lower bounds.
- ◇ Be able to intuit the supremum and infimum of simple sets, when they exist.
- ◇ Be able to use the Archimedean property of $\mathbb{R}$ and the density of $\mathbb{Q}$ in $\mathbb{R}$ in basic proofs.

Definition 3.1. Let $\mathbb{F}$ be an ordered field, and let $E \subset \mathbb{F}$ be a set.

- (i) An **upper bound** for E is an element $M \in \mathbb{F}$ such that $x \leq M$ for all $x \in E$. We say that a set E with an upper bound is **bounded above**.
- (ii) A **lower bound** for E is an element $m \in \mathbb{F}$ such that $m \leq x$ for all $x \in E$. We say that a set E with a lower bound is **bounded below**.
- (iii) We say that a set E is **bounded** if it has both an upper bound and a lower bound.

Remark 3.2. An equivalent characterization of a set being bounded in (iii) is the existence of a single element $M \in \mathbb{F}$ such that $|x| \leq M$ for all $x \in E$. Think about why this is equivalent!

Example 3.3. Here are two simple examples of sets of rational numbers.

- (1) Let $E := (-\infty, 0) \subset \mathbb{Q}$.
 - ◇ This set is bounded above. There are many upper bounds. For example, 0, 1, 2, 1000, $\frac{734}{119}$ are all upper bounds.
 - ◇ This set is *not* bounded below. (The symbol $-\infty$ is not a number!)
- (2) Let $E := (1, \frac{5}{4}]$.
 - ◇ This set is bounded above. There are many upper bounds. For example, $\frac{5}{4}$, 2, 1000 are all upper bounds.
 - ◇ This set is bounded below. There are many lower bounds. For example, 1, 0, -1000 are all lower bounds.
 - ◇ Because this set is bounded above and bounded below, it is bounded.

Definition 3.4. Let $\mathbb{F}$ be an ordered field, and let $E \subset \mathbb{F}$ be a subset.

- (i) A **supremum** of E , if it exists, is an element $\alpha \in \mathbb{F}$ such that
 - ◊ α is an upper bound for E , and
 - ◊ if $\beta < \alpha$, then β is *not* an upper bound for E .
- (ii) An **infimum** of E , if it exists, is an element $\alpha \in \mathbb{F}$ such that
 - ◊ α is a lower bound for E , and
 - ◊ if $\beta > \alpha$, then β is *not* a lower bound for E .

Remark 3.5. It is an immediate consequence of the definitions of a supremum and infimum that, if they exist, they are unique. We prove this here for the case of the supremum. *Indeed, suppose that $\alpha, \alpha' \in \mathbb{F}$ are both supremums for a set E . Because α' is an upper bound for E , and since α is a supremum for E , it cannot be the case that $\alpha' < \alpha$. Thus,*

$$\alpha' \geq \alpha.$$

Likewise, since α is an upper bound for E , and since α' is a supremum for E , it cannot be the case that $\alpha < \alpha'$. Thus,

$$\alpha \geq \alpha'.$$

Together, these inequalities imply that $\alpha = \alpha'$, and so the supremum of E , if it exists, is unique. I'll leave it as an exercise to argue that infimums, when they exist, are also unique. Because supremums and infimums are unique, we can now speak of *the* supremum of E , and we will denote this element by $\sup E$. Similarly, we can speak of *the* infimum, denoted $\inf E$.

Remark 3.6. Some texts use the term *least upper bound* in place of *supremum* and *greatest lower bound* in place of *infimum*.

Example 3.7. We revisit the two simple examples from above.

- (1) The set $E = (-\infty, 0) \subset \mathbb{Q}$ of rational numbers does have a supremum, namely, $\sup E = 0$. Note that $\sup E \notin E$. The set E does not have an infimum, since E has no lower bounds at all.
- (2) The set $E = (1, \frac{5}{4}] \subset \mathbb{Q}$ has both a supremum and an infimum. In particular, $\sup E = \frac{5}{4}$ and $\inf E = 1$.

Example 3.8. Here is a very important example. Consider the set of rational numbers

$$E := \{ q \in \mathbb{Q} \mid 0 < q^2 < 2, q > 0 \} \subset \mathbb{Q}.$$

Note that we are viewing E as a subset of the *rational*s, since we don't even know what real numbers are yet. (In your head, you should secretly think of this set as $(0, \sqrt{2})$, but at the moment this doesn't mean anything.)

- ◊ The set E is bounded. A lower bound is 0, and an upper bound is 1000.
- ◊ The set E does have an infimum, which is $\inf E = 0$.
- ◊ Importantly, the set E *does not have a supremum*.

A more familiar notion is that of the maximum and minimum of a set.

Definition 3.9. Let $E \subset \mathbb{F}$ be a subset of an ordered field.

- (i) A **maximum** of E , if it exists, is an element $M \in E$ such that $x \leq M$ for all $x \in E$.
- (ii) A **minimum** of E , if it exists, is an element $m \in E$ such that $x \geq m$ for all $x \in E$.

Remark 3.10. Compare and contrast this definition with the definition of an upper/lower bound, and the supremum/infimum. In particular, note that maxes and mins *do* have to be in the set E , whereas upper/lower bounds and sups/infs do not have to be in the set (but they can be).

Remark 3.11. Much like sups and infs, if a maximum or minimum exists, one can easily show that it is unique. Therefore, we can talk about *the* maximum and *the* minimum of a set.

Example 3.12. Returning once more to the simple sets above, we have:

- (1) The set $E = (-\infty, 0) \subset \mathbb{Q}$ of rational numbers has neither a maximum nor a minimum.
- (2) The set $E = (1, \frac{5}{4}] \subset \mathbb{Q}$ has a maximum, $\frac{5}{4}$, which coincides with the supremum. It has no minimum.

Finally, we state the punchline that rigorously describes the set of real numbers.

Theorem 3.13 (Existence and uniqueness of the real numbers). *There exists a unique^a ordered field $\mathbb{R}$, called the real numbers, satisfying the following properties.*

- (i) *The rational numbers are a subfield of $\mathbb{R}$, that is, $\mathbb{Q} \subset \mathbb{R}$.*
- (ii) *If $E \subset \mathbb{R}$ is a nonempty set which is bounded above, then $\sup E \in \mathbb{R}$ exists.*

^aHere, “unique” means up to isomorphism of fields, which is something we won’t discuss here.

Remark 3.14. Property (ii) is often called the *least upper bound property* or the *completeness axiom*.

Proof. See Subsection 3.1 below. □

Theorem 3.15 (Archimedean property of $\mathbb{R}$). *Let $x, y \in \mathbb{R}$. If $x > 0$, then there exists an $n \in \mathbb{N}$ such that $nx > y$.*

Proof. Fix $x, y \in \mathbb{R}$ such that $x > 0$. Let

$$E := \{ nx \mid n \in \mathbb{N} \}.$$

We claim that y is not an upper bound for E . Indeed, suppose for the sake of contradiction that y is an upper bound for E . Then, in particular, E is bounded above, and so by the least upper bound property of $\mathbb{R}$ we can say that $\sup E \in \mathbb{R}$ exists.

Since $x > 0$, it follows that $-x + \sup E < \sup E$. By definition of the supremum, $-x + \sup E$ is not an upper bound for E . Thus, there is an $n_0 \in \mathbb{N}$ such that

$$-x + \sup E < n_0 x.$$

This means that $\sup E < (n_0 + 1)x$. Since $(n_0 + 1)x \in E$, this contradicts the fact that $\sup E$ is an upper bound for E . Therefore, y is *not* an upper bound for E , and so there is some $n \in \mathbb{N}$ such that $nx > y$. □

Remark 3.16. The field $\mathbb{Q}$ does not satisfy the least upper bound property, but it does satisfy the Archimedean property. Can you prove the Archimedean property for $\mathbb{Q}$ directly?

Theorem 3.17 (Density of $\mathbb{Q}$ in $\mathbb{R}$). *Let $x, y \in \mathbb{R}$. If $x < y$, then there is a $q \in \mathbb{Q}$ such that $x < q < y$.*

Remark 3.18. In words, this theorem says that, given any two real numbers, you can always find a rational number between them.

Proof. Our goal is to find integers $m, n \in \mathbb{Z}$ such that

$$x < \frac{m}{n} < y.$$

Since $x < y$, it follows that $y - x > 0$. By the Archimedean property, there is an $n \in \mathbb{N}$ such that $n(y - x) > 1$. This means that $ny - nx > 1$.

For the rest of the argument, I will do something a bit informal (hence, written in black). It is possible to make this precise with the Archimedean property, but I don't really care about doing that and I won't expect you to be able to do stuff like this. Anyway, since $ny - nx > 1$, the distance between nx and ny is bigger than 1. This means that there will be an integer $m \in \mathbb{Z}$ between them, that is, an integer $m \in \mathbb{Z}$ such that $nx < m < ny$. Dividing by n gives

$$x < \frac{m}{n} < y$$

as desired. If you are unconvinced by the last part of this proof, you can check out the above referenced sections in Ross and Rudin.

□

Remark 3.19. It is a fair question to ask what the point of these two theorems are, or what they tell us about the real numbers. They tell us that $\mathbb{R}$ “completes” $\mathbb{Q}$ by filling in all the holes, but not *too* much. There are other wacky ordered fields out there (for example, check out the Wikipedia page for the [surreal number system](#)) that have some weird properties. For example, in the surreal real number system there are numbers that are *so* small that they are truly infinitesimal, and in particular they break the Archimedean property. Thus, these theorems tell us that $\mathbb{R}$ is well-behaved in ways that we would expect, and that there isn't anything too crazy that gets added in when we complete $\mathbb{Q}$.

Another fair question is about the apparent asymmetry in Theorem 3.13. Why does it only require the *least upper bound property*? What about the analogous *greatest lower bound property*? It turns out that we can actually prove this using only the least upper bound property.

Proposition 3.20 (Greatest lower bound property of $\mathbb{R}$). *The real numbers satisfy the greatest lower bound property. That is, if $E \subset \mathbb{R}$ is a nonempty set which is bounded below, then $\inf E \in \mathbb{R}$ exists.*

Proof. Let

$$L := \{ \alpha \in \mathbb{R} \mid \alpha \text{ is a lower bound of } E \}.$$

Since E is bounded below by assumption, L is nonempty. Next, we claim that L is bounded above. Indeed, let $x \in E$. We claim that x is an upper bound for L . To see this, let $\alpha \in L$ be an arbitrary element of L . Since α is a lower bound for E , it follows that $\alpha \leq x$. This shows that x is an upper bound for L , and hence L is bounded above.

Since $\mathbb{R}$ satisfies the least upper bound property, this implies that $\sup L \in \mathbb{R}$ exists. We claim that $\inf E = \sup L$.

First, we will show that $\sup L$ is a lower bound for E . Suppose for the sake of contradiction that it was not. Then there would be an element $x \in E$ such that $x < \sup L$. By definition of $\sup L$, x is not an upper bound

for L , and so there would be an element $\alpha \in L$ such that $\alpha > x$. But this contradicts the fact that α is a lower bound for E . Thus, $\sup L$ is in fact a lower bound for E .

Next, we show that $\sup L$ is in fact the greatest lower bound of E . Suppose that $\gamma \in \mathbb{R}$ is a number satisfying $\gamma > \sup L$. Since $\sup L$ is an upper bound for L , it follows that $\gamma \notin L$. In particular, γ is not a lower bound for E . Thus, $\sup L$ is in fact the greatest lower bound for E , and so $\inf E = \sup L$, as desired.

□

3.1. **Proof of Theorem 3.13.** ADD

Part 2

Sequences

4. LECTURE 4: LIMITS OF SEQUENCES

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◊ Ross, *Elementary Analysis*: Chapter 2, Section 7 and Section 8.
- ◊ Rudin, *Principles of Mathematical Analysis*: Chapter 3, some of “Convergent Sequences”, but be warned that the language here is more advanced than what we are using.
- ◊ Abbott, *Understanding Analysis*: Chapter 2, Section 2.

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◊ Understand intuitively the formal ε -definition of the limit of a sequence.
- ◊ Be able to prove that simple sequences converge to a given limit.
- ◊ Be able to prove simple abstract statements about limits using the ε -definition.

From one perspective, calculus is the study of derivatives and integrals. Both of these ideas are based on the idea of *limits*. When you take a calculus class, the concept of a limit is usually not studied completely rigorously; typically it is presented with the intuitive idea of numbers “getting close.” We will spend the next few weeks discussing limits of sequences in a completely rigorous way.

Informally, a *sequence* is an ordered infinite list of real numbers, indexed by the natural numbers. We use various different notations to describe a sequence:

$$(a_n)_{n=0}^{\infty}, \quad (a_n), \quad (a_0, a_1, a_2, \dots), \quad \text{and} \quad a(n)$$

all describe the ordered list of numbers $a_0, a_1, a_2, \dots$. More formally, we have the following definition.

Definition 4.1. A **sequence** is a function $a : \mathbb{N} \rightarrow \mathbb{R}$. We use the notation $a_n := a(n)$ to denote the value of the function a at $n \in \mathbb{N}$. We use parentheses to list the ordered set of values of the sequence as follows:

$$(a_0, a_1, a_2, \dots).$$

In contrast, the notation $\{a_0, a_1, a_2, \dots\}$ refers to the set of values of the sequence.

Remark 4.2. We also allow sequences to be defined on any domain of the form $\{n \in \mathbb{N} \mid n \geq n_0\}$ for some fixed $n_0 \in \mathbb{N}$. Usually the initial value of a sequence doesn’t matter and I won’t make a big deal out of it.

Example 4.3. Let $a_n := \frac{1}{n}$ for $n \geq 1$. We can also describe this sequence as the ordered list

$$\left(1, \frac{1}{2}, \frac{1}{3}, \dots\right).$$

The set of values of the sequence is

$$\left\{1, \frac{1}{2}, \frac{1}{3}, \dots\right\}.$$

Example 4.4. Let $b_n := (-1)^n$ for $n \geq 0$. We can also describe this sequence as the ordered list

$$(1, -1, 1, -1, \dots).$$

The set of values of the sequence is

$$\{1, -1\}.$$

Example 4.5. Let $c_n := 0$ for $n \geq 0$. This is a perfectly good (constant) sequence. We can also describe this sequence as the ordered list

$$(0, 0, 0, 0, \dots).$$

The set of values of the sequence is

$$\{0\}.$$

Given a sequence, we are interested in its *limit*, if such a thing exists. Informally, the limit of a sequence is the number that the values tend to as n gets larger and larger. You secretly know from a former mathematical life that the limit of the sequence $\frac{1}{n}$ is 0, and the sequence $(-1)^n$ does not have a limit, but we want to define the concept of a limit in a completely rigorous way. The following is probably the most fundamental and important definition I will state the entire quarter.

Definition 4.6. Let a_n be a sequence of real numbers. We say that $L \in \mathbb{R}$ is the **limit** of a_n or that a_n **converges to** L , denoted

$$L = \lim_{n \rightarrow \infty} a_n \quad \text{or} \quad a_n \rightarrow L,$$

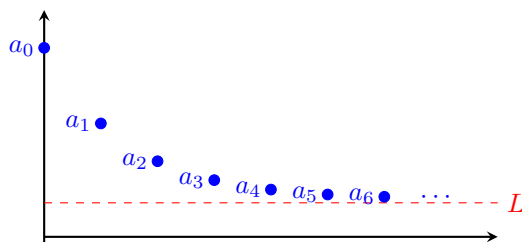
if the following condition holds:

For every $\varepsilon > 0$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $|a_n - L| < \varepsilon$.

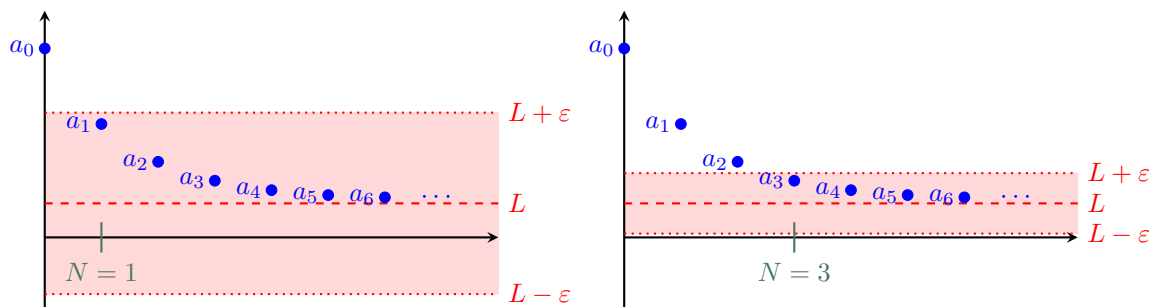
I've highlighted the above condition in blue just because it is extremely important! It is also difficult to understand and wield the first time you see it. We will spend the rest of this lecture (and the rest of this week) diving into this definition to understand what it means and how to use it and verify it in formal proofs. In fact, if there is any one concept or technique you should learn in an analysis class, it is a thorough understanding of this definition. We will see how things go, but I don't mind sacrificing some later topics and even spending up to three weeks just discussing limits of sequences.

4.1. Understanding the definition of a limit of a sequence. Here are a number of remarks about how to think of this definition. Afterwards, we will look at our first example.

Remark 4.7. Here are some pictures to help visualize the meaning of Definition 4.6. Let a_n for $n \in \mathbb{N}$ be a sequence. We can visualize this as a “graph” where there is a single dot sitting over each natural number $n \in \mathbb{N}$ on the x -axis. The height of each dot is the value $a_n \in \mathbb{R}$. I put “graph” in quotes but this is literally the graph of the function $a : \mathbb{N} \rightarrow \mathbb{R}$. For example, see the figure below:



Visually, it seems like the values of the sequence are approaching the height of the dashed red line, which is L . Suppose we fix a small number $\varepsilon > 0$. This choice of $\varepsilon > 0$ defines a band of vertical radius $\varepsilon > 0$ centered at height L ; see the figures below, where the left and right represent two choices of $\varepsilon > 0$.



Note that for both choices of $\varepsilon > 0$ in the above figures, the blue dots are eventually (permanently!) captured by the red zone. For the choice of $\varepsilon > 0$ on the left, once we reach $N = 1$ all of the subsequent blue dots are in the red zone (for all $n \geq N$, we have $|a_n - L| < \varepsilon$). For the choice of $\varepsilon > 0$ on the right, this doesn't happen until $N = 3$. Though I have only drawn two choices of $\varepsilon > 0$, it should be clear from the picture (assuming the rest of the blue dots that I can't draw behave as expected) that no matter how small we choose $\varepsilon > 0$, we will always be able to identify a point N in the sequence, such that, beyond that point, all blue dots are trapped in the red zone. Definition 4.6 defines " $a_n \rightarrow L$ " to mean exactly this. That is, a_n converges to L exactly when, given any $\varepsilon > 0$, we can always find a point N in the sequence such that all blue dots are trapped in the red zone thereafter.

Remark 4.8. One way to think about this definition, informally, is as a game between two players. You are Player 1, and you are playing against me, Player 2, aka the alien-robot game master. I give you a sequence a_n , and your goal as Player 1 is to try to convince me that $\lim_{n \rightarrow \infty} a_n = L$. My role in the game is to be as skeptical and combative as possible. Here is hypothetical typical round of the game.

- ◇ Player 1: I think the limit of the sequence a_n is L .
- ◇ Player 2: I don't believe you. If the limit truly was L , then you could find a point in the sequence² such that, beyond that point, every element of the sequence is within a distance of $\frac{1}{100}$.³ Can you find such a point?
- ◇ Player 1: Yes, I can find such a point. Let $N = 5000000$. Once you get beyond this point in the sequence, I can prove to you that *all* elements of the sequence are within $\frac{1}{100}$ of L .
- ◇ Player 2: I admit defeat. You win this round.

To describe what happened in this round, I (Player 2) gave you a challenge by picking a specific value of $\varepsilon > 0$. In particular, I picked $\varepsilon = \frac{1}{100}$. To complete my challenge, you (Player 1) had to find a point in the

²This is N .

³This is one specific choice of $\varepsilon > 0$, chosen by Player 2.

sequence (N) such that, beyond that point, *all* elements of the sequence are ε -close to L . If you can do that, you win the round.

Now, this is only one round. If you win the round I can say *okay fine, you win this time* but then I can just initiate another round and choose ε to be even smaller in attempt to make it harder for you. If you win that round (you'll have to find a new N , probably bigger than the one from the previous round) then I can just keep going forever. If you ever fail one of my challenges — that is, if I give you a certain $\varepsilon > 0$ and you *cannot* find a point N such that, beyond that point, all elements of the sequence are ε -close to L , then you lose the game. This means that you were wrong, and the limit of a_n is actually not L .

How do you win the game? I can keep picking smaller and smaller choices of $\varepsilon > 0$ for the rest of time. Eventually you get fed up and you decide to put a stop to the repetition of rounds. You can do this by saying *look, stop picking explicit stupidly small numbers for ε — I know that I will always be able to win, no matter what ε you pick. Let me prove it to you.* If you can then win a round of the game where $\varepsilon > 0$ is an abstract quantity, this proves that you will always be able to win any given round, and I will be forced to admit complete defeat. You win the game, which means you have convinced me that $\lim_{n \rightarrow \infty} a_n = L$.

Remark 4.9. By the Archimedean property, we can always assume that N is a natural number, but this is also a thing that doesn't really matter and I probably won't make a big deal out of it.

4.2. On proving statements with quantifiers. The quantifiers in the blue statement in Definition 4.6 are extremely important. Here is a brief discussion. There are two common quantifiers in logical sentences that appear often:

- ◇ There exists...
- ◇ For every...

For example, the statement

For every $\varepsilon > 0$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $|a_n - L| < \varepsilon$.

contains three quantifiers, followed by some mathematical condition. The three quantifiers are

- ◇ For every $\varepsilon > 0$...
- ◇ ...there is an $N \geq 0$ such that...
- ◇ ...for all $n \geq N$...

and the mathematical condition is the inequality $|a_n - L| < \varepsilon$. The mathematical condition is some statement involving all of the variables that are quantified in the preceding quantifiers. The first and third quantifiers are “for every” quantifiers, and the second quantifier is a “there exists” quantifier.

For every quantifier in a statement you want to prove, there is a corresponding quantification you write in the proof.

- ◇ There exists $x \in E$ such that...

What do you write in a proof?

If you want to prove a statement that contains a “there exists $x \in E$...” quantifier, it is up to you to provide such an $x \in E$ that satisfies whatever follows after the quantifier. Thus, in your proof you should write

Let $x = \text{SOMETHING}$.

or

Let $x \in E$ satisfy SOMETHING.

where the “SOMETHING” is dependent on whatever follows after the quantifier. Identifying the correct “SOMETHING” is usually the difficult part in proving a statement with a “there exists” quantifier. But whatever that “SOMETHING” is, you have to provide it.

◇ For every $y \in E$...

What do you write in a proof?

If you want to prove a statement that contains a “for every $y \in E$...” quantifier, you have to prove that whatever follows the quantifier is true *for all* $y \in E$. This means that you don’t get a choice as to what $y \in E$ is, because you have to prove it for *all* of them. Thus, the corresponding statement in your proof would simply be

Let $y \in E$.

Unlike with a “there exists” quantifier, you are not picking or providing an explicit $y \in E$. You are initializing an arbitrary $y \in E$ so that you can prove the statement for all such elements.

For example, a proof of the statement

For every $\varepsilon > 0$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $|a_n - L| < \varepsilon$.

would be of the following form.

Proof. Let $\varepsilon > 0$.

Let $N \in \mathbb{N}$ satisfy SOMETHING.

Let $n \geq N$.

Note that SOMETHING and therefore $|a_n - L| < \varepsilon$. □

The hardest part of the proof will be figuring out how to choose $N \in \mathbb{N}$ so that you can show in the last step that $|a_n - L| < \varepsilon$.

Remark 4.10. In a formal proof, **you should not use the logical symbols for “there exists” or “for all” or “therefore”**. Use words. We are humans, not robots. While there is a balance between using prose and symbols (a balance that is hard to learn, and will come with experience), the logical symbols for “there exists” and “for all” and “therefore” are ones you should avoid using in formal mathematical writing. No mathematician uses those symbols in papers they write, because they make things harder to read and understand. Just use words. Using the symbols in scratch work is fine, but do not use them in a final draft solution. I will repeat one last time: **Do not use logical symbols for “there exists”, “for all”, or “therefore” in submitted proofs.**

4.3. Baby’s first example of a limit of a sequence.

Example 4.11. Let’s now formally prove that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, keeping in mind everything we have discussed above. Before we write down a formal proof, we have some brainstorming and scratch work to do to figure out how the proof will go. Once we do that scratch work, then we will reformulate it into a rigorous and formal argument.

Scratchwork.

In order to verify that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, we need to verify the following statement:

For every $\varepsilon > 0$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $|\frac{1}{n} - 0| < \varepsilon$.

Because $n > 0$ we can simplify rewrite this as:

For every $\varepsilon > 0$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $\frac{1}{n} < \varepsilon$.

Because the sentence we need to verify begins with the quantifier “for every $\varepsilon > 0$ ” we know that we should begin our argument by initializing an arbitrary $\varepsilon > 0$. Because $\varepsilon > 0$ is arbitrary, if we can verify the rest of the statement for that arbitrary choice of $\varepsilon > 0$ then we will have verified something “for every $\varepsilon > 0$ ”. Thus, we will fix an arbitrary $\varepsilon > 0$. The next quantifier is “there exists an N such that...”. Because we need to verify a quantifier of the form “there exists and N ”, we know that the goal of our proof is to eventually write down “Let $N := \text{something}$ ” and we want to choose the “something” so that the stuff after “such that” in the blue sentence is true. How do we know what to declare for the value of N ? Well, the stuff after “such that” says “for all $n \geq N$, we have $\frac{1}{n} < \varepsilon$.” There is another “for all” quantifier, so — even though we haven’t determined what N is yet — we know that we should write “Let $n \geq N$.” Then, ultimately, the thing that we want to be true for this choice of n is

$$\frac{1}{n} < \varepsilon.$$

Rewriting this inequality gives $n > \frac{1}{\varepsilon}$. Therefore, as long as we pick N such that $N > \frac{1}{\varepsilon}$ (which is possible by the Archimedean property), then $n \geq N$ will imply $n > \frac{1}{\varepsilon}$ and thus $\frac{1}{n} < \varepsilon$ as desired.

Formal proof that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Let $\varepsilon > 0$.

Let $N \in \mathbb{N}$ be an integer such that $N > \frac{1}{\varepsilon}$.

Let $n \geq N$.

Note that

$$\begin{aligned} \left| \frac{1}{n} - 0 \right| &= \left| \frac{1}{n} \right| \\ &= \frac{1}{n} \\ &\leq \frac{1}{N} \\ &< \frac{1}{1/\varepsilon} \\ &= \varepsilon. \end{aligned}$$

Therefore, for all $\varepsilon > 0$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $|\frac{1}{n} - 0| < \varepsilon$. This implies that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Remark 4.12. Here is a more to-the-point way to describe how I went about finding my desired N , but I wouldn’t rely too heavily on this perspective (see, for example, the next lecture). Ultimately, the inequality I want to be true is of the form

$$|a_n - L| < \varepsilon$$

and so I algebraically “solve for” n in terms of ε to figure how big n needs to be in order to satisfy the inequality. This then dictates my choice of N . This perspective can sometimes be helpful when you are working with concrete sequence (that is, if you have a formula for a_n) but often we will be working with abstract quantities and this won’t do you any good.

Remark 4.13. I will point out the general structure of my proof above — both in content and physical presentation — and you’ll begin to notice that (up to some perturbation) any kind of limit proof follows the same general structure. I have three quantifiers that correspond to the three quantifiers in the definition of convergence. After quantifying everything, I then verify the desired inequality with a chain of aligned

inequalities. Finally, I have a summarizing statement, which can be a helpful expository tool for the reader of a proof.

Remark 4.14. Chapter 2, Section 8 of Ross has many more examples of informal discussions (what I'm calling my scratchwork) leading into formal proofs. That section would be very good to read alongside this lecture.

5. LECTURE 5: MORE ON LIMITS AND NEGATING QUANTIFIERS

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 2, Section 7 and Section 8.
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 3, some of “Convergent Sequences”, but be warned that the language here is more advanced than what we are using.
- ◇ Abbott, *Understanding Analysis*: Chapter 2, Section 2.

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Be able to use inequalities to your advantage when proving that limits exist (or do not exist).
- ◇ Understand how to negate the quantifiers in the ε -definition of a limit.
- ◇ Be able to prove that simple sequences do *not* converge.
- ◇ Be able to prove or disprove that simple sequences diverge to $\pm\infty$.

Example 5.1. Let $a_n = 3$. That is, let a_n be the constant sequence $(3, 3, 3, 3, \dots)$. We know intuitively that the limit of this sequence should be 3. Let’s warm up for this lecture by formally proving this using the ε -definition!

Formal proof that $\lim_{n \rightarrow \infty} 3 = 3$.

Let $\varepsilon > 0$.

Let $N = 1$.

Let $n \geq N$.

Note that

$$\begin{aligned} |a_n - 3| &= |3 - 3| \\ &= |0| \\ &= 0 \\ &< \varepsilon. \end{aligned}$$

Therefore, for all $\varepsilon > 0$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $|a_n - 3| < \varepsilon$. This implies that $\lim_{n \rightarrow \infty} a_n = 3$.

Example 5.2. Consider the sequence $a_n = \frac{n^2}{n^2+n+1}$. Let’s formally prove that $\lim_{n \rightarrow \infty} \frac{n^2}{n^2+n+1} = 1$.

Scratchwork.

As usual, we will begin by fixing $\varepsilon > 0$. Our goal is to find an N such that $n \geq N$ implies $|a_n - 1| < \varepsilon$. Rewriting this the left side of the inequality gives

$$\left| \frac{n^2}{n^2+n+1} - 1 \right| = \left| \frac{n^2 - (n^2+n+1)}{n^2+n+1} \right| = \left| \frac{-(n+1)}{n^2+n+1} \right| = \frac{n+1}{n^2+n+1}.$$

Thus, we need to determine how big N such be in order to ensure that

$$\frac{n+1}{n^2+n+1} < \varepsilon.$$

Unlike in the example we did in the previous lecture, it is difficult to “solve for n ” here. I’m not saying it’s impossible, but it’s not worth the trouble. Fortunately, in analysis, **we don’t care about equalities**. This is an extremely important principle! If we can make our lives easier with clever comparisons, we should absolutely do so. There are many ways to proceed, and that’s one of the fun parts about analysis. Here is one way.

Note that $n+1 \leq n+n=2n$, for all $n \geq 1$. Also note that $n+1 \geq 0$. This means that

$$\frac{n+1}{n^2+n+1} = \frac{(n+1)}{n^2+(n+1)} \leq \frac{2n}{n^2+(n+1)} \leq \frac{2n}{n^2} = \frac{2}{n}.$$

Why am I doing this? I am trying to find a quantity which is **bigger and simpler** than the thing I care about. In this case, I have discovered that $\frac{2}{n}$ is bigger than $\frac{n+1}{n^2+n+1}$, which is ultimately the thing we care about. If we can make $\frac{2}{n}$ small, which is much easier, then this will imply that we have made $\frac{n+1}{n^2+n+1}$ small. Anyway, $\frac{2}{n} < \varepsilon$ when $n > \frac{2}{\varepsilon}$. Thus, we can let N be an integer which satisfies $N > \frac{2}{\varepsilon}$.

Formal proof that $\lim_{n \rightarrow \infty} \frac{n^2}{n^2+n+1} = 1$.

Let $\varepsilon > 0$.

Let $N \in \mathbb{N}$ be an integer such that $N > \frac{2}{\varepsilon}$.

Let $n \geq N$.

Note that

$$\begin{aligned} \left| \frac{n^2}{n^2+n+1} - 1 \right| &= \left| \frac{n^2 - (n^2+n+1)}{n^2+n+1} \right| \\ &= \left| \frac{-(n+1)}{n^2+n+1} \right| \\ &= \frac{n+1}{n^2+n+1}. \end{aligned}$$

Next, note that $n+1 \leq n+n=2n$ and $n+1 \geq 0$. Thus,

$$\frac{n+1}{n^2+n+1} \leq \frac{2n}{n^2+n+1} \leq \frac{2n}{n^2} = \frac{2}{n}.$$

As $n \geq N > \frac{2}{\varepsilon}$, we have $\frac{2}{n} < \varepsilon$. Thus, in summary,

$$\begin{aligned} \left| \frac{n^2}{n^2+n+1} - 1 \right| &\leq \frac{2}{n} \\ &< \varepsilon. \end{aligned}$$

Therefore, for every $\varepsilon > 0$, there is an $N \geq 0$ such that, for all $n \geq N$, we have

$$\left| \frac{n^2}{n^2+n+1} - 1 \right| < \varepsilon.$$

It follows that $\lim_{n \rightarrow \infty} \frac{n^2}{n^2+n+1} = 1$.

Remark 5.3. My argument above is certainly not the only way to proceed! There are probably a number of other clever ways of estimating the desired quantity. That’s just what came to my mind first. For example,

someone in class⁴ suggested an even better estimate than what I've done above. Note that

$$\frac{n+1}{n^2+n+1} \leq \frac{n+1}{n^2+n} = \frac{n+1}{n(n+1)} = \frac{1}{n}.$$

So far we have only practiced proving that the limit of a sequence is some number. What if we wanted to prove that a limit doesn't exist? First, we need to formalize this concept.

Definition 5.4. A sequence a_n **diverges** if it does not converge to any real number.

5.1. A discussion on negating quantifiers. There are two ways to approach thinking about logic arguments in math (for example, understanding what it means to *not* converge to any real number by negating the definition of convergence):

- ◇ Think through the logic like a human.
- ◇ Learn formal rules for manipulating logical expressions and approach these things like a robot.

While both approaches have merit, the ideal goal when doing math is the former. Rules are like training wheels⁵ — they can help you when you are first learning, but you do not want to be dependent on them. Eventually, as you gain fluency and maturity in math, the logic (contrapositives, negations, implications, quantifiers) should come naturally. With that being said, it can be helpful as a sanity check to have some rules for how to logically negate quantified statements. For some quantity x , let $P(x)$ denote some statement about x . Like $P(x) = "x \text{ is blue}"$ or $P(x) = "x \geq 73"$ or $P(x) = "x \text{ is divisible by } 7"$. In a logical statement preceding a statement $P(x)$, there needs to be a quantifier that initializes the variable x . This means either "for all x " or "there exists x ". The negation rules are given as follows:

- ◇ The negation of

For every x , we have $P(x)$.

is

NOT(For every x , we have $P(x)$.) = There exists an x such that NOT($P(x)$).

- ◇ The negation of

There exists x such that $P(x)$.

is

NOT(There exists x such that $P(x)$.) = For all x we have NOT($P(x)$).

In words, negating a "there exists" quantifier gives you a "for all" quantifier, and vice versa, together with the negation of $P(x)$. For example, the negation of the statement

For $x \in E$, we have $x \geq 73$.

would be

NOT(For $x \in E$, we have $x \geq 73$.) = There exists $x \in E$ such that $x < 73$.

I will repeat, **I do not want you to memorize these rules and approach logic as such.** We are humans, not computers, and doing and communicating math is a human endeavor. The rules are there just to help. In general, the logic should just *make sense*. For instance, given a statement like "For $x \in E$, we have $x \geq 73$ ", I ask myself what would it mean for that to *not* be true. Well, if it were not the case that all elements of E

⁴I forget who — if you remind me, I'll cite you here!

⁵An analogy I'm stealing from Michael Andrews!

were greater than or equal to 73, then it must be the case that I can find an element of E which is smaller than 73. This just makes sense to me, and I don't have to write down rules or whatever to figure that out.

Let's look at one more example of negating statements, and then we'll go back to the concept of divergence.

Example 5.5. Consider the statement

There exists $x \in E$ such that, for all $y \geq x$, we have $y \in S$.

The negation of this statement would be

For all $x \in E$, there exists $y \geq x$ such that $y \notin S$.

You can get this by carefully applying negation rules, but it is also healthy to just think through the logic and make sure it makes sense. If you blindly apply the rules, things can get confusing. For example, why didn't I change the inequality $y \geq x$ in the negated statement?

5.2. An example proving divergence.

Example 5.6. Let's formally prove that the sequence $(-1)^n$ diverges.

Scratchwork. Let $L \in \mathbb{R}$. We will try to prove that $(-1)^n$ does not converge to L . Since L represents an arbitrary real number, this will prove that $(-1)^n$ diverges. To verify that $(-1)^n$ does *not* converge to L , we need to verify the negation of the blue statement in Definition 4.6. That is, we need to verify

NOT(For every $\varepsilon > 0$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $|(-1)^n - L| < \varepsilon$.)

Thus, we need to prove the following statement:

There is an $\varepsilon > 0$ such that, for all $N \geq 0$, there is an $n \geq N$ such that $|(-1)^n - L| \geq \varepsilon$.

In words, our first order of business is to provide an explicit $\varepsilon > 0$ such that, no matter how far we go out in the sequence, we can always find a sequence element that is at least ε far away from L . After drawing a picture, it should be clear that is impossible to squeeze all elements of the sequence into a band of total vertical thickness 1. Thus, we could choose $\varepsilon = \frac{1}{2}$ (but other choices would work!). Depending on whether L is positive or negative, we can let n be even or odd to ensure that $(-1)^n$ is at least $\frac{1}{2}$ from L . A formal proof is given below.

Formal proof that $(-1)^n$ diverges.

Let $L \in \mathbb{R}$. We will prove that $(-1)^n$ does not converge to L .

Let $\varepsilon = \frac{1}{2}$.

Let $N \geq 0$.

We consider two cases, according to whether L is nonnegative or negative. First, assume that $L \geq 0$. Let $n \geq N$ be any odd integer. Then

$$\begin{aligned} |(-1)^n - L| &= |-1 - L| \\ &= 1 + L \\ &\geq 1 \\ &> \frac{1}{2}. \end{aligned}$$

Next, assume that $L < 0$. Let $n \geq N$ be any even integer. Then

$$\begin{aligned} |(-1)^n - L| &= |1 - L| \\ &= 1 + |L| \\ &\geq 1 \\ &> \frac{1}{2}. \end{aligned}$$

In either case, there is an $n \geq N$ such that $|(-1)^n - L| \geq \frac{1}{2}$. Therefore, $(-1)^n$ does not converge to L . Since L was arbitrary, this proves that $(-1)^n$ diverges.

Remark 5.7. You could also prove that $(-1)^n$ diverges by contradiction, and you might end up with a slightly different looking argument. Ross does it this way in his book!

Remark 5.8. Later on we will discuss the theory of *subsequences*, and this will allow us to show that $(-1)^n$ diverges much more easily than the above admittedly annoying argument.

6. LECTURE 6: THEOREMS ABOUT LIMITS

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 2, Section 9
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 3, “Convergent Sequences”
- ◇ Abbott, *Understanding Analysis*: Chapter 2, Section 3

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Understand the meaning of diverge to $\pm\infty$ and be able to prove that simple sequences exhibit this behavior.
- ◇ Be able to prove abstract statements and properties about limits using the ε -definition.

First, there is also a more refined version of divergence, which is the concept of diverge to ∞ .

Definition 6.1. A sequence a_n **diverges to ∞** , written

$$\lim_{n \rightarrow \infty} a_n = \infty \quad \text{or} \quad a_n \rightarrow \infty$$

if, for every $M \geq 0$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $a_n \geq M$.

Similarly, a_n **diverges to $-\infty$** if, for every $m \leq 0$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $a_n \leq m$.

Remark 6.2. In this definition the number M (and m) take the role of ε in the usual definition of convergence. In the latter, ε defines a threshold for how close the sequence should be to the limit L . The definition of diverge to ∞ is morally supposed to be analogous to “converge to ∞ .” But ∞ is not a real number, and so it doesn’t make sense to talk about being “within ε of ∞ .” Instead, we fix this by requiring the threshold M . Roughly, this is a number that tells the sequence *this is how high you have to jump*. If, no matter how large M is, the sequence can always jump that high, then we say the sequence diverges to ∞ .

Example 6.3. Prove that $\frac{n^2}{n+1}$ diverges to ∞ .

Proof. Fix $M \geq 0$.

Let $N \in \mathbb{N}$ be an integer such that $N \geq 2M$.

Let $n \geq N$.

Note that

$$\begin{aligned}
 \frac{n^2}{n+1} &\geq \frac{n^2}{n+n} \\
 &= \frac{n^2}{2n} \\
 &= \frac{n}{2} \\
 &\geq \frac{N}{2} \\
 &\geq M.
 \end{aligned}$$

Thus, for all $M \geq 0$, there is an $N \geq 0$ such that $n \geq N$ implies $\frac{n^2}{n+1} \geq M$. This implies that $\frac{n^2}{n+1} \rightarrow \infty$. $\square$

Proposition 6.4. *Let a_n be a convergent sequence.*

- (i) *If $a_n \rightarrow L$ and $a_n \rightarrow L'$, then $L = L'$. That is, limits are unique.*
- (ii) *The sequence a_n is bounded. That is, the set of values $\{a_n\}$ is a bounded set. That is, there is an $M > 0$ such that, for all $n \geq 0$, $|a_n| \leq M$.*

Before we prove this proposition, here is a simple (and hopefully obvious) but important principle in analysis.

Principle 6.5. Let $a, b \in \mathbb{R}$. Then $a = b$ if and only if $|a - b| < \varepsilon$ for all $\varepsilon > 0$.

Proof of the principle. Fix $\varepsilon > 0$. If $a = b$, then $|a - b| = 0 < \varepsilon$.

Conversely, suppose that, for all $\varepsilon > 0$ we have $|a - b| < \varepsilon$. Suppose for the sake of contradiction that $a \neq b$. We can assume without loss of generality that $a < b$. By density of the rationals in $\mathbb{R}$, there is some $q \in \mathbb{Q}$ such that $a < q < b$. Choose $\varepsilon = b - q$. Then,

$$\begin{aligned}
 |a - b| &= b - a \\
 &> b - q \\
 &= \varepsilon.
 \end{aligned}$$

This contradicts the assumption that $|a - b| < \varepsilon$ for all $\varepsilon > 0$. Therefore, $a = b$. $\square$

Proof of Proposition 6.4.

- (i) We will show that $L = L'$ by showing that, for all $\varepsilon > 0$, we have $|L - L'| < \varepsilon$.

So, fix $\varepsilon > 0$. Since $a_n \rightarrow L$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $|a_n - L| < \frac{\varepsilon}{2}$. Since $a_n \rightarrow L'$, there is an $N' \geq 0$ such that, for all $n \geq N'$, we have $|a_n - L'| < \frac{\varepsilon}{2}$. Let n be any integer such that $n \geq N$ and $n \geq N'$. Then, by the triangle inequality,

$$\begin{aligned}
 |L - L'| &= |L - a_n + a_n - L'| \\
 &\leq |L - a_n| + |a_n - L'| \\
 &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\
 &= \varepsilon.
 \end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, we have shown that $|L - L'| < \varepsilon$ for all $\varepsilon > 0$ and so $L = L'$.

- (ii) Let $L = \lim_{n \rightarrow \infty} a_n$. Because $a_n \rightarrow L$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $|a_n - L| < 1$.
Let

$$M := \max(|a_0|, |a_1|, \dots, |a_{N-1}|, |L| + 1).$$

We need to show that $|a_n| \leq M$ for all $n \in \mathbb{N}$.

We consider two cases. First, suppose that $n \geq N$. Then

$$\begin{aligned} |a_n| &= |a_n - L + L| \\ &\leq |a_n - L| + |L| \\ &< 1 + |L| \\ &\leq M. \end{aligned}$$

Next, suppose that $0 \leq n \leq N - 1$. Then by definition of M , we have $|a_n| \leq M$.

□

Theorem 6.6 (Algebraic limit laws). *Let a_n and b_n be convergent sequences, and let $c \in \mathbb{R}$.*

- (i) *The sequence ca_n converges, and*

$$\lim_{n \rightarrow \infty} ca_n = c \lim_{n \rightarrow \infty} a_n.$$

- (ii) *The sequence $a_n + b_n$ converges, and*

$$\lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n.$$

- (ii) *The sequence $a_n b_n$ converges, and*

$$\lim_{n \rightarrow \infty} a_n b_n = \left(\lim_{n \rightarrow \infty} a_n \right) \cdot \left(\lim_{n \rightarrow \infty} b_n \right).$$

- (ii) *Suppose $\lim_{n \rightarrow \infty} b_n \neq 0$. The sequence $\frac{a_n}{b_n}$ converges, and*

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}.$$

Remark 6.7. Note that the assumption in the statement of the theorem is that a_n and b_n are convergent sequences. You cannot apply limit laws if a sequence diverges!

Proof.

- (i) Let $L := \lim_{n \rightarrow \infty} a_n$. We will prove that $ca_n \rightarrow cL$.

Fix $\varepsilon > 0$. Since $a_n \rightarrow L$, there is an $N \geq 0$ such that $|a_n - L| < \frac{\varepsilon}{|c|}$. Let $n \geq N$. Then

$$\begin{aligned} |ca_n - cL| &= |c||a_n - L| \\ &< |c| \frac{\varepsilon}{|c|} \\ &= \varepsilon. \end{aligned}$$

Thus, for all $\varepsilon > 0$, there is an $N \geq 0$ such that for all $n \geq N$ we have $|ca_n - cL| < \varepsilon$. This proves that $ca_n \rightarrow cL$.

- (ii) Let $L := \lim_{n \rightarrow \infty} a_n$ and $K := \lim_{n \rightarrow \infty} b_n$. We will prove that $a_n + b_n \rightarrow L + K$.

Let $\varepsilon > 0$.

Since $a_n \rightarrow L$, there is an N_1 such that $n \geq N_1$ implies $|a_n - L| < \frac{\varepsilon}{2}$. Similarly, since $b_n \rightarrow K$, there is an N_2 such that $n \geq N_2$ implies $|b_n - K| < \frac{\varepsilon}{2}$.

Let $N \geq \max(N_1, N_2)$. Let $n \geq N$. Then $n \geq N_1$ and $n \geq N_2$. Therefore,

$$\begin{aligned} |(a_n + b_n) - (L + K)| &= |(a_n - L) + (b_n - K)| \\ &\leq |a_n - L| + |b_n - K| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Thus, for all $\varepsilon > 0$, there is an $N \geq 0$ such that for all $n \geq N$ we have $|(a_n + b_n) - (L + K)| < \varepsilon$. This proves that $a_n + b_n \rightarrow L + K$.

(iii) Let $L := \lim_{n \rightarrow \infty} a_n$ and $K := \lim_{n \rightarrow \infty} b_n$. We will prove that $a_n b_n \rightarrow LK$.

Let $\varepsilon > 0$.

Because b_n converges, by Proposition 6.4 the sequence b_n is bounded. Thus, there is an $M > 0$ such that $|b_n| \leq M$ for all $n \in \mathbb{N}$. Since $a_n \rightarrow L$, there is an N_1 such that $n \geq N_1$ implies $|a_n - L| < \frac{\varepsilon}{2M}$. Since $b_n \rightarrow K$, there is an N_2 such that $n \geq N_2$ implies $|b_n - K| < \frac{\varepsilon}{2(|L|+1)}$.

Let $N = \max(N_1, N_2)$. Let $n \geq N$. Then $n \geq N_1$ and $n \geq N_2$. Note that

$$\begin{aligned} |a_n b_n - LK| &= |a_n b_n - Lb_n + Lb_n - LK| \\ &= |b_n(a_n - L) + L(b_n - K)| \\ &\leq |b_n||a_n - L| + |L||b_n - K| \\ &\leq M|a_n - L| + |L||b_n - K| \\ &< M \frac{\varepsilon}{2M} + |L| \frac{\varepsilon}{2(|L|+1)} \\ &= \frac{\varepsilon}{2} + \frac{|L|}{|L|+1} \frac{\varepsilon}{2} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Thus, for all $\varepsilon > 0$, there is an $N \geq 0$ such that for all $n \geq N$ we have $|a_n b_n - LK| < \varepsilon$. This proves that $a_n b_n \rightarrow LK$.

(iv) Omitted. This is one of your homework exercises!

□

Remark 6.8. In the proof of (iii), there is this funny choice of N_2 so that for $n \geq N_2$ we have $|b_n - K| < \frac{\varepsilon}{2(|L|+1)}$. Based on the previous kinds of arguments we have done, it would be more natural to choose N_2 so that $n \geq N_2$ implies $|b_n - K| < \frac{\varepsilon}{2|L|}$. Annoyingly, it may be the case that $L = 0$. For this reason, we adjust the denominator (making it *bigger!*) to account for this technicality.

7. LECTURE 7: MONOTONE SEQUENCES

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◊ Ross, *Elementary Analysis*: Chapter 2, Section 10
- ◊ Rudin, *Principles of Mathematical Analysis*: Chapter 3, some of “Cauchy sequences”
- ◊ Abbott, *Understanding Analysis*: Chapter 2, Section 4

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◊ Understand the definition of a monotone sequence, and understand the proof that a monotone bounded sequence converges.
- ◊ Be able to use this theorem to prove that certain sequences converge.

Recall the following definition, which I will state separately for convenience.

Definition 7.1. A sequence a_n is **bounded** if there is an $M \geq 0$ such that, for all $n \geq 0$, we have $|a_n| \leq M$.

Remark 7.2. In other words, a sequence is bounded if its set of values is a bounded set.

Definition 7.3 (Monotone sequences). Let a_n be a sequence.

- (i) We say that a_n is **increasing** if $a_{n+1} \geq a_n$ for all $n \geq 0$.
- (ii) We say that a_n is **strictly increasing** if $a_{n+1} > a_n$ for all $n \geq 0$.
- (iii) We say that a_n is **decreasing** if $a_{n+1} \leq a_n$ for all $n \geq 0$.
- (iv) We say that a_n is **strictly decreasing** if $a_{n+1} < a_n$ for all $n \geq 0$.

A **monotone** sequence is a sequence satisfying any of (i)-(iv) above.

Example 7.4. The sequence $a_n = \frac{1}{n^2}$ is monotonic. In particular, since

$$\frac{1}{(n+1)^2} < \frac{1}{n^2}$$

it follows that a_n is strictly decreasing.

Example 7.5. The sequence $b_n = \frac{(-1)^n}{n}$ is not monotonic.

Theorem 7.6. Let a_n be a monotonic bounded sequence. Then a_n converges.

Remark 7.7. Before we prove this, you should convince yourself that it is a very intuitive statement. A monotonic sequence, roughly, is a sequence that can only “move in one direction”. But if it is also bounded,

that means there is a wall in front of it. The sequence has nowhere to go, and so it eventually has to converge!

Proof. Let a_n be a monotonic bounded sequence. We will assume that a_n is increasing. The case where b_n is decreasing is nearly identical, and is left as an exercise.

Let

$$L := \sup\{a_n \mid n \in \mathbb{N}\}.$$

In words, L is the supremum of the set of values of the sequence. Since a_n is bounded by assumption, this supremum exists. We will show that $a_n \rightarrow L$.

Fix $\varepsilon > 0$. Since $L - \varepsilon < L$, the definition of the supremum implies that $L - \varepsilon$ is not an upper bound for $\{a_n \mid n \in \mathbb{N}\}$. Thus, there is an $N \geq 0$ such that $a_N > L - \varepsilon$. This means that $L - a_N < \varepsilon$.

Next, let $n \geq N$. Since a_n is monotonic, we have $a_n \geq a_N$ for all $n \geq N$. By definition of L , we also have $a_n \leq L$ for all $n \geq N$. Thus,

$$\begin{aligned} |a_n - L| &= L - a_n \\ &\leq L - a_N \\ &< \varepsilon. \end{aligned}$$

Therefore, for all $\varepsilon > 0$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $|a_n - L| < \varepsilon$. Thus, $a_n \rightarrow L$. $\square$

8. LECTURE 8: CAUCHY SEQUENCES

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 2, Section 10
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 3, some of “Cauchy Sequences”
- ◇ Abbott, *Understanding Analysis*: Chapter 2, Section 6

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Understand what it means for a sequence to be Cauchy, and how this differs from the definition of convergence.
- ◇ Be able to use the Cauchy definition to prove things about Cauchy sequences.

Definition 8.1 (Cauchy sequence). We say that a sequence a_n is **Cauchy** if the following condition holds.

For every $\varepsilon > 0$, there is an $N \geq 0$ such that, for all $n, m \geq N$, we have $|a_n - a_m| < \varepsilon$.

Remark 8.2. It is important to compare and contrast this definition with the usual definition of convergence, because they are similar. As a reminder, a sequence a_n **converges** if:

There is an $L \in \mathbb{R}$ such that, for every $\varepsilon > 0$, there is an $N \geq 0$ such that, for all $n \geq N$, we have $|a_n - L| < \varepsilon$.

In contrast, a sequence a_n is **Cauchy** if:

For every $\varepsilon > 0$, there is an $N \geq 0$ such that, for all $n, m \geq N$, we have $|a_n - a_m| < \varepsilon$.

The most important difference is that the definition of convergence references the limit $L \in \mathbb{R}$, and in order to verify it you need to know what L is. The definition of Cauchy (which, to be clear, is a priori completely unrelated to the notion of convergence) does not require prior knowledge of some number $L \in \mathbb{R}$. Instead, being Cauchy is a statement that says that eventually the elements of the sequence get close together, rather than saying that the elements of a sequence get close to some number L .

Example 8.3. The sequence $a_n = \frac{1}{n}$ is Cauchy. Indeed, fix $\varepsilon > 0$. Pick $N \geq 0$ such that $N > \frac{1}{\varepsilon}$. Let $n, m \geq N$ and assume without loss of generality that $m \geq n$. Then

$$\begin{aligned} |a_n - a_m| &= \left| \frac{1}{n} - \frac{1}{m} \right| \\ &= \frac{1}{n} - \frac{1}{m} \\ &\leq \frac{1}{n} \\ &\leq \frac{1}{N} \\ &< \varepsilon. \end{aligned}$$

Example 8.4. The sequence $b_n = (-1)^n$ is not Cauchy. Indeed, let $\varepsilon = \frac{1}{2}$. Let $N \geq 0$. Let $n \geq N$ and let $m := n + 1$. Then

$$\begin{aligned} |b_n - b_m| &= |b_n - b_{n+1}| \\ &= |(-1)^n - (-1)^{n+1}| \\ &= 2 \\ &> \varepsilon. \end{aligned}$$

Therefore, $(-1)^n$ is not Cauchy.

Remark 8.5. Make sure you understand the quantifiers used in this last example to negate the meaning of Cauchy!

Proposition 8.6 (Basic properties of Cauchy sequences). *Let a_n be a sequence.*

- (i) *If a_n converges, then it is Cauchy.*
- (ii) *If a_n is Cauchy, it is bounded.*

Proof.

- (i) Let $L := \lim_{n \rightarrow \infty} a_n$. We need to verify the Cauchy criterion for a_n .

Fix $\varepsilon > 0$. Since $a_n \rightarrow L$, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$, we have $|a_n - L| < \frac{\varepsilon}{2}$. Let $n, m \geq N$. Then

$$\begin{aligned} |a_n - a_m| &= |a_n - L + L - a_m| \\ &\leq |a_n - L| + |L - a_m| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Thus, for every $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that $n, m \geq N$ implies $|a_n - a_m| < \varepsilon$. Therefore, a_n is Cauchy.

- (ii) This proof is very similar to the proof that convergent sequences are bounded.

Since a_n is Cauchy, there is an $N \in \mathbb{N}$ such that for all $n \geq N$ we have $|a_n - a_m| < 1$. Let

$$M := \max(|a_0|, |a_1|, \dots, |a_{N-1}|, |a_N| + 1).$$

We claim that $|a_n| \leq M$ for all $n \in \mathbb{N}$. If $0 \leq n \leq N$, then clearly $|a_n| \leq M$. If $n > N$, then

$$\begin{aligned} |a_n| &= |a_n - a_N + a_N| \\ &\leq |a_n - a_N| + |a_N| \\ &< 1 + |a_N| \\ &\leq M. \end{aligned}$$

Thus, a_n is bounded.

□

Remark 8.7. The definition of a Cauchy sequence is nice, exactly because it doesn't reference any previously known quantity L . It is only a statement about the elements of the sequence, and so in practice it is easier to work with. In Lecture 11, we will prove that being Cauchy is actually completely equivalent to being convergent! We don't know this yet (we have only proven one direction of that implication, namely, convergent implies Cauchy) but once we do it will often be convenient to check whether a sequence is Cauchy in order to check whether it converges.

9. LECTURE 9: SUBSEQUENCES

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 2, Section 11
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 3, “Subsequences”
- ◇ Abbott, *Understanding Analysis*: Chapter 2, Section 5

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Understand what a subsequence is and how to prove simple things about them.
- ◇ Be able to use the divergence criterion to show that certain sequences diverge.

Definition 9.1. Let a_n be a sequence of real numbers. Let

$$n_0 < n_1 < n_2 < \cdots$$

be an increasing sequence of natural numbers. The sequence $(a_{n_k})_{k=0}^{\infty}$, i.e., the sequence

$$(a_{n_0}, a_{n_1}, a_{n_2}, \dots)$$

is called a **subsequence** of a_n .

Remark 9.2. Note that the sequence of natural numbers $(n_0, n_1, n_2, \dots)$ in the above definition diverges to ∞ .

Example 9.3. Consider the sequence $a_n = \frac{(-1)^n}{n}$. Let $n_k = 2k$. Then a_{n_k} is a valid subsequence of a_n given by

$$a_{n_k} = a_{2k} = \frac{(-1)^{2k}}{2k} = \frac{1}{2k}.$$

Proposition 9.4. Suppose that a_n is a sequence such that $a_n \rightarrow L \in \mathbb{R}$. Then every subsequence of a_n also converges to L . That is, if a_{n_k} is a subsequence of a_n , then $a_{n_k} \rightarrow L$ as $k \rightarrow \infty$.

Remark 9.5. In words, this proposition says that if a sequence converges, then *any* subsequence of that sequence will also converge to the same limit.

Proof. Fix $\varepsilon > 0$. We need to find a $K \in \mathbb{N}$ such that, for all $k \geq K$, we have $|a_{n_k} - L| < \varepsilon$.

Since $a_n \rightarrow L$, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$, we have $|a_n - L| < \varepsilon$. Let $K \in \mathbb{N}$ be large enough so that $n_K \geq N$. Let $k \geq K$. Then $n_k \geq n_K \geq N$, and so $|a_{n_k} - L| < \varepsilon$ as desired. □

Corollary 9.6 (Divergence criterion). Let a_n be a sequence. Suppose that a_{n_k} and $a_{n'_k}$ are subsequences of a_n such that $a_{n_k} \rightarrow L$ and $a_{n'_k} \rightarrow L'$ where $L \neq L'$. Then a_n diverges.

Example 9.7. Consider the sequence $a_n = (-1)^n$. Earlier we showed that this sequence diverges by negating the definition of convergence, which was kind of annoying. With the divergence criterion it is much easier to prove divergence. Consider the subsequences $(a_{2k})_{k=0}^{\infty}$ and $(a_{2k+1})_{k=0}^{\infty}$. Note that

$$a_{2k} = (-1)^{2k} = 1 \quad \text{for all } k \in \mathbb{N}$$

so that $a_{2k} \rightarrow 1$. Also note that

$$a_{2k+1} = (-1)^{2k+1} = -1 \quad \text{for all } k \in \mathbb{N}$$

so that $a_{2k+1} \rightarrow -1$. Since $1 \neq -1$, it follows that a_n diverges.

This will be the end of the content that is covered on the first midterm exam.

10. LECTURE 10: LIM INFS AND LIM SUPS

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

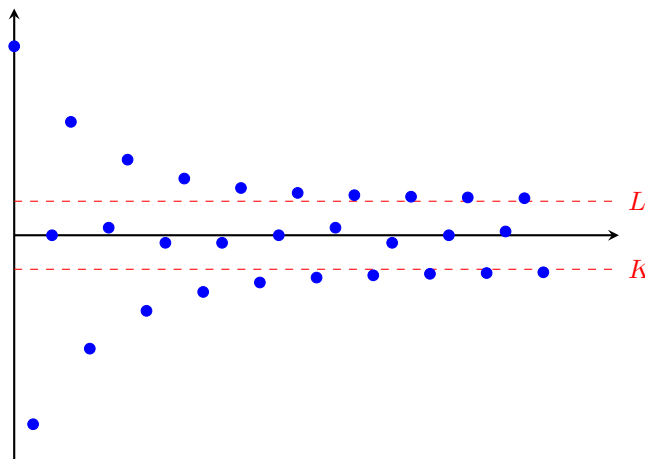
- ◊ Ross, *Elementary Analysis*: Chapter 2, Section 10
- ◊ Rudin, *Principles of Mathematical Analysis*: Chapter 3, some of “Upper and Lower Limits”
- ◊ Abbott, *Understanding Analysis*: Chapter 2, Section 5

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◊ Understand the meaning of lim infs and lim sups, and be able to intuit them in basic examples.
- ◊ Be able to prove basic properties of lim infs and lim sups.

Let a_n be a sequence, and for now we will assume that a_n is bounded (so that $\sup\{a_n \mid n \in \mathbb{N}\}$ and $\inf\{a_n \mid n \in \mathbb{N}\}$ exist). For example, consider the sequence depicted by the blue dots in the figure below.



Note that the sequence does not seem to converge. However, there are two quantities that seem to be of interest: L and K . Currently, we do not have words to describe what these numbers are, but vaguely, L is an “asymptotic supremum” and K is an “asymptotic infimum.” The goal of this lecture is to develop these concepts carefully.

We can construct a new sequence, indexed over a new variable $N \in \mathbb{N}$, by taking the supremum after further and further cutoffs of the sequence a_n . By that I mean the following: For any $N \in \mathbb{N}$, we can consider

$$\sup\{a_n \mid n \geq N\}.$$

For each choice of N , we can give this number a name:

$$U_N := \sup\{a_n \mid n \geq N\}.$$

The collection of numbers

$$(U_0, U_1, U_2, \dots)$$

is a sequence, indexed by N . We claim that this is a decreasing sequence. Indeed, note that

$$\{a_n \mid n \geq N+1\} \subset \{a_n \mid n \geq N\}.$$

Thus, any upper bound for $\{a_n \mid n \geq N\}$ is an upper bound for $\{a_n \mid n \geq N+1\}$. This implies that

$$\sup\{a_n \mid n \geq N+1\} \leq \sup\{a_n \mid n \geq N\}$$

and so $U_{N+1} \leq U_N$. We can also show that (U_N) is a bounded sequence. Indeed, note that

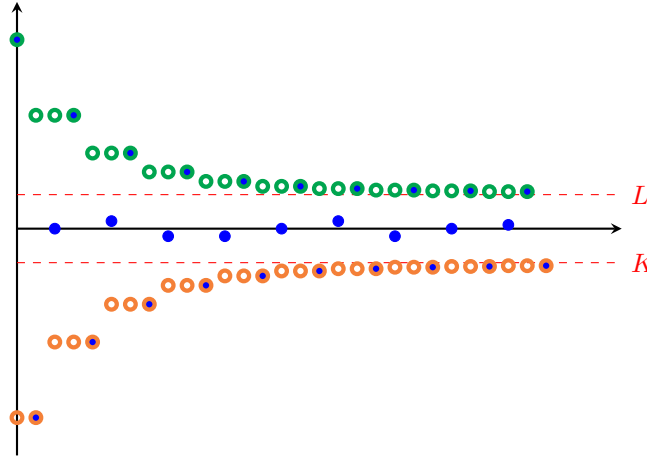
$$U_N = \sup\{a_n \mid n \geq N\} \geq \inf\{a_n \mid n \geq N\} \geq \inf\{a_n \mid n \in \mathbb{N}\}$$

since $\{a_n \mid n \geq N\} \subset \{a_n \mid n \in \mathbb{N}\}$. This shows that the number $\inf\{a_n \mid n \in \mathbb{N}\}$ is a lower bound for the sequence U_N . By Theorem 7.6, it follows that $\lim_{N \rightarrow \infty} U_N$ exists.

We can repeat the same story by looking at the sequence of infimums. In particular, let

$$V_N := \inf\{a_n \mid n \geq N\}.$$

Since $\{a_n \mid n \geq N+1\} \subset \{a_n \mid n \geq N\}$, it follows that $V_{N+1} \geq V_N$, and so V_N is an increasing sequence. Furthermore, V_N is bounded above because $\{a_n \mid n \geq N\} \subset \{a_n \mid n \in \mathbb{N}\}$ and so $V_N = \inf\{a_n \mid n \geq N\} \leq \sup\{a_n \mid n \in \mathbb{N}\}$. Thus, $\lim_{N \rightarrow \infty} V_N$ exists. The figure below depicts the sequences U_N (given by the open green dots) and V_N (given by the open orange dots) on top of a_n .



Definition 10.1. Let a_n be a bounded sequence. Define

$$\limsup a_n := \lim_{N \rightarrow \infty} \sup\{a_n \mid n \geq N\}$$

and

$$\liminf a_n := \lim_{N \rightarrow \infty} \inf\{a_n \mid n \geq N\}.$$

Example 10.2. Let

$$a_n = \begin{cases} 1 + \frac{1}{n} & \text{if } n \text{ is even} \\ -1 - \frac{1}{n} & \text{if } n \text{ is odd} \end{cases}.$$

Fix $N \in \mathbb{N}$. Note that

$$\sup\{a_n \mid n \geq N\} = \begin{cases} 1 + \frac{1}{N} & \text{if } N \text{ is even} \\ 1 + \frac{1}{N+1} & \text{if } N \text{ is odd} \end{cases}.$$

In particular, for each $N \in \mathbb{N}$, we have $1 \leq \sup\{a_n \mid n \geq N\} \leq 1 + \frac{1}{N}$. Since $1 + \frac{1}{N} \rightarrow 1$ as $N \rightarrow \infty$, it follows that $\limsup a_n = 1$. One can similarly show that $\liminf a_n = -1$.

Theorem 10.3. *Let a_n be a bounded sequence.*

(i) *We have*

$$\liminf a_n \leq \limsup a_n.$$

(ii) *The sequence a_n converges if and only if $\limsup a_n = \liminf a_n$. Moreover, if a_n converges, then*

$$\lim_{n \rightarrow \infty} a_n = \limsup a_n = \liminf a_n.$$

Proof.

(i) Note that, for all $N_0 \in \mathbb{N}$, we have

$$V_{N_0} \leq U_{N_0}.$$

This implies $\sup\{V_N \mid N \in \mathbb{N}\} \leq U_{N_0}$. By the proof of Theorem 7.6, the left hand side of this inequality is exactly $\lim_{N \rightarrow \infty} V_N = \liminf a_n$, hence $\liminf a_n \leq U_{N_0}$. Since this holds for all $N_0 \in \mathbb{N}$, it follows that $\liminf a_n \leq \inf\{U_N \mid N \in \mathbb{N}\}$. The right hand side is exactly $\lim_{N \rightarrow \infty} U_N = \limsup a_n$. Thus,

$$\liminf a_n \leq \limsup a_n$$

as desired.

(ii) First, assume that $a_n \rightarrow L \in \mathbb{R}$. We need to prove that $\limsup a_n = \liminf a_n = L$. Fix $\varepsilon > 0$. Since $a_n \rightarrow L$, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$, we have $|a_n - L| < \varepsilon$. Then for all $n \geq N$, it follows that

$$a_n < L + \varepsilon \quad \text{and} \quad a_n > L - \varepsilon$$

by unraveling the definition of the absolute value. This means that

$$U_N = \sup\{a_n \mid n \geq N\} \leq L + \varepsilon \quad \text{and} \quad V_N = \inf\{a_n \mid n \geq N\} \geq L - \varepsilon.$$

Sending $N \rightarrow \infty$ and recalling that U_N is decreasing and V_N is increasing, we get

$$\lim_{N \rightarrow \infty} U_N \leq L + \varepsilon \quad \text{and} \quad \lim_{N \rightarrow \infty} V_N \geq L - \varepsilon$$

and so

$$\limsup a_n \leq L + \varepsilon \quad \text{and} \quad \liminf a_n \geq L - \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, it follows that

$$\limsup a_n \leq L \quad \text{and} \quad \liminf a_n \geq L.$$

Thus, $L \leq \liminf a_n \leq \limsup a_n \leq L$, and so $\limsup a_n = \liminf a_n = L$ as desired.

For the other direction of the if and only if, suppose that $\limsup a_n = \liminf a_n = L$. We need to prove that $a_n \rightarrow L$. Fix $\varepsilon > 0$. Because $\limsup a_n = L$, which means $U_N \rightarrow L$, there is an $N_+ \in \mathbb{N}$ such that, for all $N \geq N_+$, we have

$$|U_N - L| < \varepsilon.$$

Then $U_{N_+} < L + \varepsilon$, which is to say $\sup\{a_n \mid n \geq N_+\} < L + \varepsilon$. In particular, for all $n \geq N_+$ we have $a_n < L + \varepsilon$. Similarly, since $\liminf a_n = L$ there is an N_- such that, for all $n \geq N_-$, we have

$$|V_N - L| < \varepsilon.$$

This implies $V_{N_-} > L - \varepsilon$ and so $a_n > L - \varepsilon$ for all $n \geq N_-$. Thus, for $n \geq \max(N_-, N_+)$, we have $-\varepsilon < a_n - L < \varepsilon$, hence $|a_n - L| < \varepsilon$. This proves that $a_n \rightarrow L$. □

The proof of the above theorem suggests the following principle, which is a useful way of working with and proving things about liminfs and limsups.

Principle 10.4. Let a_n be a bounded sequence. Then

$$\limsup a_n = L \quad \Longleftrightarrow \quad \text{For all } \varepsilon > 0 \text{ there is } N \in \mathbb{N} \text{ such that for all } n \geq N \text{ we have } a_n < L + \varepsilon.$$

and

$$\liminf a_n = K \quad \Longleftrightarrow \quad \text{For all } \varepsilon > 0 \text{ there is } N \in \mathbb{N} \text{ such that for all } n \geq N \text{ we have } a_n > K - \varepsilon.$$

Writing down a full proof of this principle would be a good exercise! It essentially boils down to reusing many of the arguments in the above proof. The homework exercises will also give you more convenient ways of thinking about lim sups and lim infs.

11. LECTURE 11: THE CAUCHY-CONVERGENCE EQUIVALENCE AND THE BOLZANO-WEIERSTRASS THEOREM

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◊ Ross, *Elementary Analysis*: Chapter 2, Section 11
- ◊ Abbott, *Understanding Analysis*: Chapter 2, Section 5

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◊ Understand how to use lim sups and lim infs to prove that a Cauchy sequence converges.
- ◊ Understand the statement and proof of the Bolzano-Weierstrass theorem via the bisection method.

Sequences will play an important part in our study of analysis for the rest of the quarter, but this will be the last lecture (at least for a while) where we discuss sequences in their own right. Beginning with the next lecture we will move on to discussing continuity. To close out our study of sequences, we will state and prove two major theorems: a theorem that says that (in the reals!) convergence is equivalent to being Cauchy, and another theorem known as the Bolzano-Weierstrass theorem.

Theorem 11.1 (Cauchy-convergence equivalence). *Let a_n be a sequence of real numbers. Then a_n is Cauchy if and only if a_n converges.*

Proof. Since this is an if and only if statement, there are two directions to prove. One of them (if a sequence is convergent, then it is Cauchy) we have already proven in Lecture 8. Thus, it only remains to prove that if a sequence is Cauchy, it is convergent.

Let a_n be a Cauchy sequence. Since Cauchy sequences are bounded, $\limsup a_n$ and $\liminf a_n$ exist. By Theorem 10.3, to prove that a_n converges it suffices to prove $\liminf a_n = \limsup a_n$. By the same theorem, it is automatically the case that $\liminf a_n \leq \limsup a_n$, so we only need to prove $\liminf a_n \geq \limsup a_n$. In order to do this, we will prove that $\liminf a_n \geq (\limsup a_n) - \varepsilon$ for all $\varepsilon > 0$.

Fix $\varepsilon > 0$.

Since a_n is Cauchy, there is an $N \in \mathbb{N}$ such that, for all $n, m \geq N$, we have $|a_n - a_m| < \varepsilon$. In particular, for all $n, m \geq N$ we have $a_n < a_m + \varepsilon$. This means that for any $m \geq N$, the number $a_m + \varepsilon$ is an upper bound for $\{a_n \mid n \geq N\}$. Thus, for all $m \geq N$, we have

$$\sup\{a_n \mid n \geq N\} \leq a_m + \varepsilon.$$

So for all $m \geq N$,

$$\lim_{N \rightarrow \infty} \sup\{a_n \mid n \geq N\} \leq a_m + \varepsilon$$

which means that $\limsup a_n \leq a_m + \varepsilon$ for all $m \geq N$.

The above argument shows that $(\limsup a_n) - \varepsilon$ is a lower bound for the set $\{a_m \mid m \geq N\}$. Thus,

$$\inf\{a_m \mid m \geq N\} \geq (\limsup a_n) - \varepsilon$$

and hence

$$\lim_{N \rightarrow \infty} \inf\{a_m \mid m \geq N\} \geq (\limsup a_n) - \varepsilon$$

which gives $\liminf a_n \geq (\limsup a_n) - \varepsilon$, as desired.

Since $\varepsilon > 0$ was arbitrary, it follows that $\liminf a_n \geq \limsup a_n$, and so $\liminf a_n = \limsup a_n$ and thus a_n converges. □

Remark 11.2. Like I remarked in Lecture 8, this theorem is very useful because it gives us a way to test for convergence without having to know in advance what the potential limit of a sequence is.

Theorem 11.3 (Bolzano-Weierstrass theorem). *Let a_n be a bounded sequence. Then a_n has a convergent subsequence.*

Proof. Because a_n is bounded, there is an $M > 0$ such that $|a_n| \leq M$ for all $n \in \mathbb{N}$. Thus, $\{a_n \mid n \in \mathbb{N}\} \subset [-M, M]$.

Because (a_n) is a list of infinitely many numbers, it must be the case that either $[-M, 0]$ or $[0, M]$ contains infinitely many elements of (a_n) .⁶ Let I_0 denote (one of) the interval(s) with infinitely many elements. Note that

$$\max I_0 - \min I_0 = M.$$

We repeat this process beginning with the interval I_0 . That is, bisect the interval and let I_1 be a choice of the resulting two intervals that necessarily contains infinitely many elements of the sequence. Note that

$$\max I_1 - \min I_1 = \frac{M}{2}.$$

Continuing in this way gives us a sequence of intervals

$$I_0 \supset I_1 \supset I_2 \supset I_3 \supset \cdots$$

such that

$$\max I_k - \min I_k = \frac{M}{2^k}.$$

See Figure 1. Next, let $n_0 \in \mathbb{N}$ such that $a_{n_0} \in I_0$. Then choose $n_1 > n_0$ such that $a_{n_1} \in I_1$. Note that this is possible because I_1 has infinitely many elements of the sequence. Having (inductively) chosen n_{k-1} such that $a_{n_{k-1}} \in I_{k-1}$, choose $n_k > n_{k-1}$ such that $a_{n_k} \in I_k$. Doing this for all $k \in \mathbb{N}$ gives a subsequence

$$(a_{n_k})_{k=0}^{\infty} = (a_{n_0}, a_{n_1}, a_{n_2}, \dots)$$

of (a_n) . We claim that a_{n_k} converges. By Theorem 11.1, it suffices to prove that a_{n_k} is Cauchy.

Fix $\varepsilon > 0$. Let $K \in \mathbb{N}$ be large enough so that $2^K > \frac{M}{\varepsilon}$. Let $k, j \geq K$. Since $a_{n_k} \in I_k \subset I_K$ and $a_{n_j} \in I_j \subset I_K$, it follows that $\min I_K \leq a_{n_j}, a_{n_k} \leq \max I_K$. Thus,

$$|a_{n_k} - a_{n_j}| \leq \max I_K - \min I_K = \frac{M}{2^K} < \varepsilon.$$

⁶It could be both!

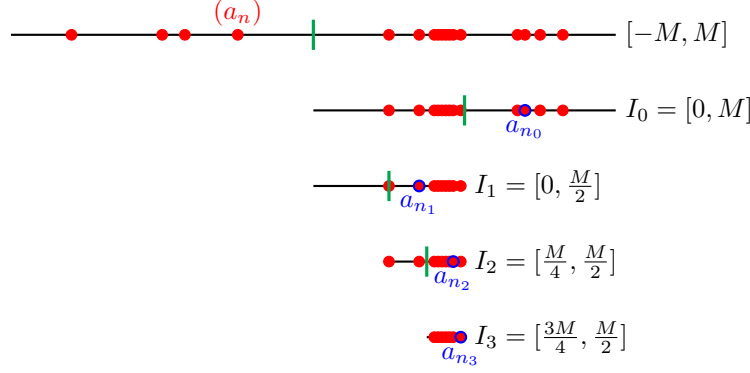


FIGURE 1. A figure depicting the proof of the Bolzano-Weierstrauss theorem. The set of values of the sequence a_n is depicted by the collection of red dots in the first line, contained in the interval $[-M, M]$. We bisect the interval $[-M, M]$, indicated by the green line, and let I_0 be the interval that contains infinitely many elements of the sequence. Roughly, we choose a sequence element $a_{n_0} \in I_0$ and then bisect *this* interval and continue this process to find a subsequence (a_{n_k}) , indicated by the open blue dots, which is Cauchy.

Thus, (a_{n_k}) is Cauchy and hence convergent, which completes the proof of the theorem. □

11.1. An alternative proof of the Bolzano-Weierstrauss theorem. There is another way to prove Theorem 11.3 by appealing to the existence of a monotone subsequence of (a_n) . Arguably, this alternative proof is shorter. However, the bisection method presented above is more fun, and it is the kind of argument that rears its head in other areas of math. For the sake of variety, here is the alternative proof. We begin with a lemma about subsequences.

Lemma 11.4. *Let (a_n) be a sequence. Then (a_n) contains a monotonic subsequence.*

Proof. We introduce a new definition. Fix $n_0 \in \mathbb{N}$. We say that the term a_{n_0} is **dominant** if, for all $n > n_0$, we have $a_n < a_{n_0}$. Given any sequence (a_n) , it is either the case that (a_n) has infinitely many dominant terms, or only finitely many dominant terms.

Case I: The sequence (a_n) has infinitely many dominant terms.

If (a_n) has infinitely many dominant terms, then we can select a subsequence (a_{n_k}) such that each a_{n_k} term is dominant. We claim that (a_{n_k}) is strictly decreasing. For each $k \in \mathbb{N}$, since $n_{k+1} > n_k$ and since a_{n_k} is dominant, we have $a_{n_{k+1}} < a_{n_k}$. Thus, (a_{n_k}) is strictly decreasing.

Case II: The sequence (a_n) has finitely many dominant terms.

Suppose (a_n) has finitely many dominant terms. Let $n_0 \in \mathbb{N}$ be any index beyond the last dominant term of (a_n) . Then a_{n_0} is not dominant. Therefore, there exists some $n_1 > n_0$ such that $a_{n_1} \geq a_{n_0}$. Since a_{n_1} is not dominant, there is an $n_2 > n_1$ such that $a_{n_2} \geq a_{n_1}$. Continuing on in this way, we get

$$n_0 < n_1 < n_2 < \dots$$

such that $a_{n_{k+1}} \geq a_{n_k}$ for each $k \in \mathbb{N}$. Thus, (a_{n_k}) is an increasing subsequence of (a_n) . □

Proof of Theorem 11.3. Let (a_n) be a bounded sequence. By the previous lemma, (a_n) contains a monotone subsequence (a_{n_k}) . Since (a_n) is bounded, the monotone subsequence (a_{n_k}) is also bounded, and hence converges. □

Part 3

Continuity

12. LECTURE 12: AN INTRODUCTION TO CONTINUITY

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◊ Ross, *Elementary Analysis*: Chapter 3, Section 17
- ◊ Rudin, *Principles of Mathematical Analysis*: Chapter 4, “Continuous functions”
- ◊ Abbott, *Understanding Analysis*: Chapter 4, Section 3

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◊ Understand the $\varepsilon - \delta$ definition of continuity of a function at a point. In particular, be able to compare and contrast the use of quantifiers in this definition with the definition of convergence of a sequence.
- ◊ Be able to verify via the definition that basic functions are continuous.

So far, we have almost exclusively studied sequences of real numbers. Analysis is mostly concerned with the study of *functions*, and this is where we will head next. In particular, we will begin by developing the theory of continuous functions at a rigorous level.

First, some notation. An *open interval* is any set of the form $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$. Sometimes open intervals will be denoted by the letter I . For the moment we will only discuss functions defined on open intervals. Note we consider $\mathbb{R}$ to be an open interval. Also recall that the notation

$$f : I \rightarrow \mathbb{R}$$

means that f is a function with domain $I \subset \mathbb{R}$ and codomain $\mathbb{R}$. That is, for each $x \in I$, there is an associated number $f(x) \in \mathbb{R}$.

Example 12.1. Let $f : (1, 2) \rightarrow \mathbb{R}$ be given by $f(x) = \frac{1}{x}$.

Example 12.2. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$g(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}.$$

Remark 12.3. When you grow up learning math, you tend to think of a function as a formula. It is important to stop thinking this way! A function is not necessarily given by a formula. Functions that are described by formulas are extremely rare in the world of all functions!

Definition 12.4 ($\varepsilon - \delta$ definition of continuity). Let $f : I \rightarrow \mathbb{R}$ be a function defined on an open interval $I \subset \mathbb{R}$. Let $x \in I$. We say that f is **continuous at** x_0 if the following condition holds:

For every $\varepsilon > 0$, there is a $\delta > 0$ such that, for all $x_0 \in I$ such that $|x - x_0| < \delta$, we have
 $|f(x) - f(x_0)| < \varepsilon$.

We say that f is **continuous** if, for all $x_0 \in I$, f is continuous at x_0 .

Just like the definition of convergence of a limit, continuity at x_0 is a triple quantified statement beginning with a “for all $\varepsilon > 0$ ” quantifier. Instead of being tasked with finding an $N \in \mathbb{N}$ satisfying some properties, we are instead tasked with finding a $\delta > 0$ satisfying some properties. Figure 2 and Figure 3 contain visualizations to help understand what the $\varepsilon - \delta$ definition of continuity is saying.

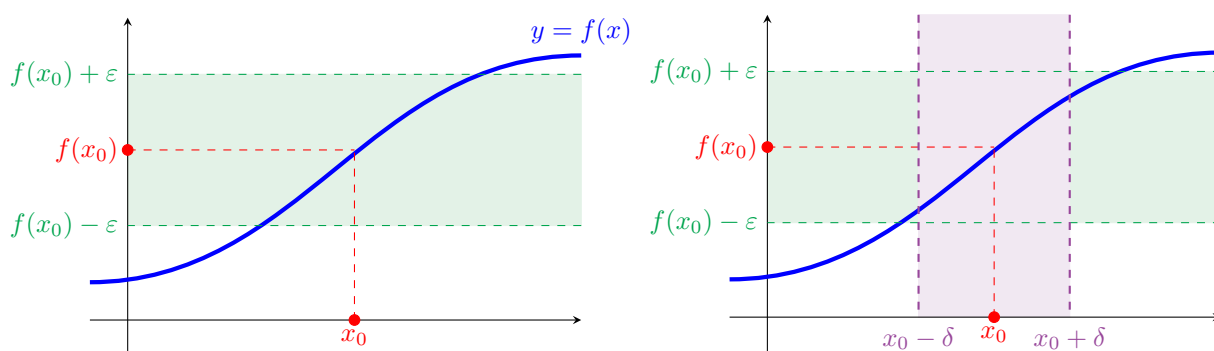


FIGURE 2. A visual depiction of the $\varepsilon - \delta$ definition of continuity. Both figures depict the graph of a function f which is continuous at a point x_0 in the domain. To verify the definition of continuity at x_0 , we are given an $\varepsilon > 0$, which determines a “winning zone” strip of vertical radius ε centered at the value $f(x_0)$. To win (to prove that f is continuous at x_0) we have to identify a horizontal radius $\delta > 0$, which gives us the purple strip on the right, such that all of the inputs in the purple region get spit out into the green region.

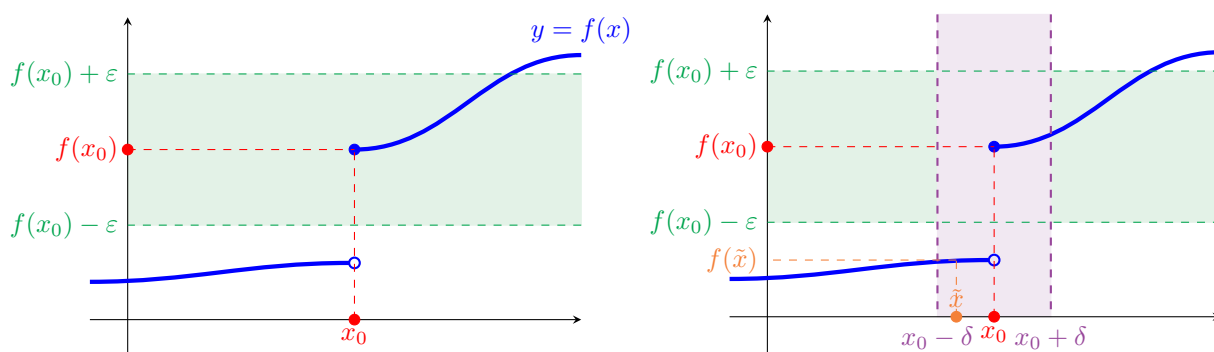


FIGURE 3. A depiction of a function which is not continuous at x_0 . An example green zone is drawn, determined by a choice of $\varepsilon > 0$. Note that, for this green region, no matter how small $\delta > 0$ is (i.e., no matter how skinny the purple region is) there will always be purple inputs that the function does not send to the green zone. For example, the orange input $\tilde{x}$ is in the purple zone, but gets mapped outside of the green zone. Thus, f is not continuous at x_0 .

Example 12.5. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$. Let's show that f is continuous.

Scratchwork.

To show that f is continuous, we need to show that f is continuous at x_0 for all $x_0 \in \mathbb{R}$. Because the first variable being quantified in this statement is x_0 , we should begin by fixing $x_0 \in \mathbb{R}$. Once we have fixed $x_0 \in \mathbb{R}$, we want to show that f is continuous at x_0 . To verify the definition of “continuous at x_0 ” we need to verify a sentence that begins with a “for all $\varepsilon > 0$ ” quantifier. Thus, the next thing we need to do is fix $\varepsilon > 0$. Our ultimate goal is to find a $\delta > 0$ such that the rest of the definition is satisfied. This is analogous to the goal of finding a $N \in \mathbb{N}$ such that something is true in the definition of sequence convergence. The thing that we want to be true is that if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \varepsilon$. In other words, the quantity we ultimately want to control is $|f(x) - f(x_0)|$, and the thing that we can impose control of by our choice of δ is $|x - x_0|$.

Let's play around with this quantity. Note that

$$\begin{aligned} |f(x) - f(x_0)| &= |x^2 - x_0^2| \\ &= |x - x_0||x + x_0|. \end{aligned}$$

I factored this way because I know via my choice of δ that we can make $|x - x_0|$ small. We ultimately want to achieve the estimate $|x - x_0||x + x_0| < \varepsilon$, but we cannot call

$$\delta := \frac{\varepsilon}{|x + x_0|}$$

for a couple reasons. First of all, the denominator could be 0. Secondly and more importantly, **we have to quantify δ before we quantify x** , so δ cannot depend on x . It can depend on x_0 , because x_0 is quantified before δ , but we choose x only after we choose δ .

Let's consider the term $|x + x_0|$ for a moment. Recall that, once we choose $\delta > 0$, we then choose x so that it satisfies $|x - x_0| < \delta$. This means that x is δ -close to x_0 . It then seems like $|x + x_0|$ can't get too big. In particular, since x is δ close to x_0 , it seems like we could prove the estimate $|x + x_0| \leq 2|x_0| + \delta$. Indeed, note that

$$\begin{aligned} |x + x_0| &\leq |x| + |x_0| \\ &= |x - x_0 + x_0| + |x_0| \\ &\leq |x - x_0| + |x_0| + |x_0| \\ &< \delta + 2|x_0|. \end{aligned}$$

There are a number of ways that we can deal with this situation carefully. Because we can choose δ , we could choose it show that $\delta \leq 1$, for example, and this would give a constant upper bound on the term $|x + x_0|$, namely, $|x + x_0| < 1 + 2|x_0|$. Once we ensure this, we could then also require δ to be small enough so that $|x - x_0|(1 + 2|x_0|) < \varepsilon$. In particular, we could choose $\delta \leq \frac{\varepsilon}{1 + 2|x_0|}$. In order to satisfy both of these estimates, we could let $\delta = \min(1, \frac{\varepsilon}{1 + 2|x_0|})$. With this choice of δ , we should be able to achieve the main estimate.

Formal proof that f is continuous.

Let $x_0 \in \mathbb{R}$.

Let $\varepsilon > 0$.

Let $\delta = \min(1, \frac{\varepsilon}{1 + 2|x_0|})$.

Let $x \in \mathbb{R}$ such that $|x - x_0| < \delta$. Then $|x - x_0| < 1$ and $|x - x_0| < \frac{\varepsilon}{1+2|x_0|}$.

Note that

$$\begin{aligned}
 |f(x) - f(x_0)| &= |x^2 - x_0^2| \\
 &= |x - x_0||x + x_0| \\
 &\leq |x - x_0|(|x| + |x_0|) \\
 &= |x - x_0|(|x - x_0 + x_0| + |x_0|) \\
 &\leq |x - x_0|(|x - x_0| + 2|x_0|) \\
 &< |x - x_0|(1 + 2|x_0|) \quad (\text{since } |x - x_0| < 1) \\
 &< \frac{\varepsilon}{1 + 2|x_0|}(1 + 2|x_0|) \quad (\text{since } |x - x_0| < \frac{\varepsilon}{1 + 2|x_0|}) \\
 &= \varepsilon.
 \end{aligned}$$

Thus, for all $\varepsilon > 0$ there is a $\delta > 0$ such that, for all $x \in \mathbb{R}$ satisfying $|x - x_0| < \delta$, we have $|f(x) - f(x_0)| < \varepsilon$. Since $x_0 \in \mathbb{R}$ was arbitrary it follows that f is continuous.

13. LECTURE 13: PROPERTIES OF CONTINUITY

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 3, Section 17 and Section 18
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 4, “Continuous functions”
- ◇ Abbott, *Understanding Analysis*: Chapter 4, Section 3 and Section 4

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Understand the equivalence between the $\varepsilon - \delta$ notion of continuity and the sequential notion of continuity, and be able to use one or the other to prove basic properties of continuous functions.

As we have defined it, continuity is defined as a triple quantified “for all $\varepsilon > 0$ ” statement. We have already done a lot heavy lifting with triple quantified “for all $\varepsilon > 0$ ” statements in our study of sequences, so it seems plausible to wonder if we can relate the notion of continuity to the notion of sequential convergence. Indeed, we can! This is the content of the next theorem, which gives an alternative characterization of what it means for a function to be in terms of sequences converging to x_0 .

Theorem 13.1 (Sequential definition of continuity). *Let $I \subset \mathbb{R}$ be an open interval and let $x_0 \in \mathbb{R}$. A function $f : I \rightarrow \mathbb{R}$ is continuous at x_0 if and only if, for all sequences (a_n) such that $a_n \rightarrow x_0$, we have $f(a_n) \rightarrow f(x_0)$.*

Proof. **Continuity implies sequential continuity.**

First, suppose that f is continuous at x_0 . Let a_n be a sequence such that $a_n \rightarrow x_0$. We wish to show that $f(a_n) \rightarrow f(x_0)$.

Fix $\varepsilon > 0$. Since f is continuous at x_0 , there is a $\delta > 0$ such that, for all $x \in I$ satisfying $|x - x_0| < \delta$ we have $|f(x) - f(x_0)| < \varepsilon$. Since $a_n \rightarrow x_0$, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$, we have $|a_n - x_0| < \delta$. Let $n \geq N$. Then $|a_n - x_0| < \delta$ and so $|f(a_n) - f(x_0)| < \varepsilon$.

Thus, for every $\varepsilon > 0$ there is an $N \in \mathbb{N}$ such that, for all $n \geq N$, we have $|f(a_n) - f(x_0)| < \varepsilon$. Therefore, $f(a_n) \rightarrow f(x_0)$.

Sequential continuity implies continuity.

To prove the converse direction, we will prove the contrapositive. That is, we will prove that if f is not continuous at x_0 , then there is a sequence a_n such that $a_n \rightarrow x_0$ but $f(a_n)$ does not converge to $f(x_0)$.

Since f is not continuous at x_0 , there is an $\varepsilon_0 > 0$ such that, for all $\delta > 0$, there is an $x \in I$ satisfying $|x - x_0| < \delta$ such that $|f(x) - f(x_0)| \geq \varepsilon_0$. In particular, for each $n \in \mathbb{N}$, there is an $x \in I$ satisfying

$|x - x_0| < \frac{1}{n}$ such that $|f(x) - f(x_0)| \geq \varepsilon_0$. For each $n \in \mathbb{N}$, denote such an element by a_n . Then $(a_n)_{n=1}^\infty$ denotes a sequence.

First, we claim that $a_n \rightarrow x_0$. This is immediate from the construction of a_n . Fix $\varepsilon > 0$ and let $N \in \mathbb{N}$ satisfy $N \geq \frac{1}{\varepsilon}$. Let $n \geq N$. Note that

$$|a_n - x_0| < \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Thus, $a_n \rightarrow x_0$.

Next, we claim that $f(a_n)$ does not converge to $f(x_0)$. By construction, $|f(a_n) - f(x_0)| \geq \varepsilon_0$ for all n , and so in particular for any $N \in \mathbb{N}$ there is an $n \geq N$ such that $|f(a_n) - f(x_0)| \geq \varepsilon_0$. Thus, $f(a_n)$ does not converge to $f(x_0)$. □

The contrapositive of the first direction of the above theorem (continuity implies sequential continuity) is worth stating separately as a corollary, as it is useful for showing that functions are not continuous.

Corollary 13.2 (A criterion for discontinuity). *Let $x_0 \in I$, and let $f : I \rightarrow \mathbb{R}$. Suppose that a_n is a sequence such that $a_n \rightarrow x_0$ and $f(a_n)$ does not converge to $f(x_0)$. Then f is not continuous at x_0 .*

Example 13.3. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) := \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}.$$

We claim that f is not continuous at 0. Let $a_n := -\frac{1}{n}$. Note that $a_n \rightarrow 0$. We have $f(a_n) = f(-\frac{1}{n}) = 0$ for all $n \in \mathbb{N}$, so that $f(a_n) \rightarrow 0$. Since $f(0) = 1 \neq 0$, it follows that f is not continuous at 0.

Example 13.4. Consider the function $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$g(x) := \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}.$$

We claim that g is not continuous at 0. Let $r_0 \in \mathbb{R} \setminus \mathbb{Q}$ be any fixed irrational number, and define $a_n := \frac{r_0}{n}$. Note that $a_n \rightarrow 0$ and that $a_n \in \mathbb{R} \setminus \mathbb{Q}$ for all $n \in \mathbb{N}$. Thus, we have $g(a_n) = g(\frac{r_0}{n}) = 0$ for all $n \in \mathbb{N}$, so that $g(a_n) \rightarrow 0$. Since $g(0) = 1 \neq 0$, it follows that g is not continuous at 0.

Theorem 13.5. *Let $f, g : I \rightarrow \mathbb{R}$ be functions that are continuous at $x_0 \in I$, and let $h : J \rightarrow \mathbb{R}$ be continuous at $f(x_0) \in J$.*

- (i) *The function $f + g$ is continuous at x_0 .*
- (ii) *The function fg is continuous at x_0 .*
- (iii) *If $g(x_0) \neq 0$, the function $\frac{f}{g}$ is continuous at x_0 .*
- (iv) *The function $|f|$ is continuous at x_0 .*
- (v) *The function $h \circ f$ is continuous at x_0 .*

Proof. Parts (i), (ii), (iii), and (v) of the theorem are straightforward to prove using the sequential characterization of continuity. For part (iv) we will appeal to the $\varepsilon - \delta$ definition of continuity.

(i) . Let $a_n \rightarrow x_0$. Note that

$$\begin{aligned}
 \lim_{n \rightarrow \infty} (f + g)(a_n) &= \lim_{n \rightarrow \infty} f(a_n) + g(a_n) && \text{by definition of } f + g \\
 &= \lim_{n \rightarrow \infty} f(a_n) + \lim_{n \rightarrow \infty} g(a_n) && \text{by algebraic limit laws} \\
 &= f(x_0) + g(x_0) && \text{by continuity of } f \text{ and } g \\
 &= (f + g)(x_0) && \text{by definition of } f + g.
 \end{aligned}$$

Thus, since $(f + g)(a_n) \rightarrow (f + g)(x_0)$ for all sequences a_n such that $a_n \rightarrow x_0$, it follows that $f + g$ is continuous at x_0 .

(ii) Exercise, similar to (i).

(iii) Exercise, similar to (i).

(iv) Fix $\varepsilon > 0$. Since f is continuous at x_0 , there is a $\delta > 0$ such that, for all $x \in I$ satisfying $|x - x_0| < \delta$, we have $|f(x) - f(x_0)| < \varepsilon$.

Let $x \in I$ satisfy $|x - x_0| < \delta$.

Note that, by the reverse triangle inequality, we have

$$\begin{aligned}
 ||f(x)| - |f(x_0)|| &\leq |f(x) - f(x_0)| \\
 &< \varepsilon.
 \end{aligned}$$

Thus, for all $\varepsilon > 0$ there is a $\delta > 0$ such that, for all $x \in I$ satisfying $|x - x_0| < \delta$ we have $||f(x)| - |f(x_0)|| < \varepsilon$. It follows that $|f|$ is continuous at x_0 .

(v) Let $a_n \rightarrow x_0$. Since f is continuous at x_0 , we have $f(a_n) \rightarrow f(x_0)$. Since h is continuous at $f(x_0)$, we have $h(f(a_n)) \rightarrow h(f(x_0))$. Thus, $(h \circ f)(a_n) \rightarrow (h \circ f)(x_0)$. Thus, since $(h \circ f)(a_n) \rightarrow (h \circ f)(x_0)$ for all $a_n \rightarrow x_0$, it follows that $h \circ f$ is continuous at x_0 .

□

Corollary 13.6. *Polynomials are continuous.*

Proof. It suffices to prove that the functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x$ and $g(x) = c$, where $c \in \mathbb{R}$ is some constant, are continuous. Indeed, once we know this then any polynomial is given by a finite number of products and sums of functions of this form, and hence Theorem 13.5 gives continuity. Continuity of $f(x) = x$ and $g(x) = c$ is immediate from sequential continuity. □

14. LECTURE 14: THE INTERMEDIATE AND EXTREME VALUE THEOREMS

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 3, Section 18
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 4, some of “Continuity and Compactness” and “Continuity and Connectedness”
- ◇ Abbott, *Understanding Analysis*: Chapter 4, Section 3 and Section 4

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Understand the statements and proofs of the extreme value theorem and the intermediate value theorem.
- ◇ Be able to use the extreme value theorem and the intermediate value theorem to prove things about continuous functions.

Given $f : I \rightarrow \mathbb{R}$, define the **image** of f to be

$$f(I) := \{ f(x) \mid x \in I \}.$$

In words, $f(I)$ is the set of all values that f spits out when you feed it everything in the domain. For example, if $f : [2, 4) \rightarrow \mathbb{R}$ is given by $f(x) = x^2$, then $f([2, 4)) = [4, 16)$, or if $g : (-1, 1) \rightarrow \mathbb{R}$ is given by $g(x) = x^4$, then $g((-1, 1)) = [0, 1)$.

Question 14.1. Suppose that I is an open interval, and let $f : I \rightarrow \mathbb{R}$ be any continuous function. Is $f(I)$ an open interval? In other words, is the property of being an *open interval* preserved under continuity?

The answer is no, not necessarily!

Example 14.2. Suppose $f : (0, 1) \rightarrow \mathbb{R}$ be given by $f(x) = 0$ for all $x \in (0, 1)$. Then $f((0, 1)) = \{0\}$, and $\{0\}$ is not an open interval.

Example 14.3. Here is an example of a continuous function $g : (0, 1) \rightarrow \mathbb{R}$ such that $g((0, 1)) = [0, 1]$. In other words, it is possible for a continuous function to turn an open interval into a closed interval! Define g by

$$g(x) = \begin{cases} 0 & 0 < x < \frac{1}{4} \\ 2(x - \frac{1}{4}) & \frac{1}{4} \leq x \leq \frac{3}{4} \\ 1 & \frac{3}{4} < x < 1 \end{cases}.$$

In math it is useful to know what properties are preserved under functions satisfying some definition (in this case, continuity). Thus, if possible we would like to identify a type of set that is preserved under continuity. This leads us to consider *compact intervals* instead of open intervals. A **compact interval**⁷ is a set

⁷The reason I’m using the language “compact interval” instead of “closed interval” is because I do not want to consider $\mathbb{R}$ a compact interval (but a mathematician would tell you that $\mathbb{R}$ is a closed interval).

of the form $K = [a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$ for some $a, b \in \mathbb{R}$. The definition of continuity of a function at x_0 still makes sense if we replace the open interval I with a compact interval K .

Theorem 14.4 (Compact preservation). *Let $K \subseteq \mathbb{R}$ be a compact interval, and let $f : K \rightarrow \mathbb{R}$ be a continuous function. Then $f(K)$ is a compact interval.*

First, we begin with a lemma.

Lemma 14.5. *Let K be a compact interval, and let $(x_n) \subset K$ be a convergent sequence. Then $\lim_{n \rightarrow \infty} (x_n) \in K$.*

Proof. Let $K = [a, b]$. Since $x_n \in K$ for all $n \in \mathbb{N}$, we have $a \leq x_n \leq b$ for all $n \in \mathbb{N}$. So for any $N \in \mathbb{N}$, we have $a \leq \inf\{x_n \mid n \geq N\}$ and $\sup\{x_n \mid n \geq N\} \leq b$. Since $V_N := \inf\{x_n \mid n \geq N\}$ is an increasing sequence, it follows that

$$\begin{aligned} \lim_{n \rightarrow \infty} x_n &= \liminf x_n \\ &= \lim_{N \rightarrow \infty} V_N \\ &\geq V_0 \\ &= \inf\{x_n \mid n \in \mathbb{N}\} \\ &\geq a. \end{aligned}$$

Similarly, since $U_N := \sup\{x_n \mid n \geq N\}$ is a decreasing sequence, we have $\lim_{n \rightarrow \infty} x_n = \limsup x_n \leq U_0 \leq b$. Thus,

$$a \leq \lim_{n \rightarrow \infty} x_n \leq b$$

and so $\lim_{n \rightarrow \infty} (x_n) \in K$. □

Now we can prove Theorem 14.4.

Proof of Theorem 14.4. Since K is a compact interval, we may write $K = [a, b]$ for some $a < b \in \mathbb{R}$. Our goal is to show that $f(K) = [\alpha, \beta]$ for some other real numbers $\alpha < \beta \in \mathbb{R}$. Because the proof is fairly involved, we first outline the argument.

- ◇ Step 1: First, we will show that $f(K)$ is a bounded set. This will allow us to use the infimum and supremum of $f(K)$. In particular, we will let $\alpha := \inf f(K)$ and $\beta := \sup f(K)$. Note that $f(K) \subseteq [\alpha, \beta]$, and so to prove $f(K) = [\alpha, \beta]$ it suffices to prove that $[\alpha, \beta] \subseteq f(K)$.
- ◇ Step 2: Next, we will show that $\alpha \in f(K)$ and $\beta \in f(K)$.
- ◇ Step 3: Finally, it remains to show that $(\alpha, \beta) \subseteq f(K)$. We will do this by selecting an arbitrary $y_0 \in (\alpha, \beta)$ and providing $x_0 \in K$ such that $f(x_0) = y_0$.

With this outline, we now proceed with the proof.

Step 1: Proving that $f(K)$ is a bounded set.

First, we claim that the set $f(K)$ is bounded. Indeed, suppose for the sake of contradiction that $f(K)$ is not bounded. Then for each $n \in \mathbb{N}$, there is an element $y_n \in f(K)$ such that $|y_n| > n$. By definition of $f(K)$, this gives us elements $x_n \in K$ such that $|f(x_n)| = |y_n| > n$. Since $x_n \in K$, the sequence (x_n) is bounded. By Bolzano-Weierstrauss, there is a convergent subsequence x_{n_k} . Let $x_0 := \lim_{k \rightarrow \infty} x_{n_k}$. By Lemma 14.5, $x_0 \in K$. By continuity of f at x_0 , $f(x_{n_k}) \rightarrow f(x_0)$, and so $|f(x_{n_k})| \rightarrow |f(x_0)|$ by Theorem 13.5. But $|f(x_{n_k})| = |y_{n_k}| > n_k$ and n_k diverges to ∞ as $k \rightarrow \infty$. This is a contradiction. Thus, $f(K)$ is bounded.

Since $f(K)$ is bounded, we may let $\alpha := \inf f(K) \in \mathbb{R}$ and $\beta := \sup f(K) \in \mathbb{R}$. Our end goal is to prove that $f(K) = [\alpha, \beta]$, which will complete the proof of the theorem. Observe that $f(K) \subseteq [\alpha, \beta]$ by definition of α and β , and so it suffices to prove that $[\alpha, \beta] \subseteq f(K)$. Note also that we may assume $\alpha < \beta$, otherwise f is constant and the theorem follows trivially.

Step 2: Proving that $\alpha \in f(K)$ and $\beta \in f(K)$.

To prove that $\alpha \in f(K)$, we need to find an element $x_0 \in K$ such that $f(x_0) = \alpha$. For each $n \in \mathbb{N}$, note that $\alpha + \frac{1}{n}$ is not a lower bound for $f(K)$. Thus, there exist elements $x_n \in K$ such that $\alpha \leq f(x_n) < \alpha + \frac{1}{n}$. Since $x_n \in K$, (x_n) is a bounded sequence. By Bolzano-Weierstrauss, there is a convergent subsequence (x_{n_k}) of (x_n) . Let $x_0 := \lim_{k \rightarrow \infty} x_{n_k}$. By Lemma 14.5, $x_0 \in K$. By continuity of f , we have

$$f(x_0) = \lim_{k \rightarrow \infty} f(x_{n_k}) = \alpha$$

where the latter inequality follows from the fact that $\alpha \leq f(x_{n_k}) < \alpha + \frac{1}{n_k}$. Since $f(x_0) = \alpha$, it follows that $\alpha \in f(K)$.

The argument to prove that $\beta \in f(K)$ is similar. For each $n \in \mathbb{N}$, note that $\beta - \frac{1}{n}$ is not an upper bound for $f(K)$. Thus, there are elements $\tilde{x}_n \in K$ satisfying $\beta - \frac{1}{n} < f(\tilde{x}_n) \leq \beta$. Let $(\tilde{x}_{n_k})$ be a convergent subsequence of $\tilde{x}_n$ provided by Bolzano-Weierstrauss, and let $\tilde{x}_0 := \lim_{k \rightarrow \infty} \tilde{x}_{n_k}$. Note that $\tilde{x}_0 \in K$ by Lemma 14.5. By continuity of f , we have

$$f(\tilde{x}_0) = \lim_{k \rightarrow \infty} f(\tilde{x}_{n_k}) = \beta$$

since $\beta - \frac{1}{n_k} < f(\tilde{x}_{n_k}) \leq \beta$. Thus, $f(\tilde{x}_0) = \beta$ and so $\beta \in f(K)$.

Step 3: Proving that $(\alpha, \beta) \subseteq f(K)$.

Let $y_0 \in (\alpha, \beta)$. We want to find an $x_0 \in K$ such that $f(x_0) = y_0$. By the previous step, there are elements $x_\alpha, x_\beta \in K$ such that $f(x_\alpha) = \alpha$ and $f(x_\beta) = \beta$. Note that, since $\alpha < \beta$, it is either the case that $x_\alpha < x_\beta$ or $x_\alpha > x_\beta$. We will assume $x_\alpha < x_\beta$; the argument for the reverse inequality is similar.

Define a set

$$E := \{x \mid x_\alpha \leq x \leq x_\beta, \text{ and } f(x) < y_0\} \subseteq [x_\alpha, x_\beta].$$

Note that $f(x_\alpha) = \alpha < y_0$ and so $E \neq \emptyset$. Let $x_0 := \sup E$. See Figure 4. Our goal is to prove that $f(x_0) = y_0$.

- ◇ Step 3(a): Showing $f(x_0) \leq y_0$. For each $n \in \mathbb{N}$, note that $x_0 - \frac{1}{n}$ is not an upper bound for E , and so there are elements $x_n \in E$ such that $x_0 - \frac{1}{n} < x_n \leq x_0$. Note that $x_n \rightarrow x_0$. Since $x_n \in E$, we have $f(x_n) < y_0$. Thus, by continuity of f at x_0 , we have

$$\begin{aligned} f(x_0) &= \lim_{n \rightarrow \infty} f(x_n) \\ &= \limsup f(x_n) \\ &\leq y_0. \end{aligned}$$

- ◇ Step 3(b): Showing $f(x_0) \geq y_0$. First, observe that $x_0 < x_\beta$. Indeed, $x_0 \leq x_\beta$ by definition and since

$$f(x_0) = y_0 < \beta = f(x_\beta)$$

it must be the case that $x_0 < x_\beta$. Thus, we may consider the sequence

$$x_0 + \frac{1}{n}$$

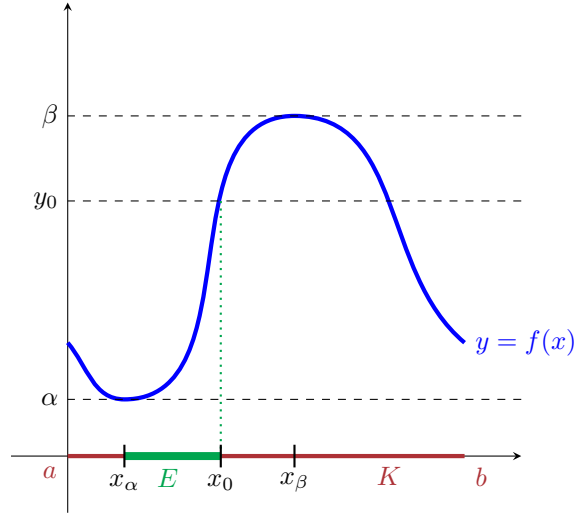


FIGURE 4. A visualization of all the sets and quantities involved in Step 3 of the proof of Theorem 14.4.

which will be a sequence in $[x_\alpha, x_\beta]$ for sufficiently large n . Note that $x_0 + \frac{1}{n} \rightarrow x_0$. Since $x_0 = \sup E$, we have $x_0 = \frac{1}{n} \notin E$. This means that $f(x_0 + \frac{1}{n}) \geq y_0$. Thus, by continuity of f at x_0 we have

$$\begin{aligned} f(x_0) &= \lim_{n \rightarrow \infty} f\left(x_0 + \frac{1}{n}\right) \\ &= \liminf f\left(x_0 + \frac{1}{n}\right) \\ &\geq y_0. \end{aligned}$$

Step 3(a) and Step 3(b) together then imply $f(x_0) = y_0$, and this completes the proof of the theorem. $\square$

Corollary 14.6 (Extreme value theorem). *Suppose that $f : K \rightarrow \mathbb{R}$ is a continuous function on a compact interval. Then there are points $x_{\max}, x_{\min} \in K$ such that, for all $x \in K$, we have*

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}).$$

In other words, f achieves its maximum and minimum on K .

Proof. By Theorem 14.4, we can write $f(K) = [\alpha, \beta]$. Then there is $x_{\max}, x_{\min} \in K$ such that $f(x_{\max}) = \beta$ and $f(x_{\min}) = \alpha$. In particular, for all $x \in K$ we have

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}).$$

$\square$

Corollary 14.7 (Intermediate value theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Suppose that $f(a) < f(b)$ and fix $y_0 \in \mathbb{R}$ such that $f(a) < y_0 < f(b)$. Then there is $c \in (a, b)$ such that $f(c) = y_0$. The analogous statement is true if $f(b) < f(a)$.*

Proof. By Theorem 14.4, $f([a, b]) = [f(a), f(b)]$. Since $f(a) < y_0 < f(b)$, this means that $y_0 \in f([a, b])$. Thus, there is $c \in (a, b)$ such that $f(c) = y_0$. $\square$

15. LECTURE 15: UNIFORM CONTINUITY

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 3, Section 19
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 4, some of “Continuity and Compactness”
- ◇ Abbott, *Understanding Analysis*: Chapter 4, Section 4

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Understand the difference between continuity and uniform continuity at the level of quantifiers and intuition.
- ◇ Be able to verify that simple functions are uniformly continuous directly from the definition.

Suppose that $f : I \rightarrow \mathbb{R}$ is a function on some interval, either open or compact. To say that f is continuous on I means that, for all $x_0 \in I$, f is continuous at x_0 . Putting all of these quantifiers into one statement gives us the following characterization of what it means for f to be continuous on I :

For all $x_0 \in I$ and for all $\varepsilon > 0$, there is a $\delta > 0$ such that, for all $x \in I$ with $|x - x_0| < \delta$, we have $|f(x) - f(x_0)| < \varepsilon$.

The important thing to point out is that *the choice of δ may be dependent on x_0* . We can see this because x_0 is quantified before δ , so necessarily δ may depend on x_0 , but this also makes sense intuitively. The definition of continuity at x_0 is based around the desire to control the variation of f near x_0 , and since the variation of a function may be more or less intense at different points in the domain, it is unsurprising that the required choice of δ will depend on the point x_0 . See, for example, Figure 5.

It turns out to be useful to consider an upgraded notion of continuity where, given $\varepsilon > 0$, a *single* choice of $\delta > 0$ works no matter what points in the domain we consider. This gives us “uniform” control of the variation in f on its domain. We will define and investigate this concept below, but it may be some time before we see its usefulness in action.

Definition 15.1. Let $f : I \rightarrow \mathbb{R}$, where I is any kind of interval. We say that f is **uniformly continuous on I** if the following condition holds:

For every $\varepsilon > 0$, there is a $\delta > 0$ such that, for all $x, y \in I$ with $|x - y| < \delta$, we have $|f(x) - f(y)| < \varepsilon$.

Remark 15.2. Note that the above definition describes what it means for a function to be uniformly continuous on a set. Unlike normal continuity, it doesn’t make sense to talk about uniform continuity at a single point. There is no point in the domain which is quantified before $\varepsilon > 0$ and $\delta > 0$!

Example 15.3. Consider the function $f : [0, 1] \rightarrow \mathbb{R}$ defined by $f(x) = \frac{1}{2-x}$. Let’s prove that f is uniformly continuous on $[0, 1]$.

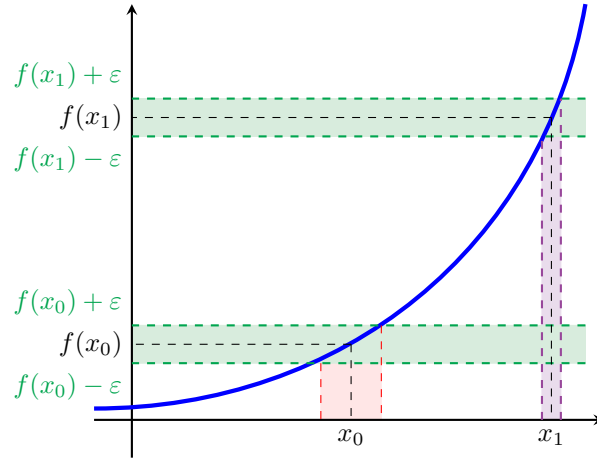


FIGURE 5. A continuous function with two specified points in the domain, x_0 and x_1 . For a single choice of $\varepsilon > 0$, the red zone indicates a viable choice of δ associated to x_0 , and the purple zone indicates a viable choice of δ associated to x_1 . Note that the purple zone is necessarily much smaller than the red zone, because the function varies more intensely near x_1 than it does at x_0 .

Informal scratchwork.

Fix $\varepsilon > 0$. Ultimately, the quantity we wish to control is $|f(x) - f(y)|$. In particular, we want to achieve $|f(x) - f(y)| < \varepsilon$ for all $x, y \in [0, 1]$ that are sufficiently close, and we need to determine exactly what “sufficiently close” means. Playing around with the quantity of interest gives

$$\begin{aligned} |f(x) - f(y)| &= \left| \frac{1}{2-x} - \frac{1}{2-y} \right| \\ &= \left| \frac{(2-y) - (2-x)}{(2-x)(2-y)} \right| \\ &= \left| \frac{x-y}{(2-x)(2-y)} \right| \\ &= \frac{|x-y|}{(2-x)(2-y)}. \end{aligned}$$

We dropped the absolute values in the denominator because $0 \leq x, y \leq 1$ and so $2-x > 0$ and $2-y > 0$. Fortunately, we have the term $|x-y|$ in the numerator, and we get to impose complete control over that term with our choice of δ . In particular, we would want to choose $\delta > 0$ so that $|x-y| < \delta$ implies

$$\frac{|x-y|}{(2-x)(2-y)} < \varepsilon$$

for all $x, y \in [0, 1]$. The fact that the denominator depends on x and y is annoying, and **it doesn't make sense to let** $\delta = \varepsilon(2-x)(2-y)$ because $x, y \in [0, 1]$ are quantified *after* δ . However, we are in luck because we know that $x, y \in [0, 1]$. This means that $2-x \geq 2-1 = 1$ and likewise $2-y \geq 2-1 = 1$. So the denominator is at least 1, and so

$$\frac{|x-y|}{(2-x)(2-y)} \leq \frac{|x-y|}{1} = |x-y|.$$

So we can actually let $\delta = \varepsilon$!

Formal proof that f is uniformly continuous on $[0, 1]$.

Fix $\varepsilon > 0$.

Let $\delta = \varepsilon$.

Let $x, y \in [0, 1]$ with $|x - y| < \delta$.

First, note that $x, y \leq 1$ and so $2 - x \geq 1$ and $2 - y \geq 1$. Then

$$\begin{aligned}
 |f(x) - f(y)| &= \left| \frac{1}{2-x} - \frac{1}{2-y} \right| \\
 &= \left| \frac{(2-y) - (2-x)}{(2-x)(2-y)} \right| \\
 &= \left| \frac{x-y}{(2-x)(2-y)} \right| \\
 &= \frac{|x-y|}{(2-x)(2-y)} && \text{since } 2-x > 0, 2-y > 0 \\
 &\leq \frac{|x-y|}{1 \cdot 1} && \text{since } 2-x \geq 1, 2-y \geq 1 \\
 &= |x-y| \\
 &< \delta \\
 &= \varepsilon.
 \end{aligned}$$

Thus, for all $\varepsilon > 0$, there is a $\delta > 0$ such that, for all $x, y \in [0, 1]$ with $|x - y| < \delta$ we have $|f(x) - f(y)| < \varepsilon$. It follows that f is uniformly continuous on $[0, 1]$.

Example 15.4. Consider the function $f : [0, 2) \rightarrow \mathbb{R}$ defined by $f(x) = \frac{1}{2-x}$. Let's prove that f is *not* uniformly continuous on $[0, 2)$.

Informal scratchwork.

We have to negate the definition of uniform continuity, which means that we have to verify the following condition:

There is an $\varepsilon > 0$ such that, for all $\delta > 0$, there are $x, y \in [0, 2)$ with $|x - y| < \delta$ such that $|f(x) - f(y)| \geq \varepsilon$.

We will exploit the “singularity” of the function at $x = 2$ to find points in the domain that are very close but with outputs that are far apart. In particular, I claim we can choose any $\varepsilon > 0$. Let's choose $\varepsilon = 1$. Ultimately, we want to achieve two inequalities:

$$|x - y| < \delta \quad \text{and} \quad \frac{|x - y|}{(2-x)(2-y)} > 1.$$

Since we get to pick x and y , and we need to ensure that they are δ -close, we may as well reduce the number of degrees of freedom that we have and once we choose x , we can just set $y = x - \frac{\delta}{2}$, for example. Note that I'm choosing $y = x - \frac{\delta}{2}$ and not $y = x + \frac{\delta}{2}$ because I know we're going to have to pick numbers really close to 2. Once I pick x , likely a number very close to 2, I don't know that $x + \frac{\delta}{2}$ will still be smaller than 2.

Anyway, with $y = x - \frac{\delta}{2}$ the quantity of interest becomes

$$\frac{|x - y|}{(2-x)(2-y)} = \frac{\frac{\delta}{2}}{(2-x)(2-x + \frac{\delta}{2})}.$$

We want to make this bigger than 1, and the only degree of freedom we have is to choose x . The denominator is a bit weird, but note that

$$2 - x + \frac{\delta}{2} \leq 2 + \frac{\delta}{2}.$$

This allows us to simplify our estimate, because

$$\frac{\frac{\delta}{2}}{(2-x)(2-x+\frac{\delta}{2})} \geq \frac{\frac{\delta}{2}}{(2-x)(2+\frac{\delta}{2})}.$$

If we can make the term on the right larger than 1, then the term we care about will certainly be larger than 1. Solving for x in the inequality

$$\frac{\frac{\delta}{2}}{(2-x)(2+\frac{\delta}{2})} > 1$$

gives

$$x > 2 - \frac{\delta}{4 + \delta}.$$

Formal proof that f is not uniformly continuous on $[0, 2)$.

Let $\varepsilon = 1$.

Fix $\delta > 0$.

Let $x = 2 - \frac{1}{2} \left(\frac{\delta}{4+\delta} \right)$ and $y = x - \frac{\delta}{2}$. Note that $x, y \in [0, 2)$, and $|x - y| = \frac{\delta}{2} < \delta$. Also note that

$$\begin{aligned} 2 - x &= 2 - \left(2 - \frac{1}{2} \left(\frac{\delta}{4+\delta} \right) \right) \\ &= \frac{1}{2} \left(\frac{\delta}{4+\delta} \right) \\ &< \frac{\delta}{4+\delta} \\ &= \frac{\frac{\delta}{2}}{2 + \frac{\delta}{2}}. \end{aligned}$$

With this, we have

$$\begin{aligned} |f(x) - f(y)| &= \left| \frac{1}{2-x} - \frac{1}{2-y} \right| \\ &= \frac{|x-y|}{(2-x)(2-y)} && \text{since } 2-x > 0, \ 2-y > 0 \\ &= \frac{\frac{\delta}{2}}{(2-x)(2-x+\frac{\delta}{2})} && \text{since } y = x - \frac{\delta}{2} \\ &\geq \frac{\frac{\delta}{2}}{(2-x)(2+\frac{\delta}{2})} && \text{since } x \geq 0 \\ &> \frac{\frac{\delta}{2}}{\left(\frac{\frac{\delta}{2}}{2+\frac{\delta}{2}} \right) (2+\frac{\delta}{2})} && \text{since } 2-x < \frac{\frac{\delta}{2}}{2+\frac{\delta}{2}} \\ &= 1. \end{aligned}$$

Thus, there is an $\varepsilon > 0$ such that, for all $\delta > 0$, there are $x, y \in [0, 2)$ with $|x-y| < \delta$ such that $|f(x)-f(y)| \geq \varepsilon$. Therefore, f is not uniformly continuous on $[0, 2)$.

16. LECTURE 16: PROPERTIES OF UNIFORM CONTINUITY

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 3, Section 19
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 4, some of “Continuity and Compactness”
- ◇ Abbott, *Understanding Analysis*: Chapter 4, Section 4

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Become more comfortable using and verifying (and verifying the negation of) uniform continuity.
- ◇ Be able to verify that simple functions are *not* uniformly continuous by the sequential criteria.

In this lecture we will investigate some general properties of uniform continuity. We begin with a statement which should hopefully be obvious, and is implied by the discussion from the previous lecture!

Proposition 16.1. *Let $f : I \rightarrow \mathbb{R}$ be uniformly continuous on I . Then f is continuous on I .*

Proof. Let $x_0 \in I$. Fix $\varepsilon > 0$. By uniform continuity of f on I , there is a $\delta > 0$ such that, for all $x, y \in I$ with $|x - y| < \delta$, we have $|f(x) - f(y)| < \varepsilon$. In particular, for all $x \in I$ with $|x - x_0| < \delta$, we have $|f(x) - f(x_0)| < \varepsilon$. Thus, f is continuous at $x_0 \in I$. Since $x_0 \in I$ was arbitrary, it follows that f is continuous on I . $\square$

Next, the following theorem gives three different possible criteria for showing that a function is not uniformly continuous, beyond just verifying the negation of the definition.

Theorem 16.2 (Criterion for failure of uniform continuity). *Let $f : I \rightarrow \mathbb{R}$ be a function.*

- (i) *Let $(x_n) \subseteq I$ be a Cauchy sequence such that $(f(x_n))$ is not a Cauchy sequence. Then f is not uniformly continuous on I .*
- (ii) *Suppose there is an $\varepsilon_0 > 0$, sequences $(x_n), (y_n) \subseteq I$, and an $N \in \mathbb{N}$ such that such that*
 - ◇ $|x_n - y_n| \rightarrow 0$, and
 - ◇ $|f(x_n) - f(y_n)| \geq \varepsilon_0$ for all $n \geq N$.*Then f is not uniformly continuous on I .*
- (iii) *Suppose there are sequences $(x_n), (y_n) \subseteq I$ such that*
 - ◇ $|x_n - y_n| \rightarrow 0$, and
 - ◇ $\limsup |f(x_n) - f(y_n)| > 0$.*Then f is not uniformly continuous on I .*

Proof.

- (i) We will prove the contrapositive, which says that if f is a uniformly continuous function and $(x_n) \subseteq I$ is a Cauchy sequence, then $(f(x_n))$ is a Cauchy sequence.
Fix $\varepsilon > 0$.

Since f is uniformly continuous on I , there is a $\delta > 0$ such that, for all $x, y \in I$ with $|x - y| < \delta$, we have $|f(x) - f(y)| < \varepsilon$.

Since (x_n) is Cauchy, there is an $N \in \mathbb{N}$ such that, for all $n, m \geq N$, we have $|x_n - x_m| < \delta$.

Thus, for all $n, m \geq N$, we have $|f(x_n) - f(x_m)| < \varepsilon$. Therefore, $(f(x_n))$ is Cauchy.

- (ii) Fix $\varepsilon_0 > 0$, sequences $(x_n), (y_n) \subseteq I$, and $N \in \mathbb{N}$ such that $|x_n - y_n| \rightarrow 0$ and $|f(x_n) - f(y_n)| \geq \varepsilon_0$ for all $n \geq N$. We will prove that f is not uniformly continuous by verifying the negation of the definition of uniform continuity.

Fix $\delta > 0$. Since $|x_n - y_n| \rightarrow 0$, there is an $N_0 \in \mathbb{N}$ such that, for all $n \geq N_0$, we have $|x_n - y_n| < \delta$. Let $n = \max(N_0, N)$. Then $|x_n - y_n| < \delta$ and $|f(x_n) - f(y_n)| \geq \varepsilon_0$. Thus, for $\varepsilon_0 > 0$, it is not the case that there is a $\delta > 0$ such that for all $x, y \in I$ with $|x - y| < \delta$ we have $|f(x) - f(y)| < \varepsilon_0$. Thus, f is not uniformly continuous.

- (iii) Let $(x_n), (y_n) \subseteq I$ be sequence such that $|x_n - y_n| \rightarrow 0$ and $\limsup |f(x_n) - f(y_n)| > 0$. Let $\alpha := \limsup |f(x_n) - f(y_n)|$. By the subsequential characterization of the limsup from the homework, there are subsequences $(x_{n_k}), (y_{n_k})$ such that $|f(x_{n_k}) - f(y_{n_k})| \rightarrow \alpha$. Note that since $|x_n - y_n| \rightarrow 0$, we also have $|x_{n_k} - y_{n_k}| \rightarrow 0$.

We will prove that f is not uniformly continuous by verifying the negation of the definition of uniform continuity. Let $\varepsilon = \frac{\alpha}{2}$. Fix $\delta > 0$. Choose K large enough so that, for $k \geq K$, we have $|x_{n_k} - y_{n_k}| < \delta$ and

$$||f(x_{n_k}) - f(y_{n_k})| - \alpha| < \frac{\alpha}{2}.$$

The latter inequality implies

$$|f(x_{n_k}) - f(y_{n_k})| > \frac{\alpha}{2} = \varepsilon.$$

Thus, for $\varepsilon = \frac{\alpha}{2}$, it is not the case that there is a $\delta > 0$ such that, for all $x, y \in I$ with $|x - y| < \delta$ we have $|f(x) - f(y)| < \varepsilon$.

□

Example 16.3. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{1}{x}$. We claim that f is not uniformly continuous on $(0, \infty)$. Indeed, let $x_n := \frac{1}{n}$. Then (x_n) is a Cauchy sequence, but $f(x_n) = n$ is not a Cauchy sequence. By (i) of Theorem 16.2, it follows that f is not uniformly continuous.

Example 16.4. Consider the function $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = x^2$. We claim that g is not uniformly continuous on $\mathbb{R}$. Indeed, consider the sequences

$$x_n = n \quad \text{and} \quad y_n = n + \frac{1}{n}.$$

Note that $|x_n - y_n| = \frac{1}{n} \rightarrow 0$. Also note that, for all $n \geq 1$, we have

$$\begin{aligned} |g(x_n) - g(y_n)| &= \left| (n)^2 - \left(n + \frac{1}{n} \right)^2 \right| \\ &= \left| n^2 - n^2 - 2 - \frac{1}{n^2} \right| \\ &= 2 + \frac{1}{n^2} \\ &\geq 2. \end{aligned}$$

By Theorem 16.2, it follows that g is not uniformly continuous on $\mathbb{R}$.

Theorem 16.5. *Let $f : [a, b] \rightarrow \mathbb{R}$. Then f is continuous on $[a, b]$ if and only if f is uniformly continuous on $[a, b]$.*

Proof. One direction is trivial by the first proposition from this lecture. Namely, if f is uniformly continuous on $[a, b]$ then it is continuous on $[a, b]$.

Next, assume $f : [a, b] \rightarrow \mathbb{R}$ is continuous. Suppose for the sake of contradiction that f is not uniformly continuous. Then there is an $\varepsilon_0 > 0$ such that, for all $\delta > 0$, there are $x, y \in [a, b]$ with $|x - y| < \delta$ and $|f(x) - f(y)| \geq \varepsilon_0$. In particular, for each $n \in \mathbb{N}$ there are elements $x_n, y_n \in [a, b]$ with $|x_n - y_n| < \frac{1}{n}$ and $|f(x_n) - f(y_n)| \geq \varepsilon_0$.

Note that $(x_n) \subseteq [a, b]$ is a bounded sequence, and hence by Bolzano-Weierstrauss there is a convergent subsequence (x_{n_k}) . Let $x_\infty := \lim_{k \rightarrow \infty} x_{n_k}$ and observe that $x_\infty \in [a, b]$ by Lemma 14.5.

We claim that $y_{n_k} \rightarrow x_\infty$ as well. Note that, by construction, $y_{n_k} - x_{n_k}$ converges to 0, and

$$y_{n_k} = (y_{n_k} - x_{n_k}) + x_{n_k}.$$

Thus,

$$\lim_{k \rightarrow \infty} y_{n_k} = \lim_{k \rightarrow \infty} (y_{n_k} - x_{n_k}) + \lim_{k \rightarrow \infty} x_{n_k} = 0 + x_\infty = x_\infty.$$

Thus, by continuity of f at x_∞ ,

$$\lim_{k \rightarrow \infty} (f(x_{n_k}) - f(y_{n_k})) = f(x_\infty) - f(x_\infty) = 0.$$

But this contradicts the fact that $|f(x_{n_k}) - f(y_{n_k})| \geq \varepsilon_0$ for all $k \in \mathbb{N}$. Thus, f is uniformly continuous on $[a, b]$. □

Example 16.6. Let $p : [a, b] \rightarrow \mathbb{R}$ be any polynomial. Then since p is continuous, p is uniformly continuous on $[a, b]$ by Theorem 16.5.

Part 4

Sequences and series of functions

17. LECTURE 17: SEQUENCES OF FUNCTIONS

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◊ Ross, *Elementary Analysis*: Chapter 4, Section 24
- ◊ Rudin, *Principles of Mathematical Analysis*: Chapter 7, some of “Discussion of Main Problem,” some of “Uniform Convergence,” and some of “Uniform Convergence and Continuity”
- ◊ Abbott, *Understanding Analysis*: Chapter 6, Section 2

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◊ Understand the difference between pointwise convergence and uniform convergence, and be able to prove or disprove both for simple sequences of functions.

In the first part of this course, we studied sequences of numbers. In particular, we thoroughly developed what it means for a sequence of numbers to converge, and we explored the consequences of this notion. After that we began studying functions. In particular, we carefully developed what it means for a function to be continuous. Now, we will combine these two concepts! We will begin studying *sequences of functions*. Studying sequences of functions is especially interesting because there are multiple notions of “convergence,” unlike with numerical sequences. While this may seem convoluted, studying the limit of sequences of functions is incredibly fruitful in both pure and applied math. In many situations you want to solve some problem, and the solution is a function which is the limit of some sequences of functions (for example, you might try to solve a complicated differential equation by using a limiting process) and so a key feature of analysis is developing the tools to study the behavior of sequences of functions.

Definition 17.1. Let $I \subseteq \mathbb{R}$. A **sequence of functions on I** , denoted $(f_n)_{n=0}^\infty$, (f_n) , or just f_n , is formally a function $\mathbb{N} \rightarrow \mathcal{F}(I, \mathbb{R})$, where $\mathcal{F}(I, \mathbb{R}) := \{f \mid f : I \rightarrow \mathbb{R}\}$. Less formally, a sequence of functions on I is an ordered list of functions

$$(f_1, f_2, f_3, \dots)$$

where each $f_n : I \rightarrow \mathbb{R}$.

Example 17.2. Let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f_n(x) = 2x + n$. This is a sequence of functions, whose graphs can be visualized by a sequence of lines with slope 2 with increasingly higher y intercepts.

Definition 17.3 (Pointwise convergence). Let (f_n) be a sequence of functions on $I \subset \mathbb{R}$, and let $f : I \rightarrow \mathbb{R}$. We say that f_n **converges to f pointwise**, written $f_n \rightarrow f$ **pointwise**, if the following condition holds:

For every $x \in I$, we have $f_n(x) \rightarrow f(x)$.

That is, expanding out the quantifiers packed into the statement $f_n(x) \rightarrow f(x)$, this means that $f_n \rightarrow f$ pointwise if the following condition holds:

For every $x \in I$ and for every $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$, we have
 $|f_n(x) - f(x)| < \varepsilon$.

In words, a sequence of functions f_n converges pointwise to a limiting function f if, for each fixed point x in the domain, the numerical sequence $f_n(x)$ converges to the fixed number $f(x)$.

Example 17.4. Let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f_n(x) = \frac{x}{1+nx^2}$. Let $0 : \mathbb{R} \rightarrow \mathbb{R}$ denote the constant 0 function, defined by $0(x) = 0$ for all $x \in \mathbb{R}$. We claim that $f_n \rightarrow 0$ pointwise. Indeed, fix $x \in \mathbb{R}$. There are two cases to consider.

- ◇ Case I: $x = 0$. If $x = 0$, then since $f_n(0) = 0$ for all $n \in \mathbb{N}$, we have $\lim_{n \rightarrow \infty} f_n(0) = \lim_{n \rightarrow \infty} 0 = 0$.
- ◇ Case II: $x \neq 0$. If $x \neq 0$, note that $1 + nx^2$ diverges to ∞ as $n \rightarrow \infty$. By a homework exercise, this implies that $\frac{x}{1+nx^2} \rightarrow 0$.

Thus, for all $x \in \mathbb{R}$, we have $f_n(x) \rightarrow 0$. It follows that $f \rightarrow 0$ pointwise.

A central theme of analysis is understanding what properties are preserved under taking limits. For example, here is a natural question.

Question 17.5. Suppose that $f_n \rightarrow f$ pointwise, and each f_n is continuous. Is f necessarily continuous?

Annoyingly but interestingly, the answer to this question is actually no!

Example 17.6. Consider the sequence of functions $f_n : [0, 1] \rightarrow \mathbb{R}$ given by $f(x) = x^n$, and define $f : [0, 1] \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} 0 & 0 \leq x < 1 \\ 1 & x = 1 \end{cases}.$$

We claim that $f_n \rightarrow f$ pointwise. First, suppose $x < 1$. Then

$$f_n(x) = f_n(1) = 1^n = 1 \rightarrow 1 = f(1).$$

Next, suppose that $x \in [0, 1)$. Then $f_n(x) = x^n$. Since $|x| < 1$, we have $\lim_{n \rightarrow \infty} x^n = 0 = f(x)$. Thus, $f_n \rightarrow f$ pointwise. Finally, note that each f_n is continuous, but f is not continuous at $x = 1$ (for example, because the sequence $1 - \frac{1}{n} \rightarrow 1$ but $f(1 - \frac{1}{n}) = 0 \rightarrow 0 \neq 1$.)

This kind of phenomenon is annoying, and it would be desirable to have a different type of convergence that preserves continuity, in the sense that if f_n is a sequence of continuous functions then the limiting function is also continuous. Fortunately, there is a notion of convergence that does this, given by the following definition.

Definition 17.7 (Uniform convergence). Let (f_n) be a sequence of functions on $I \subset \mathbb{R}$, and let $f : I \rightarrow \mathbb{R}$. We say that f_n **converges to f uniformly**, written $f_n \rightarrow f$ **uniformly**, if the following condition holds:
 For every $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$ and all $x \in I$, we have $|f_n(x) - f(x)| < \varepsilon$.

It is important to compare and contrast the definition of *uniform* convergence with the definition of *pointwise* convergence, both formally (i.e. at the level of quantifiers), and intuitively. For reference, here are the two statements:

◇ $f_n \rightarrow f$ pointwise:

For every $x \in I$ and for every $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$, we have $|f_n(x) - f(x)| < \varepsilon$.

◇ $f_n \rightarrow f$ uniformly:

For every $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$ and all $x \in I$, we have $|f_n(x) - f(x)| < \varepsilon$.

Formally, the difference is essentially the order of the four quantifiers. In the definition of pointwise convergence, the quantifiers are *for all x* then *for all ε* then *there exists N* then *for all n* . In the definition of uniform convergence, it is *for all ε* then *there exists N* then *for all n* then *for all x* . In pointwise convergence, the *for all x* quantifier comes first, and in uniform convergence the *for all x* quantifier comes last. This makes a huge difference!

Intuitively, pointwise convergence says that all of the values of the functions f_n converge to the values of f , but the rate at which the convergence happens could depend on the point x in the domain. That is, for $x_1 \neq x_2 \in I$, the rate at which $f_n(x_1) \rightarrow f(x_1)$ could be much different than the rate at which $f_n(x_2) \rightarrow f(x_2)$. As such, the choice of $N_i \in \mathbb{N}$ that gives control of $|f_n(x_i) - f(x_i)|$ could depend on x_i (and this is seen since $x \in I$ is quantified first in the definition of pointwise convergence). In contrast, uniform convergence roughly requires $f_n(x) \rightarrow f(x)$ at the same rate, in the sense that we can gain a *uniform* control of the variation $|f_n(x) - f(x)|$ with a single choice of N that works for all $x \in I$.

Example 17.8. Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be given by $f_n(x) = (x - \frac{1}{n})^2$. Let $f : [0, 1] \rightarrow \mathbb{R}$ be given by $f(x) = x^2$. We claim that $f_n \rightarrow f$ uniformly.

Informal scratchwork.

Fix $\varepsilon > 0$. To prove that $f_n \rightarrow f$ uniformly, we ultimately have to gain uniform control of the term $|f_n(x) - f(x)|$. In particular, we need to show $|f_n(x) - f(x)| < \varepsilon$. Expanding this quantity out gives

$$\begin{aligned} |f_n(x) - f(x)| &= \left| \left(x - \frac{1}{n} \right)^2 - x^2 \right| \\ &= \left| x^2 - \frac{2}{n}x + \frac{1}{n^2} - x^2 \right| \\ &= \left| -\frac{2}{n}x + \frac{1}{n^2} \right|. \end{aligned}$$

Now I'm thinking about how to gain uniform control of this term on the interval $[0, 1]$. The most robust estimation tool we have is the triangle inequality, so we may as well try that first. We can conclude

$$\left| -\frac{2}{n}x + \frac{1}{n^2} \right| \leq \frac{2}{n}|x| + \frac{1}{n^2}.$$

Then I remember that my domain is $[0, 1]$, and so I have a uniform bound on $|x|$. In particular, I know that $|x| \leq 1$. So

$$\frac{2}{n}|x| + \frac{1}{n^2} \leq \frac{2}{n} + \frac{1}{n^2}.$$

Next, I know the sequence $\frac{2}{n} + \frac{1}{n^2}$ converges to 0 by limit laws, so we can always pick an N large enough to ensure that this sequence is less than ε .

Formal proof that $f_n \rightarrow f$ uniformly.

Fix $\varepsilon > 0$.

Note that, by algebraic limit laws, $\lim_{n \rightarrow \infty} \frac{2}{n} + \frac{1}{n^2} = 0$. Thus, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$, we have $\frac{2}{n} + \frac{1}{n^2} < \varepsilon$.

Let $n \geq N$ and $x \in [0, 1]$.

Note that

$$\begin{aligned} |f_n(x) - f(x)| &= \left| \left(x - \frac{1}{n}\right)^2 - x^2 \right| \\ &= \left| x^2 - \frac{2}{n}x + \frac{1}{n^2} - x^2 \right| \\ &= \left| -\frac{2}{n}x + \frac{1}{n^2} \right| \\ &\leq \frac{2}{n}|x| + \frac{1}{n^2} && \text{by the triangle inequality} \\ &\leq \frac{2}{n} + \frac{1}{n^2} && \text{since } x \in [0, 1] \text{ and so } |x| \leq 1 \\ &< \varepsilon. \end{aligned}$$

Thus, for all $\varepsilon > 0$ there is an $N \in \mathbb{N}$ such that, for all $n \geq N$ and all $x \in [0, 1]$, we have $|f_n(x) - f(x)| < \varepsilon$. Thus, $f_n \rightarrow f$ uniformly.

Example 17.9. Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be defined by $f_n(x) = x^n$. Note that $f_n \rightarrow 0$ pointwise. We claim that f_n does not converge to 0 uniformly.

Informal scratchwork.

In order to prove that f_n does not converge to 0 uniformly, we need to verify the negation of the definition of uniform convergence. Thus, the statement we need to verify is:

There is an $\varepsilon > 0$ such that, for all $N \in \mathbb{N}$, there is an $n \geq N$ and an $x \in [0, 1]$ with $|f_n(x) - f(x)| \geq \varepsilon$.

Thus, the first thing we need to do is choose an $\varepsilon > 0$. Since ε always determines some threshold of variation in the codomain, and $0 \leq f_n(x) \leq 1$ for all $x \in [0, 1]$, we should not pick $\varepsilon > 1$. It seems tempting to pick $\varepsilon = \frac{1}{2}$ (though any choice of $0 < \varepsilon < 1$ would work).

Then we don't get to choose N , so we'll fix that. Then we will somehow have to pick both an $n \geq N$ and also an $x \in [0, 1]$ such that $|f_n(x) - f(x)| \geq \frac{1}{2}$. Let's play around with the quantity of interest to see what we can discover. Note that

$$|f_n(x) - f(x)| = |x^n - 0| = x^n$$

since $x \geq 0$. Thus, we want to choose $n \geq N$ and $x \in [0, 1]$ such that $x^n \geq \frac{1}{2}$. Since x^n is continuous on $\mathbb{R}$ for any fixed n , and since $1^n = 1$, we know that, for any n , we can always choose an x close enough to 1 such that $x^n \geq \frac{1}{2}$ by appealing to continuity. So the choice of n probably doesn't matter, so we will let $n = N$. Then, we can choose x by a continuity argument. This will conclude the proof.

Formal proof that f_n does not converge to 0 uniformly.

Let $\varepsilon = \frac{1}{2}$.

Let $N \in \mathbb{N}$.

Let $n = N$.

Because $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = x^N$ is continuous, and in particular continuous at 1, there exists a $\delta > 0$ such that, for all $x \in \mathbb{R}$ with $|x - 1| < \delta$ we have $|x^N - 1^N| < \frac{1}{2}$.

Let $x = 1 - \frac{\delta}{2}$. Note that $|x - 1| = \frac{\delta}{2} < \delta$. Thus, by the previous remark, we have

$$\left| \left(1 - \frac{\delta}{2}\right)^N - 1^N \right| < \frac{1}{2}$$

and in particular

$$\left(1 - \frac{\delta}{2}\right)^N - 1 > -\frac{1}{2}.$$

This implies $x^N = \left(1 - \frac{\delta}{2}\right)^N > \frac{1}{2}$.

Finally, note that

$$\begin{aligned} |f_n(x) - 0(x)| &= |x^n - 0| \\ &= x^n \\ &= x^N && \text{(since } n = N\text{)} \\ &> \frac{1}{2} \end{aligned}$$

where the final inequality comes from the reasoning from above that $x^N = \left(1 - \frac{\delta}{2}\right)^N > \frac{1}{2}$. Thus, there is an $\varepsilon > 0$, namely $\varepsilon = \frac{1}{2}$, such that for all $N \in \mathbb{N}$ there is a $n \geq N$ and an $x \in [0, 1)$ such that $|f_n(x) - 0(x)| \geq \varepsilon$. Therefore, we have verified the negation of the definition of uniform convergence, and so f does not converge uniformly to 0.

18. LECTURE 18: MORE ON UNIFORM CONVERGENCE

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◊ Ross, *Elementary Analysis*: Chapter 4, some of Section 25
- ◊ Rudin, *Principles of Mathematical Analysis*: Chapter 7, some of “Uniform Convergence” and some of “Uniform Convergence and Continuity”
- ◊ Abbott, *Understanding Analysis*: Chapter 6, Section 2

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◊ Be more comfortable with proving things about uniform limits of functions, for example proving continuity of a uniform limit.

One of the major punchlines related to uniform convergence is that the uniform limit of a sequence of continuous functions is continuous. This is one of the many properties that uniform convergence preserves.

Theorem 18.1. *Let $f_n : I \rightarrow \mathbb{R}$ be a sequence of continuous functions. Let $f : I \rightarrow \mathbb{R}$. Suppose that $f_n \rightarrow f$ uniformly. Then f is continuous.*

Informal scratchwork before the proof. The proof of Theorem 18.1 is such a good proof that I want to include the informal discussion that leads to the discovery of the proof. The formal purple proof is below.

To prove that f is continuous, we have to fix an arbitrary anchor point $x_0 \in I$ and then prove that f is continuous at x_0 . By appealing to the $\varepsilon - \delta$ definition of continuity at a point, we know that the structure of our proof will be as follows:

- ◊ Fix $\varepsilon > 0$.
- ◊ Let δ be something (we don’t know what this is yet, and one of the hard parts of the proof will be to determine this δ).
- ◊ Let $x \in I$ satisfy $|x - x_0| < \delta$.
- ◊ Then, with all of these variables quantified, we will try to show that $|f(x) - f(x_0)| < \varepsilon$.

How will we gain control of $|f(x) - f(x_0)|$? We should think about the quantities that we can control, given the assumptions of the theorem.

- ◊ Since $f_n \rightarrow f$ uniformly, we know that we can gain control of quantities of the form $|f_n(y) - f(y)|$ for any $y \in I$. In particular, with x and x_0 quantified as above, we can gain control of $|f_n(x) - f(x)|$ and $|f_n(x_0) - f(x_0)|$, and crucially we can control them at the same time (i.e., independent of x and x_0) by the *uniform* convergence assumption.
- ◊ Since each f_n is continuous, we know that for any n we can gain control of $|f_n(x) - f_n(x_0)|$. Necessarily, we will have to choose an n before we gain control of such a term.

Thus, we need to figure out how to connect the term $|f(x) - f(x_0)|$ that we ultimately care about with the terms $|f_n(x) - f(x)|$, $|f_n(x_0) - f(x_0)|$, and $|f_n(x) - f_n(x_0)|$. We can do this with an enhanced version of the

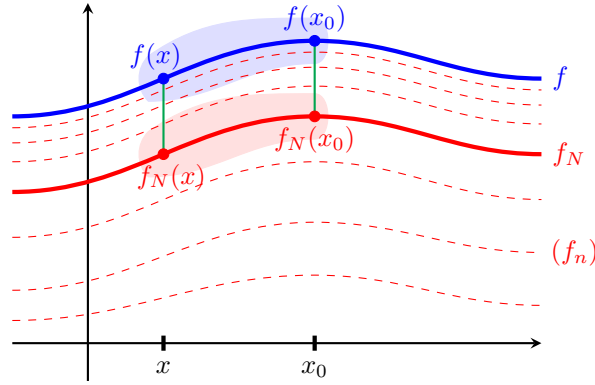


FIGURE 6. The goal is to prove that f is continuous at x_0 , which means that the distance from $f(x_0)$ and $f(x)$ needs to be controlled. This desired control is represented by the shaded blue region. We achieve this control via a chain with three links: two of the links, depicted by the green lines, represent the control we gain from uniform convergence $f_n \rightarrow f$. The middle link of the chain is given by continuity of f_N at x_0 , and is represented by the shaded red region.

trick we have seen many times in this course:

$$\begin{aligned} |f(x) - f(x_0)| &= |f(x) - f_N(x) + f_N(x) - f_N(x_0) + f_N(x_0) - f(x_0)| \\ &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)|. \end{aligned}$$

We can control the first and third terms by appealing to uniform convergence and choosing an n sufficiently large. Then once we have chosen a sufficiently large n , we can control the middle term by appealing to continuity of that function f_n .

The idea is described by figure 6.

Proof of Theorem 18.1. Fix $x_0 \in I$. We need to prove that f is continuous at x_0 .

Fix $\varepsilon > 0$.

Because $f_n \rightarrow f$ uniformly, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$ and all $x \in I$ we have $|f_n(x) - f(x)| < \frac{\varepsilon}{3}$.

Because f_N is continuous at x_0 , there is a $\delta > 0$ such that, for all $x \in I$ with $|x - x_0| < \delta$ we have $|f_N(x) - f_N(x_0)| < \frac{\varepsilon}{3}$.

Let $x \in I$ satisfy $|x - x_0| < \delta$.

Note that

$$\begin{aligned} |f(x) - f(x_0)| &= |f(x) - f_N(x) + f_N(x) - f_N(x_0) + f_N(x_0) - f(x_0)| \\ &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon. \end{aligned}$$

Thus, for all $\varepsilon > 0$, there is a $\delta > 0$ such that, for all $x \in I$ with $|x - x_0| < \delta$ we have $|f(x) - f(x_0)| < \varepsilon$.

Thus, f is continuous at $x_0 \in I$. Since x_0 was arbitrary, it follows that f is continuous. $\square$

Example 18.2. Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be given by $f_n(x) = x^n$. Recall that $f_n \rightarrow f$ pointwise, where

$$f(x) = \begin{cases} 0 & 0 \leq x < 1 \\ 1 & x = 1 \end{cases}.$$

Since each f_n is continuous on $[0, 1]$, but f is not continuous on $[0, 1]$, it follows that f_n does *not* converge to f uniformly.

Just like with numerical sequences, we would like a way to investigate whether a sequence of functions converges (pointwise, or uniformly) without having to necessarily identify the candidate limiting function in advance. With numerical sequences, we then introduced the property of being Cauchy (which we eventually proved is equivalent to being convergent). We can do the same thing with sequences of functions.

Definition 18.3. Let $f_n : I \rightarrow \mathbb{R}$ be a sequence of functions.

- (1) We say that f_n is **pointwise Cauchy** if the following condition holds:

For all $x \in I$, the sequence $f_n(x)$ is Cauchy.

That is, f_n is pointwise Cauchy if the following condition holds:

For all $x \in I$ and for all $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that, for all $n, m \geq N$, we have

$$|f_n(x) - f_m(x)| < \varepsilon.$$

- (2) We say that f_n is **uniformly Cauchy** if the following condition holds:

For all $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that, for all $n, m \geq N$ and for all $x \in I$, we have

$$|f_n(x) - f_m(x)| < \varepsilon.$$

The same comments can be made about the difference between pointwise Cauchy and uniformly Cauchy as with pointwise convergence and uniform convergence, both formally (the order of the quantifiers) and intuitively.

Also, because the property of being Cauchy is equivalent to being convergent for numerical sequences, it should be clear that a sequence f_n is pointwise Cauchy if and only if it pointwise convergent to some limiting function f . The more interesting part of the above definition is the *uniformly Cauchy* property. If a sequence is uniformly Cauchy, does it necessarily converge uniformly to some limiting function? We would hope the answer is yes, and fortunately it is!

Theorem 18.4. Let $f_n : I \rightarrow \mathbb{R}$ be a sequence of functions. Then f_n is uniformly Cauchy if and only if there is a function $f : I \rightarrow \mathbb{R}$ such that $f_n \rightarrow f$ uniformly.

Proof. Uniform convergence implies uniformly Cauchy.

There are two directions to prove. We begin with the easier one. Suppose that $f : I \rightarrow \mathbb{R}$ is a function such that $f_n \rightarrow f$ uniformly. We will prove that f_n is uniformly Cauchy.

Let $\varepsilon > 0$. Since $f_n \rightarrow f$ uniformly, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$ and all $x \in I$, we have $|f_n(x) - f(x)| < \frac{\varepsilon}{2}$. Fix $n, m \geq N$ and $x \in I$. Note that

$$\begin{aligned} |f_n(x) - f_m(x)| &= |f_n(x) - f(x) + f(x) - f_m(x)| \\ &\leq |f_n(x) - f(x)| + |f(x) - f_m(x)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Thus, for all $\varepsilon > 0$ there is an $N \in \mathbb{N}$ such that, for all $n, m \geq N$ and $x \in I$, we have $|f_n(x) - f_m(x)| < \varepsilon$. It follows that f_n is uniformly Cauchy.

Uniformly Cauchy implies uniform convergence.

Suppose that f_n is uniformly Cauchy. We need to prove that there is a function $f : I \rightarrow \mathbb{R}$ such that $f_n \rightarrow f$ uniformly.

First, we will define the function f . Fix $x_0 \in I$. We claim that the numerical sequence $f_n(x_0)$ is Cauchy. Indeed, fix $\varepsilon > 0$. Because $f_n \rightarrow f$ uniformly, there is an $N \in \mathbb{N}$ such that, for all $n, m \geq N$ and all $x \in I$, we have $|f_n(x) - f_m(x)| < \varepsilon$. In particular, for all $n, m \geq N$ we have $|f_n(x_0) - f_m(x_0)| < \varepsilon$. Thus, the sequence $f_n(x_0)$ is Cauchy. By the Cauchy-convergence equivalence (Theorem 11.1) the sequence $f_n(x_0)$ converges. Thus, we may define $f : I \rightarrow \mathbb{R}$ as follows: For each $x_0 \in I$, set

$$f(x_0) := \lim_{n \rightarrow \infty} f_n(x_0).$$

Now we claim that $f_n \rightarrow f$ uniformly.

Let $\varepsilon > 0$. Here we are initiating a different instance of $\varepsilon > 0$ than the one used to prove that $f_n(x_0)$ is a Cauchy sequence above.

Since f_n is uniformly Cauchy, there is an $N \in \mathbb{N}$ such that, for all $n, m \geq N$ and $x \in I$, we have $|f_n(x) - f_m(x)| < \frac{\varepsilon}{2}$.

Let $n \geq N$, and let $x \in I$. Our goal is to show that $|f_n(x) - f(x)| < \varepsilon$. Because $f_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$, there is an $N_1 \in \mathbb{N}$ such that, for all $m \geq N_1$, we have $|f_m(x) - f(x)| < \frac{\varepsilon}{2}$. Let $m = \max(N, N_1)$. Then $m \geq N$ and $m \geq N_1$. Thus,

$$\begin{aligned} |f_n(x) - f(x)| &= |f_n(x) - f_m(x) + f_m(x) - f(x)| \\ &\leq |f_n(x) - f_m(x)| + |f_m(x) - f(x)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Thus, for all $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$ and $x \in I$, we have $|f_n(x) - f(x)| < \varepsilon$. It follows that $f_n \rightarrow f$ uniformly. $\square$

Remark 18.5. This theorem will be especially useful when we study infinite sums of functions.

This will be the end of the material that is tested on midterm 2.

19. LECTURE 19: INFINITE SERIES

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◊ Ross, *Elementary Analysis*: Chapter 2, Section 14
- ◊ Rudin, *Principles of Mathematical Analysis*: Chapter 3, “Series”
- ◊ Abbott, *Understanding Analysis*: Chapter 2, Section 4 and Section 7

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◊ Understand the definition of an infinite via the sequence of partial sums.
- ◊ Be able to prove that simple series converge or diverge using the partial sum definition.
- ◊ Understand the Cauchy criterion for infinite sums.

One of the classic features of a calculus course is the study of infinite sums. For example, can we make sense of the never ending sum of numbers

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots .$$

Do we get a finite number? If so, what does it equal? Beginning in this lecture we will treat infinite sums with the rigor of real analysis.

First, a reminder on sigma summation notation is in order. Given a sequence of numbers $(a_0, a_1, a_2, \dots)$, we have the following shorthand notation to represent a sum of (some of) the elements of the sequence: For $N_0 \leq N_1 \in \mathbb{N}$, we define

$$\sum_{n=N_0}^{N_1} a_n := a_{N_0} + a_{N_0+1} + \cdots + a_{N_1} .$$

For example,

$$\sum_{n=3}^6 \frac{1}{n^2} = \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} .$$

The goal of today’s lecture is to make give meaning to the collection of symbols

$$“ \sum_{n=0}^{\infty} a_n ”$$

which currently doesn’t make sense (hence the quotes), as ∞ is not a number.

Remark 19.1. A simple remark, but an important one. Note that

$$\sum_{n=N_0}^{N_1} a_n = \sum_{k=N_0}^{N_1} a_k = \sum_{p=N_0}^{N_1} a_p .$$

In other words, the *index* of the summation is just a dummy variable.

The way that we will give precise meaning to an infinite sum is by recasting it as a sequence in its own right. This is made formal by the following definition.

Definition 19.2 (Sequence of partial sums.). Let (a_n) be a sequence. Define the **sequence of partial sums of (a_n)** to be the sequence (S_N) defined as follows:

$$S_N := \sum_{n=0}^N a_n.$$

Example 19.3. Let $a_n = 1$ be the constant sequence $(1, 1, 1, \dots)$. Then the sequence of partial sums is given by

$$S_N = \sum_{n=0}^N 1 = \underbrace{1 + \dots + 1}_{N+1 \text{ times}} = N + 1.$$

That is, the sequence of partial sums is

$$(S_N)_{N=0}^\infty = (1, 2, 3, 4, \dots).$$

Example 19.4. Let $b_n = (-1)^n$. The sequence of partial sums is then given by

$$S_N = \begin{cases} 1 & \text{if } N \text{ is even} \\ 0 & \text{if } N \text{ is odd} \end{cases}.$$

Definition 19.5 (Infinite series). Let (a_n) be a sequence, and let (S_N) be the sequence of partial sums. If the sequence (S_N) converges, then we define

$$\sum_{n=0}^{\infty} a_n := \lim_{N \rightarrow \infty} S_N$$

and we say that $\sum_{n=0}^{\infty} a_n$ **converges**. If the sequence (S_N) diverges, then we say $\sum_{n=0}^{\infty} a_n$ **diverges**. In either case we refer to the collection of symbols

$$\sum_{n=0}^{\infty} a_n$$

as an **infinite series**, an **infinite sum**, or just a **series**.

Remark 19.6. Make sure you are very careful about distinguishing *sequences* and *series*, as this is a very common mistake in calculus classes!

Remark 19.7. Note that an infinite series

$$\sum_{n=0}^{\infty} a_n$$

converges if and only if

$$\sum_{n=N_0}^{\infty} a_n$$

for some N_0 . This should hopefully be obvious, but I'll leave it as an exercise. I will use this fact frequently, often without comment!

Example 19.8. Let $a_n = 1$ be the constant sequence. By Example 19.3, we know that $S_N = N + 1$. This sequence diverges to ∞ , and so

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} 1$$

diverges. This makes sense intuitively, because if we add 1 to itself over and over again, the resulting sum grows uncontrollably.

Example 19.9. Let $b_n = (-1)^n$. By Example 19.4, the sequence of partial sums is given by

$$S_N = \begin{cases} 1 & \text{if } N \text{ is even} \\ 0 & \text{if } N \text{ is odd} \end{cases}.$$

The sequence S_N diverges because the subsequence $S_{2K} \rightarrow 1$ and the subsequence $S_{2K+1} \rightarrow 0$. Thus, the sum

$$\sum_{n=0}^{\infty} b_n = \sum_{n=0}^{\infty} (-1)^n$$

diverges.

Example 19.10. Let $c_n = 0$ be the constant 0 sequence. Then $S_N = 0$ is also the constant 0 sequence. Thus, since $S_N \rightarrow 0$, we have

$$\sum_{n=0}^{\infty} c_n = 0.$$

Before we see some more interesting examples of convergent infinite sums, we will state and prove some simple but useful facts regarding the convergence of infinite sums.

Theorem 19.11 (Cauchy criterion for series). *An infinite sum $\sum_{n=0}^{\infty} a_n$ converges if and only if the following condition holds:*

For every $\varepsilon > 0$, there is an $N_0 \in \mathbb{N}$ such that, for all $N > M \geq N_0$, we have

$$\left| \sum_{n=M+1}^N a_n \right| < \varepsilon.$$

Proof. This is just a rephrasing of the Cauchy-convergence equivalence applied to the sequence of partial sums. Indeed, by definition the sum $\sum_{n=0}^{\infty} a_n$ converges if and only if the sequence (S_N) converges, which by Theorem 11.1 happens if and only if (S_N) is Cauchy. To say that (S_N) is Cauchy is to say that, for every $\varepsilon > 0$, there is an $N_0 \in \mathbb{N}$ such that, for all $N > M \geq N_0$ we have $|S_N - S_M| < \varepsilon$. Note that

$$\begin{aligned} S_N - S_M &= \sum_{n=0}^N a_n - \sum_{n=0}^M a_n \\ &= \sum_{n=M+1}^N a_n. \end{aligned}$$

This completes the proof. □

Theorem 19.12 (Divergence test). *Let (a_n) be a sequence. Suppose that $\lim_{n \rightarrow \infty} a_n \neq 0$. Then the series $\sum_{n=0}^{\infty} a_n$ diverges.*

Proof. We prove the contrapositive, which says that if $\sum_{n=0}^{\infty} a_n$ converges, then $a_n \rightarrow 0$. Fix $\varepsilon > 0$. By Theorem 19.11, there is an $N_0 \in \mathbb{N}$ such that, for $N > M \geq N_0$ we have

$$\left| \sum_{n=M+1}^N a_n \right| < \varepsilon.$$

In particular, for $n \geq N_0 + 1$ we have

$$|a_n| = \left| \sum_{k=n}^n a_k \right| < \varepsilon.$$

Thus, $a_n \rightarrow 0$. □

Example 19.13. Consider the sum

$$\sum_{n=1}^{\infty} \left(1 - \frac{1}{n} \right).$$

Since $1 - \frac{1}{n} \rightarrow 1 \neq 0$, the divergence test implies that the series diverges.

Perhaps the simplest, most useful, and most well understood family of infinite sums is that of *geometric series*.

Theorem 19.14 (Geometric series). *The infinite series*

$$\sum_{n=0}^{\infty} r^n$$

converges if and only if $|r| < 1$. Moreover, if $|r| < 1$ then

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

Remark 19.15. Geometric series are extremely special because we can determine the value that one converges to. This is, in general, very rare in the world of math. It is one thing to ask whether an infinite sum converges, and it is a very different and much more difficult question to ask what a series converges to.

Remark 19.16. There is a more general formula (that I will have you prove as a homework exercise) for the value of a convergent geometric series, depending on the starting index. In particular, if $|r| < 1$ then

$$\sum_{n=N_0}^{\infty} cr^n = \frac{cr^{N_0}}{1-r}.$$

I remember this with the mantra that any kind of geometric sum converges to

$$\frac{\text{The first term of the sum}}{1 - (\text{the ratio})}.$$

This is a useful way to remember the formula, because the exact form of a geometric sum (the starting index, the powers of the ratio, any constants) can vary in the wild.

Proof. First, note that if $r = \pm 1$ then the series diverges by the above examples. Thus, it remains to consider the cases where $|r| < 1$ and $|r| > 1$. Let

$$S_N = \sum_{n=0}^N r^n = 1 + r + r^2 + \cdots + r^N$$

denote the sequence of partial sums. Note that

$$\begin{aligned} S_N - rS_N &= \sum_{n=0}^N r^n - r \sum_{n=0}^N r^n \\ &= \sum_{n=0}^N r^n - \sum_{n=0}^N r^{n+1} \\ &= \sum_{n=0}^N r^n - \sum_{n=1}^{N+1} r^n \\ &= \left(1 + \sum_{n=1}^N r^n \right) - \left(r^{N+1} + \sum_{n=1}^N r^n \right) \\ &= 1 - r^{N+1}. \end{aligned}$$

Thus, for $r \neq 1$ we have

$$S_N = \frac{1 - r^{N+1}}{1 - r}.$$

Note that if $|r| < 1$, then $r^{N+1} \rightarrow 0$ and so S_N converges to $\frac{1}{1-r}$. This means that $\sum_{n=0}^{\infty} r^n$ converges and

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

as desired. If $|r| > 1$, then r^{N+1} diverges (to ∞) and so S_N diverges. □

Example 19.17. Consider the infinite sum

$$\sum_{n=2}^{\infty} \frac{2 \cdot 3^{n+1}}{4^n}.$$

We claim that this is a convergent geometric series. **Indeed, note that**

$$\frac{2 \cdot 3^{n+1}}{4^n} = 6 \left(\frac{3}{4} \right)^n.$$

Thus, this is a geometric series with $r = \frac{3}{4}$, and since $|r| < 1$ it follows that the sum converges. Moreover,

$$\sum_{n=2}^{\infty} \frac{2 \cdot 3^{n+1}}{4^n} = \frac{6 \left(\frac{3}{4} \right)^2}{1 - \frac{3}{4}}.$$

Theorem 19.18 (p -series). *The infinite series*

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if $p > 1$.

Remark 19.19. There are many ways to prove this theorem. Common proofs involve the integral test, or comparisons with a geometric series. In the exercises I will have you work through a different proof using something called the Cauchy condensation test.

Proof. Let $S_N = \sum_{n=1}^N \frac{1}{n^p}$ denote the sequence of partial sums.

First, suppose that $p > 1$.⁸ Note that $S_{N+1} - S_N = \frac{1}{N+1} > 0$ and so $S_{N+1} \geq S_N$, so that (S_N) is an increasing sequence. By Theorem 7.6, to conclude that the sum converges it suffices to show that (S_N) is bounded above. Fix $N \in \mathbb{N}$. We begin by considering the partial sum S_{2N+1} . Note that

$$\begin{aligned} S_{2N+1} &= \sum_{n=1}^{2N+1} \frac{1}{n^p} \\ &= \frac{1}{1^p} + \left(\frac{1}{2^p} + \frac{1}{3^p} \right) + \left(\frac{1}{4^p} + \frac{1}{5^p} \right) + \cdots + \left(\frac{1}{(2N)^p} + \frac{1}{(2N+1)^p} \right) \\ &= 1 + \sum_{k=1}^N \left(\frac{1}{(2k)^p} + \frac{1}{(2k+1)^p} \right) \\ &< 1 + \sum_{k=1}^N \left(\frac{1}{(2k)^p} + \frac{1}{(2k)^p} \right) \\ &= 1 + \sum_{k=1}^N \frac{2}{(2k)^p} \\ &= 1 + \frac{1}{2^{p-1}} \sum_{k=1}^N \frac{1}{k^p} \\ &= 1 + \frac{1}{2^{p-1}} S_N. \end{aligned}$$

Since $2N+1 > N$ and (S_N) is increasing, we have $S_N \leq S_{2N+1}$. This inequality together with the above estimate gives

$$S_N < 1 + \frac{1}{2^{p-1}} S_N$$

which yields

$$\left(1 - \frac{1}{2^{p-1}} \right) S_N < 1.$$

Because $p > 1$, we have $2^{p-1} > 1$ and hence $1 - \frac{1}{2^{p-1}} > 0$. Thus,

$$S_N < \left(1 - \frac{1}{2^{p-1}} \right)^{-1}.$$

Thus, (S_N) is bounded above and so $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges.

Next, suppose that $p \leq 1$. Note that $n^p \leq n$ and hence $\frac{1}{n^p} \geq \frac{1}{n}$. Thus,

$$\sum_{n=1}^N \frac{1}{n^p} \geq \sum_{n=1}^N \frac{1}{n}$$

⁸This very clever proof is due to [this StackExchange solution](#).

and so it suffices to prove that S_N diverges to ∞ when $p = 1$. Fix $N \in \mathbb{N}$. Note that

$$\begin{aligned}
 S_{2N} &= \sum_{n=1}^{2N} \frac{1}{n} \\
 &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \cdots + \left(\frac{1}{2N-1} + \frac{1}{2N}\right) \\
 &\geq 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \cdots + \left(\frac{1}{2N} + \frac{1}{2N}\right) \\
 &= 1 + \frac{1}{2} + \left(\frac{2}{4}\right) + \cdots + \left(\frac{2}{2N}\right) \\
 &= \frac{1}{2} + \left(1 + \frac{1}{2} + \cdots + \frac{1}{N}\right) \\
 &= \frac{1}{2} + S_N.
 \end{aligned}$$

Thus, for all $N \in \mathbb{N}$ we have $S_{2N} - S_N \geq \frac{1}{2}$. This implies that the sequence (S_N) is not Cauchy, and hence S_N diverges. Since S_N is an increasing sequence, it follows that S_N diverges to ∞ , as desired. Thus, the sum⁹

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

diverges, and therefore so does $\sum_{n=1}^{\infty} \frac{1}{n^p}$ for $p \leq 1$.

□

⁹Known as the *harmonic series*.

20. LECTURE 20: MORE ON INFINITE SERIES (THE ROOT, RATIO, AND COMPARISON TESTS)

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 2, Section 14
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 3, “The Root and Ratio Test”
- ◇ Abbott, *Understanding Analysis*: Chapter 2, Section 7

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Understand the statement of the root test and the ratio test, and be able to use them to verify that certain series converge or diverge.
- ◇ Understand the statement of the direct comparison test, and be able to use it to verify that certain series converge or diverge.
- ◇ Understand why absolute convergence implies convergence, and more generally be more comfortable with using the Cauchy criterion for series to prove statements about them.

In this lecture we will introduce and prove a number of concepts that are likely familiar from calculus (absolute convergence, the comparison tests, and the root and ratio tests). First, we actually state a theorem that gives the existence of functions of the form $x \mapsto x^\alpha$ for any real number $\alpha > 0$. If you have been reading carefully, you’ll have noticed that I haven’t yet used functions like this (just as I haven’t used trigonometric functions or exponential functions). This is completely intentional, because it is nontrivial to even construct such functions. Now, we will need to use such functions.

Theorem 20.1. For all $\alpha > 0$ and all $x \geq 0$, there is a real number x^α satisfying the following properties.

- ◇ For all $\alpha > 0$, $0^\alpha = 0$ and $1^\alpha = 1$.
- ◇ For all $n \in \mathbb{N}$ and $x > 0$, $y := x^{\frac{1}{n}} \in \mathbb{R}$ is the unique real number such that $y^n = x$.
- ◇ For all $\alpha > 0$ and all $0 \leq x < y$, we have $x^\alpha < y^\alpha$.
- ◇ For all $\alpha > 0$, the function $x \mapsto x^\alpha$ is continuous on $[0, \infty)$.
- ◇ For all $\alpha, \beta > 0$ and $x > 0$, we have $x^\alpha x^\beta = x^{\alpha+\beta}$ and $(x^\alpha)^\beta = x^{\alpha\beta}$.

Proof. Optional; see Subsection 20.2 below. □

Definition 20.2. We say that $\sum_{n=0}^{\infty} a_n$ **converges absolutely** if $\sum_{n=0}^{\infty} |a_n|$ converges.

Theorem 20.3 (Absolute convergence implies convergence). Suppose that $\sum_{n=0}^{\infty} |a_n|$ converges. Then $\sum_{n=0}^{\infty} a_n$ converges. In other words, if $\sum_{n=0}^{\infty} a_n$ converges absolutely, then it converges.

Proof. Fix $\varepsilon > 0$. Since $\sum_{n=0}^{\infty} |a_n|$ converges, it satisfies the Cauchy criterion. Thus, there is an $N_0 \in \mathbb{N}$ such that, for $N > M \geq N_0$ we have

$$\left| \sum_{n=M+1}^N |a_n| \right| < \varepsilon.$$

Note that

$$\begin{aligned} \left| \sum_{n=M+1}^N a_n \right| &\leq \sum_{n=M+1}^N |a_n| \\ &= \left| \sum_{n=M+1}^N |a_n| \right| \\ &< \varepsilon. \end{aligned}$$

Thus, $\sum_{n=0}^{\infty} a_n$ satisfies the Cauchy criterion for series and hence converges. $\square$

Theorem 20.4 (Comparison test). *Let $(a_n), (b_n)$ be sequences. Suppose that there is an $N_0 \in \mathbb{N}$ such that, for $n \geq N_0$ we have $a_n \geq b_n \geq 0$.*

- $\diamond$ *If $\sum_{n=0}^{\infty} a_n$ converges, then $\sum_{n=0}^{\infty} b_n$ converges.*
- $\diamond$ *If $\sum_{n=0}^{\infty} b_n$ diverges, then $\sum_{n=0}^{\infty} a_n$ diverges.*

Proof. Let S_N^a and S_N^b denote the sequence of partial sums for the sums $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$, respectively. As usual we can assume without loss of generality that $N_0 = 0$.

First, suppose that $\sum_{n=0}^{\infty} a_n$ converges. Since $b_n \geq 0$, S_N^b is an increasing sequence. Since $b_n \leq a_n$ for all $n \in \mathbb{N}$, we have, for all $N \in \mathbb{N}$,

$$\begin{aligned} S_N^b &= \sum_{n=0}^N b_n \\ &\leq \sum_{n=0}^N a_n \\ &\leq S_N^a. \end{aligned}$$

Since $\sum_{n=0}^{\infty} a_n$ converges, S_N^a converges and hence S_N^a is bounded. By the above, $S_N^b \leq S_N^a \leq \sup S_N^a$ and so (S_N^b) is bounded above. Since (S_N^b) is an increasing sequence which is bounded above, Theorem 7.6 implies that S_N^b converges. Thus, $\sum_{n=0}^{\infty} b_n$ converges.

Next, suppose that $\sum_{n=0}^{\infty} b_n$ diverges. Because (S_N^b) is an increasing sequence, this means that (S_N^b) diverges to ∞ . Since $S_N^a \geq S_N^b$, it follows that S_N^a diverges to ∞ , and thus $\sum_{n=0}^{\infty} a_n$ diverges. $\square$

Theorem 20.5 (Root test). *Let $\sum_{n=0}^{\infty} a_n$ be an infinite series, and let $\alpha := \limsup |a_n|^{\frac{1}{n}}$.*

- $\diamond$ *If $\alpha < 1$, then $\sum_{n=0}^{\infty} a_n$ converges absolutely.*
- $\diamond$ *If $\alpha > 1$, then $\sum_{n=0}^{\infty} a_n$ diverges.*

Proof. First, suppose that $\alpha < 1$. Set $\varepsilon := \frac{1-\alpha}{2}$, so that $\alpha < \alpha + \varepsilon < 1$. Since $\alpha = \limsup |a_n|^{\frac{1}{n}}$, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$, we have $|a_n|^{\frac{1}{n}} < \alpha + \varepsilon$. Thus, $|a_n| < (\alpha + \varepsilon)^n$ for all $n \geq N$. Since

$0 < \alpha + \varepsilon < 1$, the sum

$$\sum_{n=0}^{\infty} (\alpha + \varepsilon)^n$$

is a convergent geometric series. Since $0 \leq |a_n| < (\alpha + \varepsilon)^n$, the comparison test implies

$$\sum_{n=1}^{\infty} |a_n|$$

converges.

Next, suppose that $\alpha > 1$. By the subsequential characterization of the $\limsup$, there is a subsequence a_{n_k} such that $|a_{n_k}|^{\frac{1}{n_k}} \rightarrow \alpha$. In particular, there is a $K \in \mathbb{N}$ such that, for $k \geq K$, we have $|a_{n_k}|^{\frac{1}{n_k}} > 1$. This implies that

$$|a_{n_k}| > 1^{n_k} = 1$$

for $k \geq K$. In particular, $\limsup |a_n| \geq 1 > 0$. This implies that (a_n) cannot converge to 0. By the divergence test, it follows that $\sum_{n=0}^{\infty} a_n$ diverges. □

Theorem 20.6 (Ratio test). *Let $\sum_{n=0}^{\infty} a_n$ be an infinite series with $a_n \neq 0$.*

◇ *If*

$$\limsup \left| \frac{a_{n+1}}{a_n} \right| < 1$$

then $\sum_{n=0}^{\infty} a_n$ converges.

◇ *If*

$$\liminf \left| \frac{a_{n+1}}{a_n} \right| > 1$$

then $\sum_{n=0}^{\infty} a_n$ diverges.

Proof. First, assume that $\limsup \left| \frac{a_{n+1}}{a_n} \right| < 1$. Let $\beta \in \mathbb{R}$ be any number such that

$$\limsup \left| \frac{a_{n+1}}{a_n} \right| < \beta < 1.$$

There is an $N \in \mathbb{N}$ such that, for $n \geq N$, we have $\left| \frac{a_{n+1}}{a_n} \right| < \beta$. This means that for all $n \geq N$ we have $|a_{n+1}| < \beta |a_n|$. In particular,

$$\begin{aligned} |a_{N+k}| &< \beta |a_{N+k-1}| \\ &< \beta^2 |a_{N+k-2}| \\ &< \beta^3 |a_{N+k-3}| \\ &\vdots \\ &< \beta^k |a_N|. \end{aligned}$$

Note that $\sum_{k=0}^{\infty} |a_N| \beta^k$ converges, since this is a geometric sum with ration $\beta \in (-1, 1)$. Thus, since $|a_{N+k}| < |a_N| \beta^k$ for all $k \in \mathbb{N}$, the comparison test implies that

$$\sum_{k=0}^{\infty} |a_{N+k}|$$

converges. Reindexing the sum gives convergence of $\sum_{n=N}^{\infty} |a_n|$, and hence of $\sum_{n=0}^{\infty} |a_n|$ as desired.

Next, suppose that $\liminf \left| \frac{a_{n+1}}{a_n} \right| > 1$. There is an $N \in \mathbb{N}$ such that, for all $n \geq N$, we have $\left| \frac{a_{n+1}}{a_n} \right| > 1$. This means that, for $n \geq N$, we have

$$|a_{n+1}| \geq |a_n|$$

and so $|a_n| \geq |a_N|$. This implies that $\liminf |a_n| \geq |a_N| \neq 0$. Thus, a_n does not converge to 0, and so the divergence test implies that $\sum_{n=0}^{\infty} a_n$ diverges. $\square$

20.1. Examples.

Example 20.7. Determine whether the following series converges or diverges.

$$\sum_{n=1}^{\infty} \frac{(n^3 - 1)^{\frac{1}{2}}}{2n^7 + n + 1}.$$

Solution. Note that

$$0 \leq \frac{(n^3 - 1)^{\frac{1}{2}}}{2n^7 + n + 1} \leq \frac{(n^3)^{\frac{1}{2}}}{2n^7} = \frac{1}{2n^{\frac{11}{2}}}.$$

Since the series

$$\sum_{n=1}^{\infty} \frac{1}{2n^{\frac{11}{2}}} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n^{\frac{11}{2}}}$$

is a convergent p -series. ($p = \frac{11}{2} > 1$), the direct comparison test implies that

$$\sum_{n=1}^{\infty} \frac{(n^3 - 1)^{\frac{1}{2}}}{2n^7 + n + 1}$$

converges. $\square$

Example 20.8. Determine whether the following series converges or diverges.

$$\sum_{n=1}^{\infty} \frac{n^2 3^n}{(2n)!}.$$

Solution. We use the ratio test. Observe that

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \frac{(n+1)^2 3^{n+1}}{(2(n+1))!} \cdot \frac{(2n)!}{n^2 3^n} = \lim_{n \rightarrow \infty} \frac{(2n)!}{(2n+2)!} \cdot \frac{3(n+1)^2}{n^2} \\ &= \lim_{n \rightarrow \infty} \frac{(2n)!}{(2n+2)(2n+1)(2n)!} \cdot 3 \left(1 + \frac{1}{n}\right)^2 \\ &= \lim_{n \rightarrow \infty} \frac{1}{(2n+2)(2n+1)} \cdot 3 \left(1 + \frac{1}{n}\right)^2. \end{aligned}$$

Since $3 \left(1 + \frac{1}{n}\right)^2 \rightarrow 3$, it follows that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 0 \cdot 3 = 0.$$

Since $0 < 1$, the ratio test implies that the series converges. $\square$

20.2. Proof of Theorem 20.1. In this subsection we prove Theorem 20.1, which gives the existence and properties of functions of the form $x \mapsto x^\alpha$. We begin with a lemma.

Lemma 20.9. Fix $n \in \mathbb{N}$ and $x \geq 0$. There is a unique real number $y \geq 0$ such that $y^n = x$.

Proof. **ADD** $\square$

Definition 20.10. Fix $x \geq 0$ and $\alpha > 0$. Define

$$x^\alpha := \sup\{x^q \mid q \in \mathbb{Q} \text{ satisfies } q < \alpha\}.$$

Proof of Theorem 20.1. **ADD**

□

21. LECTURE 21: SERIES OF FUNCTIONS

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 4, Section 25 and Section 23
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 3, “Power Series”
- ◇ Abbott, *Understanding Analysis*: Chapter 6, Section 1 and Section 4

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Understand the definition of pointwise and uniform convergence of an infinite series of functions.
- ◇ Be more comfortable with using and verifying concepts like uniformly Cauchy, the Cauchy criterion for series, uniform convergence of sequences of functions, and so on.
- ◇ Be able to use the Weierstrauss M -test to prove that simple infinite sums of functions converge uniformly.

We began 131A by studying sequences of numbers. Then we moved on to study functions and their properties, particularly continuity. Then we combined these endeavors and studied *sequences of functions*, which led to concepts like pointwise convergence and uniform convergence. After that, we introduced series of numbers, which are just infinite sums of numbers. We introduce the extra layer of functions again and in this lecture we start considering *infinite sums of functions*.

While this seems like it could be a complicated and ridiculous thing to do, it turns out to be very natural and extremely important. For example, there are many important functions that are ubiquitous in math that we have yet to discuss. These are functions like the exponential, e^x , and trigonometric functions like $\sin x$ and $\cos x$. We have not discussed or used these functions at all in analysis, because we have not had the ability to discuss them. When we start from the clean slate of abstract fields, we only know how to add, subtract, multiply, and divide. With these operations one can very naturally discuss polynomials and rational functions, but little else. For example, it took a great deal of effort in Lecture 20 to even discuss functions of the form x^α for any real number $\alpha > 0$. How in the world does one begin to describe a function like $\sin x$ with the level of rigor that we seek in 131A?

This is where infinite sums of functions becomes very natural. You will recall from your calculus class that we have “power series expansions” of many important functions. This gave formulas like

$$\begin{aligned}
 e^x &= \sum_{n=0}^{\infty} \frac{1}{n!} x^n \\
 \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \\
 \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}
 \end{aligned}$$

and so on. In the story of 131A, where we have forgotten all math beyond what we have covered so far, we will use these formulas to *define* these functions. For example, in Lecture 22 we will *define* the trigonometric function $\sin x$ to be exactly the infinite sum above. But in order to make sense of this — and then in order to prove desirable properties about such functions — we need to discuss infinite sums of functions.

Definition 21.1. Let I be an interval (open, compact, or neither) and let $f_n : I \rightarrow \mathbb{R}$ be a sequence of functions. For each $x \in I$, let

$$S_N(x) := \sum_{n=0}^N f_n(x)$$

be the sequence of partial sums of the infinite series $\sum_{n=0}^{\infty} f_n(x)$. Note that this defines a sequence of functions $S_N : I \rightarrow \mathbb{R}$.

- (1) If the sequence of functions S_N converges pointwise to some limiting function $f : I \rightarrow \mathbb{R}$, we write

$$f(x) := \sum_{n=0}^{\infty} f_n(x) \quad \text{and} \quad f = \sum_{n=0}^{\infty} f_n$$

and say that the infinite sum $\sum_{n=0}^{\infty} f_n$ **converges pointwise to f** .

- (2) If the sequence of functions S_N converges uniformly to some limiting function $f : I \rightarrow \mathbb{R}$, we write

$$f(x) := \sum_{n=0}^{\infty} f_n(x) \quad \text{and} \quad f = \sum_{n=0}^{\infty} f_n$$

and say that the infinite sum $\sum_{n=0}^{\infty} f_n$ **converges uniformly to f** .

Proposition 21.2. Let $f_n : I \rightarrow \mathbb{R}$ be a sequence of continuous functions. Suppose that the sum

$$\sum_{n=0}^{\infty} f_n(x)$$

converges uniformly. Then $\sum_{n=0}^{\infty} f_n$ is a continuous function.

Proof. Since each f_n is continuous, Theorem 13.5 implies that each partial sum function

$$S_N(x) := \sum_{n=0}^N f_n(x)$$

is continuous. Since $\sum_{n=0}^{\infty} f_n(x)$ converges uniformly, this means that $S_N \rightarrow \sum_{n=0}^{\infty} f_n$ uniformly. By Theorem 18.1, it follows that f is continuous. $\square$

This fact is especially useful for proving continuity of functions that are *defined* as an infinite sum, which we will soon do. To make use of this fact, it will be convenient to have a simpler criterion for determining when a series of functions converges uniformly. This is given by the following theorem, which goes by the name of the M -test.

Theorem 21.3 (Weierstrauss M -test). Let $f_n : I \rightarrow \mathbb{R}$ be a sequence of functions, and let $(M_n)_{n \in \mathbb{N}} \subseteq \mathbb{R}$ be a sequence of real numbers such that, for all $n \in \mathbb{N}$ and all $x \in I$, we have $|f_n(x)| \leq M_n$.

If $\sum_{n=0}^{\infty} M_n$ converges, then $\sum_{n=0}^{\infty} f_n$ converges uniformly.

Proof. Let $S_N := \sum_{n=0}^N f_n$ denote the partial sum functions. We need to prove that S_N converges uniformly. To do this, we will prove that S_N is uniformly Cauchy.

Fix $\varepsilon > 0$. Since $\sum_{n=0}^{\infty} M_n$ converges, it satisfies the Cauchy criterion for series. Thus, there is an $N \in \mathbb{N}$ such that, for $N_0 > M_0 \geq N$ we have

$$\left| \sum_{n=M_0+1}^{N_0} M_n \right| < \varepsilon.$$

Let $N_0 > M_0 \geq N$. Let $x \in I$. Note that

$$\begin{aligned} |S_{N_0}(x) - S_{M_0}(x)| &= \left| \sum_{n=M_0+1}^{N_0} f_n(x) \right| \\ &\leq \sum_{n=M_0+1}^{N_0} |f_n(x)| && \text{by the triangle inequality} \\ &\leq \sum_{n=M_0+1}^{N_0} M_n && \text{since } |f_n(x)| \leq M_n \text{ for all } n \in \mathbb{N}, x \in I \\ &= \left| \sum_{n=M_0+1}^{N_0} M_n \right| && \text{since each } M_n \geq 0 \\ &< \varepsilon. \end{aligned}$$

Thus, for all $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that, for all $N_0 > M_0 \geq N$ and $x \in I$ we have $|S_{N_0}(x) - S_{M_0}(x)| < \varepsilon$. This means that the sequence of functions S_N is uniformly Cauchy, and hence $S_N \rightarrow \sum_{n=0}^{\infty} f_n$ uniformly, as desired. $\square$

Example 21.4. Consider the infinite series of functions on $[-1, 1]$ given by

$$\sum_{n=1}^{\infty} \frac{1}{n^n} x^n.$$

We claim that this series converges uniformly. Indeed, note that, for all $x \in [-1, 1]$ and all $n \geq 2$ we have

$$0 \leq \left| \frac{1}{n^n} x^n \right| = \frac{1}{n^n} |x|^n \leq \frac{1}{n^n} \leq \frac{1}{n^2}.$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges by the p -test, Theorem 21.3 implies that $\sum_{n=1}^{\infty} \frac{1}{n^n} x^n$ converges uniformly on $[-1, 1]$. Note that, for each $n \in \mathbb{N}$, the function $x \mapsto \frac{1}{n^n} x^n$ is continuous. Since the sum converges uniformly, it follows that the function

$$\sum_{n=1}^{\infty} \frac{1}{n^n} x^n$$

is continuous.

Remark 21.5. It is worth remarking on how cool this example is. The function defined by $\sum_{n=1}^{\infty} \frac{1}{n^n} x^n$ is not something that we can otherwise describe. It doesn't have a previously known name; it isn't a polynomial or rational function. It is a perfectly good function, but it is *defined* as the limit of some sequence of functions, and this is the only way that we can describe it. In other words, we have birthed into existence a new function that we *cannot* describe with conventional or previously used formulas! It's a brand new thing. And yet, using the theory we have developed, we can prove that this brand new function is continuous, among other things.

22. LECTURE 22: POWER SERIES, AND A CAST OF SPECIAL FUNCTIONS

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 4, Section 23
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 3, “Power Series”
- ◇ Abbott, *Understanding Analysis*: Chapter 6, Section 5

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Understand when a power series converges (absolutely) and when a power series converges uniformly as a series of functions.
- ◇ Be able to compute the radius of convergence for simple power series.

The example from the previous lecture of a series of functions took a very special form. This special type of infinite sum of functions will be the star of this lecture.

Definition 22.1. A **power series** is a sum of functions of the form

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n$$

where (a_n) is some sequence of numbers. The **center** of the power series x_0 .

Remark 22.2. As usual, we allow a different starting index other than $n = 0$.

Remark 22.3. Informally, you can think about a power series as being a polynomial of potentially infinite degree. For example, expanding on the formal infinite sum of a typical power series with center $x_0 = 0$ gives

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

Remark 22.4. Note that a power series is a series of functions as discussed in the previous lecture. Indeed, a power series is of the form

$$\sum_{n=0}^{\infty} f_n(x)$$

where $f_n : I \rightarrow \mathbb{R}$ is defined by $f_n(x) = a_n(x - x_0)^n$. Note that in the definition of a power series, we have not specified the domain of the functions.

Example 22.5. Consider the series

$$\sum_{n=0}^{\infty} x^n.$$

This is a power series with $a_n = 1$ and center $x_0 = 0$.

Example 22.6. Consider the series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (x-1)^{2n}.$$

This is a power series, but we have to be a little careful in how we describe it because of the fact that the nonzero powers of x exist only in even indices. In other words, this series is represented by the formal sum

$$\frac{(-1)^0}{0!} (x-1)^0 + 0x^1 + \frac{(-1)^1}{2!} (x-1)^2 + 0x^3 + \frac{(-1)^2}{4!} (x-1)^4 + \cdots.$$

It may clarify to use a different indexing. We can rewrite the original series as a power series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (x-1)^{2n} = \sum_{k=0}^{\infty} a_k (x-1)^k$$

with center $x_0 = 1$ and coefficient sequence

$$a_k = \begin{cases} 0 & \text{if } k \text{ is odd} \\ \frac{(-1)^{\frac{k}{2}}}{k!} & \text{if } k \text{ is even} \end{cases}.$$

Example 22.7. The series

$$\sum_{n=0}^{\infty} n^2 x^{\frac{n}{2}}$$

is *not* a power series, even after reindexing of any kind. This is because the powers of x are not (always) natural numbers. By definition, a power series must involve only powers of x that are natural numbers. The above series is still a perfectly good series of functions, it just isn't what we would call a *power series*.

Example 22.8. Consider the power series

$$\sum_{n=0}^{\infty} x^n.$$

This is a power series, but we also recognize it as a geometric series with ratio x . We know that such a series converges if and only if $|x| < 1$. Thus, if we want to treat this power series as a series of functions, we would have to specify the data of each function as $f_n : (-1, 1) \rightarrow \mathbb{R}$ given by $f_n(x) = x^n$, or more generally any domain I where $I \subseteq (-1, 1)$.

The previous example demonstrates that, given a power series, it may or may not converge for different values of x . This is to say that if we take a power series to be a series of functions, we may get different statements about convergence depending on the domains of the functions. Fortunately, there is a well-understood way to identify the range of x -values for which a power series converges. This is given by the next theorem, which is the main result of this lecture.

Theorem 22.9. *Let*

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n$$

be a power series. There is an element $R \in [0, \infty) \cup \{\infty\}$ satisfying the following properties.

(i) *If $|x - x_0| < R$, then the series*

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n$$

converges absolutely. If $R = \infty$, then the series converges for all $x \in \mathbb{R}$.

(ii) *If $|x - x_0| > R$, then the series*

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n$$

diverges.

(iii) *Suppose $R \neq 0$ and let $0 \leq R_0 < R$, where if $R = \infty$ this just means $R_0 \in [0, \infty)$. Then the series of functions*

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n$$

converges uniformly on $[x_0 - R_0, x_0 + R_0]$.

Remark 22.10. In words, this theorem gives an interval

$$(x_0 - R, x_0 + R)$$

on which the power series converges (absolutely) and such that if $x > x_0 + R$ or $x < x_0 - R$ then the series diverges. Note that the theorem does not address what happens at the boundary points of this interval, i.e., at $x = x_0 \pm R$. In general, in these notes we will not study what happens at these boundary points (though there are many interesting things one can say and prove — for example, there is a well-known result called *Abel's theorem* that you could look up if you're curious).

Proof. Let

$$R = \frac{1}{\limsup |a_n|^{\frac{1}{n}}}$$

where we take $R = \infty$ if $\limsup |a_n|^{\frac{1}{n}} = 0$ and we take $R = 0$ if the sequence

$$U_N = \sup \left\{ |a_n|^{\frac{1}{n}} \mid n \geq N \right\}$$

diverges to ∞ .

(i) First suppose that $R \in (0, \infty)$. Suppose that $|x - x_0| < R$. We will prove that the series

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n$$

converges absolutely using the root test. Note that

$$|a_n(x - x_0)^n|^{\frac{1}{n}} = |a_n|^{\frac{1}{n}} |x - x_0|.$$

This means that

$$\begin{aligned} \limsup |a_n(x - x_0)^n|^{\frac{1}{n}} &\leq |x - x_0| \limsup |a_n|^{\frac{1}{n}} \\ &< 1. \end{aligned}$$

Thus, the root test implies that the series converges absolutely. If $R = \infty$, then $\limsup |a_n|^{\frac{1}{n}} = 0$ and so

$$\begin{aligned} \limsup |a_n(x - x_0)^n|^{\frac{1}{n}} &\leq |x - x_0| \limsup |a_n|^{\frac{1}{n}} \\ &= 0 \\ &< 1 \end{aligned}$$

and the root test again gives absolute convergence.

- (ii) Similarly, the root test gives divergence. This part is left as an exercise.
- (iii) Let $0 \leq R_0 < R_0$. To prove that the power series converges uniformly on $[x_0 - R_0, x_0 + R_0]$, we will use the Weierstrauss M -test. Let $M_n = |a_n|R_0^n$. Note that, for all $x \in [x_0 - R_0, x_0 + R_0]$, we have $|x - x_0| \leq R_0$ and so

$$|a_n(x - x_0)^n| \leq M_n.$$

Thus, by Theorem 21.3, to conclude that the power series converges uniformly on $[x_0 - R_0, x_0 + R_0]$ it suffices to prove that the series $\sum_{n=0}^{\infty} M_n$ converges.

Let $x = x_0 + R_0$. Note that

$$|x - x_0| = |R_0| = R_0 < R.$$

By (i), the series

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n$$

converges absolutely, which is to say that the series

$$\sum_{n=0}^{\infty} |a_n| |x - x_0|^n = \sum_{n=0}^{\infty} |a_n| R_0^n = \sum_{n=0}^{\infty} M_n$$

converges, as desired.

□

The number R in the above theorem has a name that is (secretly) familiar to you from calculus.

Definition 22.11. Let

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n$$

be a power series. Let R be as in Theorem 22.9. We call R the **radius of convergence** of the power series.

The proof of Theorem 22.9 gave one way to compute the radius of convergence, essentially using the data of the root test. For some power series it is also possible to compute the radius of convergence using the data of the ratio test. This is summarized by the following theorem.

Theorem 22.12. *Let*

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n$$

be a power series.

(i) *The radius of convergence of the power series is given by*

$$R = \frac{1}{\limsup |a_n|^{\frac{1}{n}}}$$

where we take $R = \infty$ if $\limsup |a_n|^{\frac{1}{n}} = 0$ and we take $R = 0$ if the sequence

$$U_N = \sup \left\{ |a_n|^{\frac{1}{n}} \mid n \geq N \right\}$$

diverges to ∞ .

(ii) *Assume that $a_n \neq 0$ for all $n \in \mathbb{N}$, and that the sequence $\left| \frac{a_n}{a_{n+1}} \right|$ converges or diverges to ∞ . The radius of convergence of the power series is given by*

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

where we take $R = \infty$ in the case the sequence diverges to ∞ .

Proof. The proof of (i) is given by the proof of Theorem 22.9, so it suffices to prove (ii). We leave this as an exercise, which is immediate from applying the ratio test. $\square$

Corollary 22.13. *Let*

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n$$

be a power series with radius of convergence R . Then the function $P : (x_0 - R, x_0 + R) \rightarrow \mathbb{R}$ defined by

$$P(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^n$$

is continuous. Here we understand $(x_0 - R, x_0 + R) = \mathbb{R}$ in the case $R = \infty$.

Proof. Fix $\tilde{x} \in (x_0 - R, x_0 + R)$. We want to prove that P is continuous at $\tilde{x}$. Let $R_0 := |\tilde{x} - x_0|$ and note that $R_0 < R$ and $\tilde{x} \in [x_0 - R_0, x_0 + R_0]$. By Theorem 22.9, the series of functions

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n$$

converges uniformly on $[x_0 - R_0, x_0 + R_0]$. Note that each function

$$x \mapsto a_n(x - x_0)^n$$

is a polynomial and hence continuous on $[x_0 - R_0, x_0 + R_0]$. Thus, by Proposition 21.2, it follows that $P(x)$ is continuous on $[x_0 - R_0, x_0 + R_0]$ and hence continuous at $\tilde{x}$. Since $\tilde{x} \in (x_0 - R, x_0 + R)$ was arbitrary, it follows that $P(x)$ is continuous on $(x_0 - R, x_0 + R)$. $\square$

22.1. Introducing some special functions. Having discussed series of functions and in particular power series, we can now introduce some special functions: the exponential function, the sine function, and the cosine function. We will define these from a purely analytic point of view. First, a theorem.

Theorem 22.14. *Each of the three power series*

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{1}{n!} x^n \\ \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \\ \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \end{aligned}$$

has radius of convergence $R = \infty$.

Proof. Note that the first power series is a power series with coefficient sequence $a_n = \frac{1}{n!}$. We have

$$\begin{aligned} \left| \frac{a_n}{a_{n+1}} \right| &= \left| \frac{1}{n!} \cdot (n+1)! \right| \\ &= n+1. \end{aligned}$$

This sequence diverges to ∞ , and so $R = \infty$, as desired.

Strictly speaking, we cannot compute the radius of convergence of the second power series using the formula

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

because the coefficient sequence is 0 for all even indices. In particular, we have

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = \sum_{k=0}^{\infty} b_k x^k$$

where

$$b_k = \begin{cases} \frac{(-1)^{\frac{k-1}{2}}}{k!} & \text{if } k \text{ is odd} \\ 0 & \text{if } k \text{ is even} \end{cases}.$$

However, the erroneous ratio test argument still does give the correct behavior, so we leave it as an exercise to verify this. $\square$

With this theorem, we can then make the following definition.

Definition 22.15. Define three functions $\exp, \sin, \cos : \mathbb{R} \rightarrow \mathbb{R}$ by

$$\begin{aligned} \exp(x) &:= \sum_{n=0}^{\infty} \frac{1}{n!} x^n \\ \sin(x) &:= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \\ \cos(x) &:= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \end{aligned}$$

Remark 22.16. We also use the shorthand notation $e^x := \exp(x)$, $\sin x = \sin(x)$, and $\cos x = \cos(x)$.

The following corollary is an immediate consequence of the theorems from this lecture.

Corollary 22.17. *The functions $\exp, \sin, \cos : \mathbb{R} \rightarrow \mathbb{R}$ are continuous.*

We can also define the number e , finally.

Definition 22.18. Define the real number $e \in \mathbb{R}$ as follows:

$$e := \exp(1) = \sum_{n=0}^{\infty} \frac{1}{n!}.$$

Remark 22.19. We now have rigorous definitions of functions like the exponential function and some basic trigonometric functions, but these definitions tell us almost nothing about these functions besides the fact that they are continuous. In theory, it is possible to conclude many of the familiar properties of these functions from the power series definitions. The exercises will give some indications of how to do this. Other familiar properties will be unattainable until we've discussed the derivative.

Part 5

Differentiation

23. LECTURE 23: LIMITS OF FUNCTIONS

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 3, Section 20
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 4, “Limits of Functions”
- ◇ Abbott, *Understanding Analysis*: Chapter 4, Section 2

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Understand the definition of the limit of a function, both in terms of ε s and δ s and also sequences.
- ◇ Be able to compute simple functional limits.

You’ll notice that at no point in these notes have I ever written something of the form

$$\lim_{x \rightarrow x_0} f(x).$$

Although this a familiar concept from calculus, we have yet to discuss limits of functions; we have only discussed limits of sequences. In order to define the derivative, which is the goal of the final part of this course, we will need to discuss limits of functions.

The following definition is one which is important for technical reasons. In particular, if a function is only defined at a single point x_0 , it doesn’t make sense to talk about the limit of that function as you approach that point.

Definition 23.1. Let $E \subset \mathbb{R}$ be a set. A **limit point of E** is a point $x_0 \in \mathbb{R}$ (not necessarily in E) such that there is a sequence $(a_n) \subseteq E$ with $a_n \rightarrow x_0$ and $a_n \neq x_0$ for all $n \in \mathbb{N}$.

Example 23.2. Consider the set

$$E = [0, 1).$$

Every point of E is a limit point of E , but also 1 is a limit point of E .

Example 23.3. Let $E = \mathbb{R} \setminus \{x_0\}$. Then x_0 is a limit point of E .

Definition 23.4. Let $f : E \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be a function, and let $x_0 \in \mathbb{R}$ be a limit point of E . We say that

$$\lim_{x \rightarrow x_0} f(x) = L,$$

said **the limit of f as x approaches x_0 is L** if the following condition holds.

For every $\varepsilon > 0$, there is a $\delta > 0$ such that, for all $x \in E$ with $0 < |x - x_0| < \delta$, we have $|f(x) - L| < \varepsilon$.

Remark 23.5. Take note of some subtleties in the above definition. The point x_0 must be a limit point of E , which means it could be in E but could also not be in E . Secondly, unlike all of the $\varepsilon - \delta$ definitions we have seen before, we only consider points $x \in E$ with a distance *strictly* larger than 0. This is picked up by the inequality $0 < |x - x_0| < \delta$.

Theorem 23.6. Let $f : E \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be a function, and let $x_0 \in \mathbb{R}$ be a limit point of E . Then

$$\lim_{x \rightarrow x_0} f(x) = L$$

if and only if for all sequences $(a_n) \subseteq E$ such that $a_n \rightarrow x_0$ and $a_n \neq x_0$, we have $f(a_n) \rightarrow L$.

Proof. First, assume that $\lim_{x \rightarrow x_0} f(x) = L$. Let $(a_n) \subseteq E$ be a sequence such that $a_n \rightarrow x_0$ and $a_n \neq x_0$. Fix $\varepsilon > 0$. Since $\lim_{x \rightarrow x_0} f(x) = L$, there is a $\delta > 0$ such that, for all $x \in E$ with $0 < |x - x_0| < \delta$, we have $|f(x) - L| < \varepsilon$. Since $a_n \rightarrow x_0$, there is an $N \in \mathbb{N}$ such that, for all $n \geq N$, we have $|a_n - x_0| < \delta$. Thus, for all $n \geq N$, we have $|f(a_n) - L| < \varepsilon$. This shows that $f(a_n) \rightarrow L$.

For the other direction of the if and only if statement, we will prove the contrapositive. That is, suppose that $\lim_{x \rightarrow x_0} f(x) \neq L$. We need to produce a sequence $(a_n) \subseteq E \setminus \{x_0\}$ such that $a_n \rightarrow x_0$ but that $f(a_n)$ does not converge to L . Since $\lim_{x \rightarrow x_0} f(x) \neq L$, there is an $\varepsilon_0 > 0$ such that, for every $n \in \mathbb{N}$, there is an $a_n \in E \setminus \{x_0\}$ with $0 < |a_n - x_0| < \frac{1}{n}$ and $|f(a_n) - L| \geq \varepsilon_0$. This follows from directly negating the definition of a functional limit. By construction, $f(a_n)$ does not converge to L . □

An immediate consequence of the above theorem is a host of unsurprising limit laws.

Theorem 23.7 (Algebraic limit laws). Let $f, g : E \rightarrow \mathbb{R}$ be functions, and let x_0 be a limit point of E . Suppose that $\lim_{x \rightarrow x_0} f(x) = L$ and $\lim_{x \rightarrow x_0} g(x) = K$. Then

(i) for any $\lambda \in \mathbb{R}$,

$$\lim_{x \rightarrow x_0} (\lambda f)(x) = \lambda L,$$

(ii)

$$\lim_{x \rightarrow x_0} (f + g)(x) = L + K,$$

(ii)

$$\lim_{x \rightarrow x_0} (fg)(x) = LK,$$

(ii) and if $K \neq 0$, then

$$\lim_{x \rightarrow x_0} (f/g)(x) = \frac{L}{K}.$$

Proof. Apply the previous theorem, together with the sequential algebraic limit laws described in Theorem 6.6. $\square$

The following result is one of the main theorems involving functional limits.

Theorem 23.8. *Let $f : E \rightarrow \mathbb{R}$ be a function, and let $x_0 \in \mathbb{R} \setminus E$ be a limit point of E which is not in E . Then*

$$\lim_{x \rightarrow x_0} f(x) = L$$

if and only if there is a function $\tilde{f} : E \cup \{x_0\} \rightarrow \mathbb{R}$ which is continuous at x_0 such that $\tilde{f}|_E = f$ and $\tilde{f}(x_0) = L$.

The more down to earth version corollary of this theorem is a characterization of continuity at a point using functional limits.

Corollary 23.9. *Let $f : E \rightarrow \mathbb{R}$ be a function, and let $x_0 \in E$ be a limit point of E . Then f is continuous at x_0 if and only if*

$$\lim_{x \rightarrow x_0} f(x) = f(x_0).$$

Proof of Theorem 23.8. First, suppose that $\lim_{x \rightarrow x_0} f(x) = L$. Define $\tilde{f} : E \cup \{x_0\} \rightarrow \mathbb{R}$ by

$$\tilde{f}(x) = \begin{cases} f(x) & x \in E \\ L & x = x_0 \end{cases}.$$

By definition we have $\tilde{f}|_E = f$ and $\tilde{f}(x_0) = L$, so we simply need to prove that $\tilde{f}$ is continuous at x_0 . Let $(a_n) \subseteq E \cup \{x_0\}$ be any sequence such that $a_n \rightarrow x_0$. Since $\lim_{x \rightarrow x_0} f(x) = L$ we have $f(a_n) \rightarrow L = \tilde{f}(x_0)$. By the sequential characterization of continuity, it follows that $\tilde{f}$ is continuous at x_0 .

For the other direction of the if and only if, suppose that $\tilde{f} : E \cup \{x_0\} \rightarrow \mathbb{R}$ is a function which is continuous at x_0 such that $\tilde{f}|_E = f$ and $\tilde{f}(x_0) = L$. We need to prove that $\lim_{x \rightarrow x_0} f(x) = L$. Let $(a_n) \subseteq E$ be any sequence such that $a_n \rightarrow x_0$. Since $\tilde{f}$ is continuous at x_0 , we have $\tilde{f}(a_n) \rightarrow \tilde{f}(x_0) = L$, and this implies $\lim_{x \rightarrow x_0} f(x) = L$, as desired. $\square$

Proof of Corollary 23.9. Consider the function $g : E \setminus \{x_0\} \rightarrow \mathbb{R}$ defined by $g(x) = f(x)$. Define $\tilde{g} : E \rightarrow \mathbb{R}$ by $\tilde{g}(x) = f(x)$. Theorem 23.8 implies that

$$\lim_{x \rightarrow x_0} g(x) = f(x_0)$$

if and only if $\tilde{g}$ is continuous at x_0 . This gives the corollary. $\square$

Example 23.10. Consider the function $f : \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$ defined by

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

We are interested in $\lim_{x \rightarrow 1} f(x)$, if it exists. Note that f is not defined at 1, and so it is not continuous at 1. However, we have

$$f(x) = \frac{(x-1)(x+1)}{x-1} = x+1.$$

It is still the case that f is not defined at 1. This is because the domain of f is $\mathbb{R} \setminus \{1\}$. However, with this algebraic manipulation we can define $\tilde{f} : \mathbb{R} \rightarrow \mathbb{R}$ by

$$\tilde{f}(x) = \begin{cases} x+1 & x \neq 1 \\ 2 & x = 1 \end{cases}.$$

Then $\tilde{f}$ agrees with f on the domain of the latter, and $\tilde{f}$ is continuous. Thus,

$$\lim_{x \rightarrow 1} f(x) = \tilde{f}(1) = 2.$$

Example 23.11. Here is a simple example that demonstrates the necessity of some of the technical details in discussing limits of functions (for example, why we state everything in terms of sequences $a_n \rightarrow x_0$ that satisfy $a_n \neq x_0$). Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases}.$$

Note that for any sequence $(a_n) \subseteq \mathbb{R}$ with $a_n \neq 0$, we have $f(a_n) = 0 \rightarrow 0$ and so $\lim_{x \rightarrow 0} f(x) = 0$. However, it is *not* the case that for *all* sequences $(a_n) \subseteq \mathbb{R}$ with $a_n \rightarrow 0$ we have $f(a_n) \rightarrow 0$. Indeed, the sequence $a_n = 0$ gives $f(a_n) = 1 \rightarrow 1 \neq 0$. Thus, we must strictly consider sequences that never equal the limit point.

24. LECTURE 24: THE DERIVATIVE

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 5, Section 28
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 5, “The Derivative of Real Valued Functions”
- ◇ Abbott, *Understanding Analysis*: Chapter 5, Section 2

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Understand the limit definition of differentiability and be able to verify that simple functions are differentiable.
- ◇ Be able to prove basic properties of differentiable functions from the definition (differentiation rules, implied continuity, etc.).

The concept of the derivative needs no introduction. We all know of its importance and utility in all aspects of math, physics, engineering, and so on. We also already know all of the intuition that guides the familiar definition from calculus (approximating the instantaneous slope of a function at a point with secant lines). Instead of spending time describing and motivating this intuition, we will dive into the definition.

Definition 24.1. Let $f : I \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be a function defined on an (open, compact, or neither) interval I . Let $x_0 \in I$.

- ◇ We say that f is **differentiable at x_0** if the following limit exists:

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

where on the right hand side we view $x \mapsto \frac{f(x) - f(x_0)}{x - x_0}$ as a function on $I \setminus \{x_0\}$.

- ◇ If f is differentiable at x_0 , then we use the notation

$$f'(x_0) := \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

and call the number $f'(x_0)$ the **derivative of f at x_0** .

- ◇ Let $J \subseteq I$ denote the set points of points $x_0 \in I$ such that f is differentiable at x_0 . **The derivative of f** is the function $f' : J \rightarrow \mathbb{R}$ by^a

$$f'(x) := f'(x).$$

^aAdmittedly, the notation used in this definition is kind of silly and tautological. On the left is the name of the function f' applied to an input $x \in J$, and on the right is the single collection of symbols “ $f'(x)$ ” as defined in the previous bullet point.

Example 24.2. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$. We claim that f is differentiable at every point in $\mathbb{R}$. Indeed, fix $x_0 \in \mathbb{R}$. Note that, for $x \neq x_0$, we have

$$\begin{aligned} \frac{f(x) - f(x_0)}{x - x_0} &= \frac{x^2 - x_0^2}{x - x_0} \\ &= \frac{(x - x_0)(x + x_0)}{x - x_0} \\ &= x + x_0. \end{aligned}$$

Since the function $x \mapsto x + x_0$ is a polynomial, it is continuous, and hence $\lim_{x \rightarrow x_0} x + x_0 = x_0 + x_0 = 2x_0$. Thus,

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} x + x_0 = 2x_0.$$

This shows that f is differentiable at x_0 and that $f'(x_0) = 2x_0$.

The following theorem is our first exposure to the strength of the property of differentiability.

Theorem 24.3. Let $f : I \rightarrow \mathbb{R}$ be a function, and let $x_0 \in I$. If f is differentiable at x_0 , then f is continuous at x_0 .

Proof. By Theorem 23.9, to prove that f is continuous at x_0 it suffices to prove that $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. Note that, for all $x \neq x_0$,

$$f(x) - f(x_0) = \frac{f(x) - f(x_0)}{x - x_0} \cdot (x - x_0).$$

By limit laws and the assumption that f is differentiable at x_0 , we have

$$\begin{aligned} \lim_{x \rightarrow x_0} (f(x) - f(x_0)) &= \left(\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \right) \cdot \left(\lim_{x \rightarrow x_0} (x - x_0) \right) \\ &= f'(x_0) \cdot (x_0 - x_0) \\ &= 0. \end{aligned}$$

The fact that $\lim_{x \rightarrow x_0} (f(x) - f(x_0)) = 0$ implies $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. Thus, f is continuous at x_0 . $\square$

Of course, it is possible to be continuous but not differentiable at a point.

Example 24.4. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = |x|$. Recall that f is continuous. We claim that f is not differentiable at 0. To prove this, we must show that the limit

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{|x|}{x}$$

does not exist. Consider the sequences $a_n = \frac{1}{n}$ and $b_n = -\frac{1}{n}$. Note that $a_n, b_n \rightarrow 0$ and $a_n, b_n \neq 0$. But

$$\lim_{n \rightarrow \infty} \frac{|a_n|}{a_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{\frac{1}{n}} = 1$$

and

$$\lim_{n \rightarrow \infty} \frac{|b_n|}{b_n} = \lim_{n \rightarrow \infty} \frac{|-\frac{1}{n}|}{-\frac{1}{n}} = -1.$$

This implies that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist, and hence f is not differentiable at 0.

Theorem 24.5 (Basic derivative rules). *Let $f, g : I \rightarrow \mathbb{R}$ be functions that are differentiable at $x_0 \in I$.*

(i) *For all $\lambda \in \mathbb{R}$, $\lambda f : I \rightarrow \mathbb{R}$ is differentiable at x_0 and*

$$(\lambda f)'(x_0) = \lambda f'(x_0).$$

(ii) *The function $f + g : I \rightarrow \mathbb{R}$ is differentiable at x_0 and*

$$(f + g)'(x_0) = f'(x_0) + g'(x_0).$$

(iii) *The function $fg : I \rightarrow \mathbb{R}$ is differentiable at x_0 and*

$$(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0).$$

(iv) *If $g(x_0) \neq 0$, then the function $\frac{f}{g} : I \rightarrow \mathbb{R}$ is differentiable at x_0 and*

$$\left(\frac{f}{g}\right)'(x_0) = \frac{g(x_0)f'(x_0) - f(x_0)g'(x_0)}{[g(x_0)]^2}.$$

Proof. We will prove (iii), and leave the rest as exercises. **Note that**

$$\begin{aligned} \frac{(fg)(x) - (fg)(x_0)}{x - x_0} &= \frac{f(x)g(x) - f(x_0)g(x) + f(x_0)g(x) - f(x_0)g(x_0)}{x - x_0} \\ &= \frac{f(x) - f(x_0)}{x - x_0} \cdot g(x) + f(x_0) \cdot \frac{g(x) - g(x_0)}{x - x_0}. \end{aligned}$$

By assumption

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \quad \text{and} \quad \lim_{x \rightarrow x_0} \frac{g(x) - g(x_0)}{x - x_0}$$

exist and equal $f'(x_0)$ and $g'(x_0)$, respectively. Furthermore, since f and g are differentiable at x_0 , they are continuous at x_0 , and hence $\lim_{x \rightarrow x_0} f(x) = f(x_0)$ and $\lim_{x \rightarrow x_0} g(x) = g(x_0)$. Thus, limit laws imply

$$\begin{aligned} \lim_{x \rightarrow x_0} \frac{(fg)(x) - (fg)(x_0)}{x - x_0} &= \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \cdot g(x) + \lim_{x \rightarrow x_0} f(x_0) \cdot \frac{g(x) - g(x_0)}{x - x_0} \\ &= f'(x_0)g(x_0) + f(x_0)g'(x_0). \end{aligned}$$

Thus, fg is differentiable at x_0 and

$$(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0).$$

□

One of the most important results about differentiability is the chain rule. This concerns differentiability of composition of functions.

Theorem 24.6 (Chain rule). *Let $I, J \subseteq \mathbb{R}$ be intervals. Let $f : I \rightarrow J$ be differentiable at $x_0 \in I$ and let $g : J \rightarrow \mathbb{R}$ be differentiable at $f(x_0) \in J$. Then the composition $g \circ f : I \rightarrow \mathbb{R}$ is differentiable at $x_0 \in I$ and*

$$(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0).$$

The proof of the chain rule is a little bit involved for technical reasons, but the basic idea is straightforward. Note that

$$\frac{(g \circ f)(x) - (g \circ f)(x_0)}{x - x_0} = \frac{g(f(x)) - g(f(x_0))}{f(x) - f(x_0)} \cdot \frac{f(x) - f(x_0)}{x - x_0}.$$

Since f is differentiable at x_0 , it is continuous, and so as $x \rightarrow x_0$ we have $f(x) \rightarrow f(x_0)$. Since g is differentiable at $f(x_0)$, this means that

$$\lim_{y \rightarrow f(x_0)} \frac{g(y) - g(f(x_0))}{y - f(x_0)} = g'(f(x_0)).$$

Since $f(x) \rightarrow f(x_0)$, this basically means (by setting $y = f(x)$) that

$$\frac{g(f(x)) - g(f(x_0))}{f(x) - f(x_0)} \rightarrow g'(f(x_0)).$$

Thus,

$$\frac{g(f(x)) - g(f(x_0))}{f(x) - f(x_0)} \cdot \frac{f(x) - f(x_0)}{x - x_0} \rightarrow g'(f(x_0)) \cdot f'(x_0).$$

The reason why this doesn't give a formal proof is because the quantity $f(x) - f(x_0)$ could be 0 for values of x arbitrarily close to x_0 . This means that the quotient $\frac{g(f(x)) - g(f(x_0))}{f(x) - f(x_0)}$ may not be defined on an interval containing $f(x_0)$.

Proof of Theorem 24.6. Define a function $Q : J \rightarrow \mathbb{R}$ for $y \in J$ by

$$Q(y) := \begin{cases} \frac{g(y) - g(f(x_0))}{y - f(x_0)} & \text{if } y \neq f(x_0) \\ g'(f(x_0)) & \text{if } y = f(x_0) \end{cases}.$$

Since g is differentiable at $f(x_0)$, we have

$$\lim_{y \rightarrow f(x_0)} Q(y) = \lim_{y \rightarrow f(x_0)} \frac{g(y) - g(f(x_0))}{y - f(x_0)} = g'(f(x_0)) = Q(f(x_0)).$$

Thus, Q is continuous at $f(x_0) \in J$.

Next, note that for all $y \in J$ we have

$$g(y) - g(f(x_0)) = Q(y)(y - f(x_0)).$$

In particular, for any $x \in I$ we have $f(x) \in J$ and so

$$g(f(x)) - g(f(x_0)) = Q(f(x))(f(x) - f(x_0)).$$

More specifically, for any $x \in I \setminus \{x_0\}$ we have

$$\frac{g(f(x)) - g(f(x_0))}{x - x_0} = Q(f(x)) \frac{f(x) - f(x_0)}{x - x_0}.$$

Since f is differentiable at x_0 , it is continuous at x_0 , and so $f(x) \rightarrow f(x_0)$ as $x \rightarrow x_0$. Continuity of Q at $f(x_0)$ then implies $\lim_{x \rightarrow x_0} Q(f(x)) = Q(f(x_0)) = g'(f(x_0))$. Since f is differentiable at x_0 , we have $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$. Thus, by limit laws we have

$$\begin{aligned} \lim_{x \rightarrow x_0} \frac{g(f(x)) - g(f(x_0))}{x - x_0} &= \lim_{x \rightarrow x_0} Q(f(x)) \frac{f(x) - f(x_0)}{x - x_0} \\ &= g'(f(x_0)) \cdot f'(x_0). \end{aligned}$$

This implies that $g \circ f$ is differentiable at x_0 and that

$$(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0).$$

□

In case we need to differentiate polynomials, the following theorem gives the well-known *power rule*.

Theorem 24.7 (Power rule). *Let $n \in \mathbb{N}$ with $n \geq 1$. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^n$ is differentiable, and $f'(x) = nx^{n-1}$.*

Remark 24.8. It is possible to upgrade the statement of this theorem to non-integral powers, but in order to do this we need to study exponential functions in more depth. For now, we will stick with polynomials.

Proof. We proceed by strong induction on n .

Base Case: $n = 0$.

When $n = 1$ we have $f(x) = x^1 = x$. Note that, for any $x \neq x_0 \in \mathbb{R}$,

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{x - x_0}{x - x_0} = 1.$$

Thus,

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = 1 = 1x^0.$$

Inductive step.

Suppose that for some $k \in \mathbb{N}$, we know that $x \mapsto x^n$ is differentiable for all $1 \leq n \leq k$ with derivative $x \mapsto nx^{n-1}$. Consider the function $f(x) = x^{k+1}$. Since $f(x) = x \cdot x^k$, and $x \mapsto x$ and $x \mapsto x^k$ are differentiable, Theorem 24.5 implies that f is differentiable and

$$f'(x) = 1 \cdot x^k + x \cdot kx^{k-1} = (k+1)x^k.$$

This completes the proof. □

Corollary 24.9. *All polynomials are differentiable.*

25. LECTURE 25: THE MEAN VALUE THEOREM

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 5, Section 29
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 5, “Mean Value Theorems”
- ◇ Abbott, *Understanding Analysis*: Chapter 5, Section 3

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Understand the statement, intuition, and proof of the mean value theorem, as well as its immediate consequences.

The goal of this lecture is to prove the *mean value theorem*, which is one of the most important theorems about differentiable functions.

Theorem 25.1 (Mean value theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be a function which is*

- ◇ *continuous on $[a, b]$, and*
- ◇ *differentiable on (a, b) .*

Then there is an $x_0 \in (a, b)$ such that

$$f'(x_0) = \frac{f(b) - f(a)}{b - a}.$$

Remark 25.2. The reason why the theorem is stated the way it is — by requiring continuity on the compact interval $[a, b]$, but differentiability only on the open interval (a, b) — is to all the possibility for functions like $f : [0, 1]$ defined by $f(x) = x^{\frac{1}{2}}$. It turns out that f is not differentiable at 0, but the mean value theorem still holds.

In order to prove the mean value theorem, we must first prove a different theorem which is also one of the more important and motivated facts about the derivative. In words, it says that local minimums and maximums of differentiable functions occur at *critical points*, i.e., points where the derivative is 0.

Theorem 25.3. *Suppose that f is differentiable on (a, b) . Let $x_0 \in (a, b)$.*

- (i) *Suppose that, for all $x \in (a, b)$, we have $f(x) \leq f(x_0)$. Then $f'(x_0) = 0$.*
- (ii) *Suppose that, for all $x \in (a, b)$, we have $f(x) \geq f(x_0)$. Then $f'(x_0) = 0$.*

Proof. We prove (ii); (i) is identical. That is, suppose that $f(x) \geq f(x_0)$ for all $x \in (a, b)$. Consider the sequences $a_n = x_0 + \frac{1}{n}$ and $b_n = x_0 - \frac{1}{n}$. Note that $a_n \rightarrow x_0$ and $b_n \rightarrow x_0$. Since f is differentiable at x_0 , we

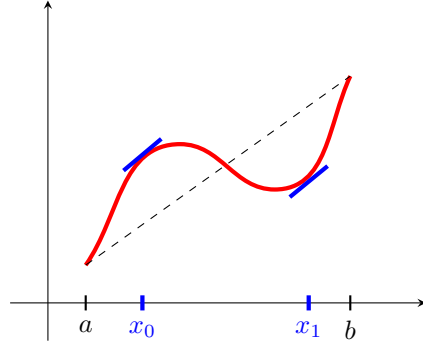


FIGURE 7. The intuition behind the mean value theorem. Given a differentiable function on an interval, there is at least one point where the instantaneous slope coincides with the average rate of change over the whole interval. In the pictured function, there are two points (x_0 and x_1) where this happens.

have

$$\begin{aligned} f'(x_0) &= \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \\ &= \lim_{n \rightarrow \infty} \frac{f(a_n) - f(x_0)}{a_n - x_0}. \end{aligned}$$

Note that $a_n - x_0 = \frac{1}{n} > 0$, and by assumption $f(a_n) - f(x_0) \geq 0$. Thus, $\frac{f(a_n) - f(x_0)}{a_n - x_0} \geq 0$ for all n , which implies

$$f'(x_0) = \lim_{n \rightarrow \infty} \frac{f(a_n) - f(x_0)}{a_n - x_0} \geq 0.$$

Similarly, we have

$$\begin{aligned} f'(x_0) &= \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \\ &= \lim_{n \rightarrow \infty} \frac{f(b_n) - f(x_0)}{b_n - x_0}. \end{aligned}$$

Note that $b_n - x_0 = -\frac{1}{n} < 0$, and by assumption $f(b_n) - f(x_0) \geq 0$. Thus, $\frac{f(b_n) - f(x_0)}{b_n - x_0} \leq 0$ for all n , which implies

$$f'(x_0) = \lim_{n \rightarrow \infty} \frac{f(b_n) - f(x_0)}{b_n - x_0} \leq 0.$$

Putting these two inequalities together gives $f'(x_0) = 0$, as desired. $\square$

Now we can prove the mean value theorem.

Proof of Theorem 25.1. Define a function $g : [a, b] \rightarrow \mathbb{R}$ by

$$g(x) := f(a) + \frac{f(b) - f(a)}{b - a}(x - a) - f(x).$$

Since f is continuous on $[a, b]$ and differentiable on (a, b) , so is g . By derivative rules, we have

$$g'(x) = \frac{f(b) - f(a)}{b - a} - f'(x).$$

Thus, to prove the theorem it suffices to find a point $x_0 \in [a, b]$ such that $g'(x_0) = 0$.

Since g is continuous on $[a, b]$, the extreme value theorem implies that there are points $x_{\max}, x_{\min} \in [a, b]$ such that, for all $x \in [a, b]$, $f(x_{\min}) \leq f(x) \leq f(x_{\max})$. If $x_{\max}, x_{\min} \in \{a, b\}$ — that is, if these points are the

endpoints of the interval — then the function g is necessarily constant and hence $g'(x) = 0$ for all $x \in (a, b)$. Thus, we may assume that at least one of $x_{\max}, x_{\min}$ is an element of (a, b) . Let x_0 denote this point. By Theorem 25.3, it follows that $g'(x_0) = 0$. This completes the proof. $\square$

One immediate consequence of the mean value theorem is the following intuitive result.

Corollary 25.4. *Let $f : (a, b) \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x) = 0$ for all $x \in (a, b)$. Then there is a $c \in \mathbb{R}$ such that $f(x) = c$ for all $x \in (a, b)$.*

Proof. Suppose for the sake of contradiction that f is not constant on (a, b) . Then there are points $x_0 < x_1 \in (a, b)$ with $f(x_0) \neq f(x_1)$. This means that

$$\frac{f(x_1) - f(x_0)}{x_1 - x_0} \neq 0.$$

Note that f is differentiable (and hence continuous) on $[x_0, x_1]$. The mean value theorem then gives a point $x_3 \in [x_0, x_1] \subseteq (a, b)$ such that

$$f'(x_3) = \frac{f(x_1) - f(x_0)}{x_1 - x_0} \neq 0.$$

This contradicts the fact that $f'(x) = 0$ for all $x \in (a, b)$. $\square$

26. LECTURE 26: DIFFERENTIABILITY AND UNIFORM CONVERGENCE

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◊ Rudin, *Principles of Mathematical Analysis*: Chapter 5, “Uniform Convergence and Differentiation”
- ◊ Abbott, *Understanding Analysis*: Chapter 6, Section 3

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◊ Appreciate how much mathematical maturity in analysis you have built up by understanding the proof of a sophisticated theorem!

So far, the little that we have studied about differentiability has not been new or surprising. After taking calculus you have seen facts like differentiability implies continuity, rules like the chain rule, and theorems like the mean value theorem. The main philosophical difference in what we’ve done in the last few lectures in comparison to a calculus class is that we have gone through the trouble of carefully and rigorously building all of the machinery necessary to make these concepts well-founded. In the remaining two lectures of the quarter, we will study some new and surprising facts about differentiability that cannot be proven (let alone discussed) in calculus.

To motivate the content of this lecture, recall that when we studied continuity and the convergence of sequences of functions, a natural question was to ask if (pointwise) convergence preserves continuity. That is, if f_n is a sequence of continuous functions and $f_n \rightarrow f$ pointwise, can we necessarily say that f is continuous? The answer was no. However, we then studied *uniform* convergence and showed that continuity is preserved under this stronger form of convergence. Now that we developed the concept of differentiability, we can ask the same kinds of questions.

Question 26.1. Suppose that $f_n : I \rightarrow \mathbb{R}$ is a sequence of differentiable functions. Suppose that $f_n \rightarrow f$ pointwise for some function $f : I \rightarrow \mathbb{R}$. Is f necessarily differentiable?

The answer to this question is, maybe unsurprisingly, no. In fact, we have already seen an example of functions that are differentiable and not even *continuous* in the limit. In particular, define $f_n : [0, 1] \rightarrow \mathbb{R}$ by $f_n(x) = x^n$. Naturally, we can upgrade the question.

Question 26.2. Suppose that $f_n : I \rightarrow \mathbb{R}$ is a sequence of differentiable functions. Suppose that $f_n \rightarrow f$ uniformly for some function $f : I \rightarrow \mathbb{R}$. Is f necessarily differentiable?

You might now expect the answer to be yes, but the answer is actually still no! Uniform convergence is *not* strong enough to preserve differentiability. Here is an example.

Example 26.3. Define a sequence of function $f_n : [-1, 1] \rightarrow \mathbb{R}$ by

$$\begin{cases} |x| & x \in [-1, \frac{1}{n}] \cup [\frac{1}{n}, 1] \\ \frac{n}{2}x^2 + \frac{1}{2n} & x \in (-\frac{1}{n}, \frac{1}{n}) \end{cases}$$

Let $f : [-1, 1] \rightarrow \mathbb{R}$ be given by $f(x) = |x|$. One can check (see the exercises) that $f_n \rightarrow f$ uniformly, though this is visually clear by looking at the graphs of the functions; see Figure ???. One can also verify that each f_n is differentiable, which is only interesting at the points $x_0 = \pm \frac{1}{n}$. However, the limiting function f is not differentiable at 0.

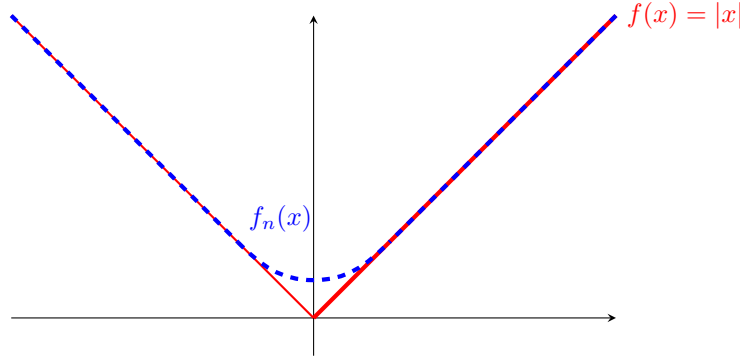


FIGURE 8. A family of differentiable function f_n that converges uniformly to a function f which is not differentiable at 0.

Because of this phenomenon, the question about preserving differentiability in the limit becomes much more delicate. An open ended question — which is the type of question at the heart of serious question in analysis — is the following.

Question 26.4. Suppose that $f_n : I \rightarrow \mathbb{R}$ is a sequence of differentiable functions. Suppose also that $f_n \rightarrow f$ under some notion of convergence. Under what conditions can we conclude that f is differentiable?

There are a number of different ways to answer this question. The following theorem, which is the main theorem of this lecture and in some sense one of the culminations of the entire body of work in this course, gives one answer.

Theorem 26.5. Let $f_n : I \rightarrow \mathbb{R}$ be a sequence of differentiable functions. Assume that

- ◊ $f_n \rightarrow f$ pointwise for some function $f : I \rightarrow \mathbb{R}$, and
- ◊ $f'_n \rightarrow g$ uniformly for some function $g : I \rightarrow \mathbb{R}$.

Then f is differentiable and $f' = g$.

Proof. Fix $x_0 \in I$. To prove the theorem we need to prove that f is differentiable at x_0 and that $f'(x_0) = g(x_0)$. Thus, we need to prove that

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = g(x_0).$$

Fix $\varepsilon > 0$.

Because $f'_n \rightarrow g$ uniformly, there is an $N_1 \in \mathbb{N}$ such that, for all $n \geq N_1$ and all $x \in I$, we have $|f'_n(x) - g(x)| < \frac{\varepsilon}{3}$. We also know that (f'_n) is uniformly Cauchy, and so there is an $N_2 \in \mathbb{N}$ such that, for all $n, m \geq N_2$ and all $x \in I$, we have $|f'_n(x) - f'_m(x)| < \frac{\varepsilon}{3}$. Let $n = \max(N_1, N_2)$.

Since f_n is differentiable at x_0 , there is a $\delta > 0$ such that, for all $x \in I$ with $0 < |x - x_0| < \delta$, we have

$$\left| \frac{f_n(x) - f_n(x_0)}{x - x_0} - f'_n(x_0) \right| < \frac{\varepsilon}{3}.$$

Let $x \in I$ satisfy $0 < |x - x_0| < \delta$.

Fix $m \geq n$. Consider the function $f_m - f_n : I \rightarrow \mathbb{R}$ and observe that this function is differentiable. In particular, we can apply the mean value theorem to $f_m - f_n$ on the interval $[x_0, x]$ (or $[x, x_0]$, in the case that $x < x_0$). Thus, there exists a $y_m \in (x_0, x)$ such that

$$f'_m(y_m) - f'_n(y_m) = \frac{[f_m(x) - f_n(x)] - [f_m(x_0) - f_n(x_0)]}{x - x_0} = \frac{f_m(x) - f_m(x_0)}{x - x_0} - \frac{f_n(x) - f_n(x_0)}{x - x_0}.$$

Note that, since $n, m \geq N_2$, we have

$$\left| \frac{f_m(x) - f_m(x_0)}{x - x_0} - \frac{f_n(x) - f_n(x_0)}{x - x_0} \right| = |f'_m(y_m) - f'_n(y_m)| < \frac{\varepsilon}{3}.$$

This implies that

$$\begin{aligned} \left| \frac{f(x) - f(x_0)}{x - x_0} - \frac{f_n(x) - f_n(x_0)}{x - x_0} \right| &= \lim_{m \rightarrow \infty} \left| \frac{f_m(x) - f_m(x_0)}{x - x_0} - \frac{f_n(x) - f_n(x_0)}{x - x_0} \right| \\ &\leq \limsup (|f'_m(y_m) - f'_n(y_m)|)_{m \in \mathbb{N}} \\ &\leq \frac{\varepsilon}{3}. \end{aligned}$$

Finally, note that

$$\begin{aligned} \left| \frac{f(x) - f(x_0)}{x - x_0} - g(x_0) \right| &= \left| \frac{f(x) - f(x_0)}{x - x_0} - \frac{f_n(x) - f_n(x_0)}{x - x_0} + \frac{f_n(x) - f_n(x_0)}{x - x_0} - f'_n(x_0) + f'_n(x_0) - g(x_0) \right| \\ &\leq \left| \frac{f(x) - f(x_0)}{x - x_0} - \frac{f_n(x) - f_n(x_0)}{x - x_0} \right| + \left| \frac{f_n(x) - f_n(x_0)}{x - x_0} - f'_n(x_0) \right| + |f'_n(x_0) - g(x_0)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon. \end{aligned}$$

Thus, for all $\varepsilon > 0$, there is a $\delta > 0$ such that, for all $x \in I$ with $0 < |x - x_0| < \delta$, we have

$$\left| \frac{f(x) - f(x_0)}{x - x_0} - g(x_0) \right| < \varepsilon.$$

This implies that

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = g(x_0).$$

Hence, f is differentiable at x_0 and $f'(x_0) = g(x_0)$. This completes the proof of the theorem. $\square$

27. LECTURE 27: A CONTINUOUS BUT NOWHERE DIFFERENTIABLE FUNCTION

Relevant textbook references:

For textbook references to supplement the material from this lecture, some options are:

- ◇ Ross, *Elementary Analysis*: Chapter 7, Section 38
- ◇ Rudin, *Principles of Mathematical Analysis*: Chapter 5, “Uniform Convergence and Differentiation”
- ◇ Abbott, *Understanding Analysis*: Chapter 5, Section 4

Learning goals:

After understanding everything in this lecture thoroughly, you will:

- ◇ Appreciate the fact that we can build and study functions that were previously indescribable with only the tools of calculus.

Exposition

Theorem 27.1 (Existence of continuous nowhere differentiable functions). *There exists a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is continuous and nowhere differentiable. That is, for all $x_0 \in \mathbb{R}$, f is continuous at x_0 and not differentiable at x_0 .*

Proof. We proceed with three steps: we will first define the function, then we will prove it is continuous, and finally we will prove that it is nowhere differentiable.

Step I: Defining $f : \mathbb{R} \rightarrow \mathbb{R}$.

First, we define the function $f : \mathbb{R} \rightarrow \mathbb{R}$. We begin by defining an auxiliary function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ as follows. Fix $k \in \mathbb{Z}$. For $x \in [2k - 1, 2k + 1)$, set $\phi(x) = |x - 2k|$. The function ϕ can be described as a sawtooth wave; see Figure ??.

ADD FIGURE

Note that ϕ is continuous and satisfies $|\phi(x)| \leq 1$ for all $x \in \mathbb{R}$.

Define $\phi_n : \mathbb{R} \rightarrow \mathbb{R}$ by $\frac{1}{2^n} \phi(2^n x)$, and then define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) := \sum_{n=0}^{\infty} \phi_n(x).$$

Convergence of the series defining f will be proven in the next step.

Step II: Proving that f is continuous.

First, note that for all n the function ϕ_n is continuous, because ϕ is continuous. Also note that, for all $x \in \mathbb{R}$, we have

$$|\phi_n(x)| = \frac{1}{2^n} |\phi(2^n x)| \leq \frac{1}{2^n}.$$

Since $\sum_{n=0}^{\infty} \frac{1}{2^n}$ is a convergent geometric series, the Weierstrass M -test implies that the sum

$$\sum_{n=0}^{\infty} \phi_n(x)$$

converges uniformly on $\mathbb{R}$. Since each of the summands is continuous, it follows that f is continuous on $\mathbb{R}$.

Step III: Proving that f is nowhere differentiable.

Fix $x_0 \in \mathbb{R}$. We will prove that f is not differentiable at x_0 . First, we claim that for every $k \in \mathbb{N}$ there is a number $q_k \in \mathbb{Z}$ such that

$$\frac{q_k}{2^k} \leq x_0 < \frac{q_k + 1}{2^k}.$$

Note that, for each $k \in \mathbb{N}$, the existence of such a number $q_k \in \mathbb{Z}$ is equivalent to the existence of a number $q_k \in \mathbb{Z}$ satisfying

$$q_k \leq 2^k x_0 < q_k + 1$$

and further

$$-2^k x_0 \geq q_k > -2^k x_0 + 1.$$

The existence of such an integer is clear since the distance between $-2^k x_0$ and $-2^k x_0 + 1$ is 1.

With $q_k \in \mathbb{Z}$ as above, let $a_k := \frac{q_k}{2^k}$ and $b_k = \frac{q_k + 1}{2^k}$. Then $a_k \leq x_0 < b_k$ and $|a_k - b_k| = \frac{1}{2^k} \rightarrow 0$ as $k \rightarrow \infty$. Note that, since $a_k \leq x_0 < b_k$ for all $k \in \mathbb{N}$, we have

$$\begin{aligned} |a_k - x_0| &= x_0 - a_k \\ &\leq b_k - a_k \\ &< \frac{1}{2^k} \\ &\rightarrow 0. \end{aligned}$$

It follows that $\lim_{k \rightarrow \infty} a_k = x_0$. Since $|a_k - b_k| = \frac{1}{2^k} \rightarrow 0$ as $k \rightarrow \infty$, we have

$$\lim_{n \rightarrow \infty} a_k = \lim_{n \rightarrow \infty} b_k = x_0.$$

To prove that f is not differentiable at x_0 , we consider the sequence of difference quotients

$$\frac{f(b_k) - f(a_k)}{b_k - a_k} = \frac{f(b_k) - f(a_k)}{\frac{1}{2^k}}.$$

First, if f were differentiable at x_0 , then

$$f'(x_0) = \lim_{k \rightarrow \infty} \frac{f(b_k) - f(a_k)}{b_k - a_k}.$$

Indeed, by definition we have

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

By Theorem 23.6, since $b_k \rightarrow x_0$ and $b_k \neq x_0$ we have

$$f'(x_0) = \lim_{k \rightarrow \infty} \frac{f(b_k) - f(x_0)}{b_k - x_0} = \lim_{k \rightarrow \infty} \frac{f(x_0) - f(b_k)}{x_0 - b_k}.$$

FINISH

Note that for $n > k$, $2^n a_k = 2^{n-k} q_k \in \mathbb{Z}$ and in particular $2^n a_k$ is an even integer. Thus, by definition of ϕ ,

$$\phi_n(a_n) = \frac{1}{2^n} \phi(2^n a_k) = 0$$

for all $n > k$. Likewise, for $n > k$ we have $2^n b_k = 2^{n-k}(q_n + 1) \in \mathbb{Z}$ and in particular $2^n b_k$ is an even integer, and so

$$\phi_n(b_n) = \frac{1}{2^n} \phi(2^n b_k) = 0.$$

All of this implies that

$$\frac{f(b_k) - f(a_k)}{\frac{1}{2^k}} = \sum_{n=0}^k \frac{\frac{1}{2^n} \phi(2^n b_k) - \frac{1}{2^n} \phi(2^n a_k)}{\frac{1}{2^k}} = \sum_{n=0}^k 2^{k-n} (\phi(2^n b_k) - \phi(2^n a_k))$$

FINISH

□

discussion about BCT

Part 6

Appendices

APPENDIX A. SETS

One of the most fundamental objects of interest in all of math is that of a *set*. A set is just a collection — possibly infinite — of things. For example,

$$S_1 = \{1, 2, -400\}$$

is a set consisting of the numbers 1, 2, and -400 . As another example,

$$S_2 = \{\text{blue}, :, \pi\}$$

is a set consisting of the color blue, a smiley face, and the number π . Believe it or not, this is perfectly valid mathematical set.

Set notation.

For larger sets that are hard to write down explicitly, we will sometimes use the following notation:

$$S_3 = \{n \mid n \text{ is a positive, even number}\}.$$

This notation is read as follows: S_3 is the set consisting of elements of the form n , such that n is any positive, even number. The vertical bar is read as “such that”, and the stuff after the vertical bar is a set of conditions that all elements of the set must satisfy. Unraveling the set S_3 above, this means that

$$S_3 = \{n \mid n \text{ is a positive, even number}\} = \{2, 4, 6, 8, \dots\}.$$

We use the symbol $\in$ to indicate if something is an element of a set. For example, recall the set $S_1 = \{1, 2, -400\}$ from above. We could write

$$2 \in S_1$$

because 2 is an element of S_1 . We could also write

$$3 \notin S_1$$

because 3 is *not* an element of S_1 . The objects of sets can even be as complicated as sets themselves: for example, we could write

$$\{1, 2\} \in \{\text{green}, \pi, \{1, 2\}, -34\}.$$

Unions and intersections.

We can define operations on sets. For example, if A and B are sets, then we define

$$A \cup B := \{x \mid x \in A \text{ or } x \in B\},$$

$$A \cap B := \{x \mid x \in A \text{ and } x \in B\}.$$

The first is the *union* of the sets A and B , and the second is the *intersection*. For example, using S_1 and S_3 from above,

$$S_1 \cup S_3 = \{-400, 1, 2, 4, 6, 8, \dots\}$$

$$S_1 \cap S_3 = \{2, -400\}.$$

The *empty set*, denoted $\emptyset$, is the set consisting of no elements. That is, $\emptyset := \{ \}$. Thus, we could write

$$S_1 \cap S_2 = \emptyset.$$

When two sets have empty intersection, we say that they are *disjoint*.

We can also discuss *subsets*. In particular, if A and B are two sets, then we say $A \subset B$ (or $A \subseteq B$, both notations confusingly mean the same thing) if every element of A is also an element of B . For example,

$$\{4, 6, e\} \subset \{4, 6, e, 10, 24\}.$$

One thing that you will have to do often in linear algebra, and in life, is show that two sets are the same. The following is a fundamental principle in math.

Principle A.1. To show that $A = B$, you should separately show that $A \subset B$ and $B \subset A$.

Here are some important sets that you are already familiar with.

$$\mathbb{N} := \{ x \mid x \text{ is a natural number} \} = \{0, 1, 2, 3, \dots\}$$

$$\mathbb{Z} := \{ x \mid x \text{ is an integer} \} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

$$\mathbb{Q} := \{ x \mid x \text{ is a rational number} \} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

APPENDIX B. FIELDS

Some sets are just simple collections of elements with no extra structure. Other sets naturally admit an extra amount of structure and interaction (i.e., *algebra*.) For example, in the set of integers, $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, we are (secretly!) familiar with the algebraic operations of addition (+) and multiplication ($\cdot$). That is, given two elements $n, m \in \mathbb{Z}$, we can construct a third element $n + m \in \mathbb{Z}$ by adding them together, and likewise a fourth element $n \cdot m \in \mathbb{Z}$. Furthermore, these algebraic operations obey a handful of rules (commutativity, distribution, etc.) that you learned when you were little.

In contrast, consider the set $S_2 = \{\text{blue square}, \cdot, \pi\}$ from before. We don't have any familiar algebraic structure on this set, so for now it will be just be an unstructured collection of random elements.

To read the rest of this section, **I want you to forget everything you ever knew about algebra**. Addition doesn't exist, multiplication doesn't exist, you don't know what the quadratic formula is, and so on. Currently, all we have at our disposal is the notion of a set. We wish to do algebra, and so it is up to us to *define* the correct notions of algebra. In other words, we get to impose our own algebraic rules on an arbitrary set. An important example of this construction is the notion of a *field*.

Definition B.1. A **field** is a set $\mathbb{F}$ with two operations, addition (+) and multiplication ($\cdot$), that take a pair of elements $x, y \in \mathbb{F}$ and produce new elements $x + y, x \cdot y \in \mathbb{F}$. Furthermore, these operations satisfy the following properties:

- (1) For all $x, y \in \mathbb{F}$,

$$x + y = y + x \quad \text{and} \quad x \cdot y = y \cdot x.$$

- (2) For all $x, y, z \in \mathbb{F}$,

$$(x + y) + z = x + (y + z) \quad \text{and} \quad (x \cdot y) \cdot z = x \cdot (y \cdot z)$$

- (3) For all $x, y, z \in \mathbb{F}$,

$$x \cdot (y + z) = x \cdot y + x \cdot z.$$

- (4) There are elements $0, 1 \in \mathbb{F}$ such that, for all $x \in \mathbb{F}$,

$$0 + x = x \quad \text{and} \quad 1 \cdot x = x.$$

The element 0 is called an **additive identity** and the element 1 is called a **multiplicative identity**.

- (5) For each $x \in \mathbb{F}$, there is an element $x' \in \mathbb{F}$, called an **additive inverse**, such that $x + x' = 0$. Similarly, for every $y \neq 0 \in \mathbb{F}$, there is an element $y' \in \mathbb{F}$, called a **multiplicative inverse**, such that $y \cdot y' = 1$.

Remark B.2. Property (1) is called *commutativity of addition and multiplication*, property (2) is called *associativity of addition and multiplication*, and property (3) is called *distributivity of multiplication over addition*. I'll emphasize that these properties do not comprise a theorem, or a claim about something you already know. We are defining the concept of algebra by arbitrarily imposing these conditions on our given set $\mathbb{F}$.

Example B.3. The main example of a field that you are (secretly!) familiar with is $\mathbb{R}$, the set of real numbers, with the usual operations of addition and multiplication. All of the above properties should look familiar to you, precisely because they are modeled after the behavior of $\mathbb{R}$.

Example B.4. Other familiar examples of fields are $\mathbb{Q}$ and $\mathbb{C}$.

Example B.5. The set of integers, $\mathbb{Z}$, with the usual operations of addition and multiplication, is *not* a field. Almost all of the field properties are satisfied, except for the multiplicative inverse property. In particular, it is *not* the case that for any $y \neq 0 \in \mathbb{Z}$, there is a $y' \in \mathbb{Z}$ such that $y \cdot y' = 1$. For example, the element $2 \in \mathbb{Z}$ does not have a multiplicative inverse. We (secretly) know that such a number would have to be $\frac{1}{2}$, but that number doesn't exist in $\mathbb{Z}$. Similarly, $\mathbb{N}$ is not a field. Not only does $\mathbb{N}$ not in general have multiplicative inverses, but it also doesn't have additive inverses!

Example B.6. Here is an example of a field that you may not have seen before. Let $\mathbb{F}_2 := \{0, 1\}$ be the set consisting of 2 elements, 0 and 1. Define addition in the following way:

$$0 + 0 := 0$$

$$0 + 1 := 1$$

$$1 + 1 := 0$$

and define multiplication as follows:

$$0 \cdot 0 := 0$$

$$0 \cdot 1 := 0$$

$$1 \cdot 1 := 1.$$

I claim that $\mathbb{F}_2$ is a field! I won't verify all of the properties here (that will be an exercise), but I will point out the most interesting aspect: each element has an additive inverse (the additive inverse of 0 is 0, and the additive inverse of 1 is 1), and each nonzero element has a multiplicative inverse (the multiplicative inverse of 1 is 1).

B.1. Properties of fields. Next, we will prove some basic properties of fields. Not only will this build up the theory of fields, but it will be our first exposure to writing proofs and communicating mathematics in a professional and rigorous way.

There are many algebraic operations in fields that will (secretly!) be familiar to you, as they will mimic the algebraic rules you learned when you were a baby. But remember — we have formally forgotten everything we have ever known about math, so we must first prove all of these properties using the definition. For example, you (secretly!) know that in an algebraic equation, if you have the same term that appears on both sides then you can “cancel it out.” I didn't say anything about this property in the definition of a field! It turns out that you *can* do this in an abstract field, but we have to prove that we can do so first.

Proposition B.7 (Cancellation laws). *Let $\mathbb{F}$ be a field. Let $x, y, z \in \mathbb{F}$.*

- (i) *If $x + y = x + z$, then $y = z$.*
- (ii) *If $x \cdot y = x \cdot z$ and $x \neq 0$, then $y = z$.*

Proof. First, we prove (i). Suppose that $x + y = x + z$. By existence of additive inverses (property (5) in the definition of a field), there exists an element x' such that $x + x' = 0$. Adding x' to both sides of the assumed equality gives

$$x' + (x + y) = x' + (x + z).$$

By associativity of addition in a field, this is equivalent to

$$(x' + x) + y = (x' + x) + z.$$

Using the fact that $x' + x = 0$, we have

$$0 + y = 0 + z \quad \text{and thus} \quad y = z.$$

Next, we prove (ii). Suppose that $x \cdot y = x \cdot z$ and $x \neq 0$. By the existence of multiplicative inverses, there is an element x' such that $x' \cdot x = 1$. Multiplying both sides of the assumed equality by x' gives

$$x' \cdot (x \cdot y) = x' \cdot (x \cdot z).$$

By associativity of multiplication, it follows that

$$(x' \cdot x) \cdot y = (x' \cdot x) \cdot z$$

and thus $1 \cdot y = 1 \cdot z$. This implies $y = z$, as desired. $\square$

As a corollary of this result, we can prove another fact that you have likely already taken for granted. Note that in the definition of a field, I never said anything about there only being *one* element called “0”, or only *one* element called “1”. In the number systems that we are used to, there is of course only one of each. But a priori, the definition of a field allows for the possibility that there are multiple 0 elements and multiple 1 elements. Fortunately, we can definitively prove, using the above properties, that these elements are unique.

Corollary B.8. *The elements 0 and 1 in a field are unique. That is, there is only one element satisfying the additive identity property, and only one element satisfying the multiplicative identity property.*

Proof. Suppose that $0' \in \mathbb{F}$ is another additive identity, so that $0' + x = x$ for all $x \in \mathbb{F}$. Then since $0 + x = x$, we have

$$0' + x = 0 + x$$

for all $x \in \mathbb{F}$. By the previous proposition, it follows that $0' = 0$.

Similarly, suppose that there is an element $1' \in \mathbb{F}$ such that $1' \cdot x = x$ for all $x \in \mathbb{F}$. Then since $1 \cdot x = x$, we have

$$1' \cdot x = 1 \cdot x$$

for all $x \in \mathbb{F}$. In particular, we may choose $x = 1$. By the previous proposition, it follows that $1 = 1'$. $\square$

This proof is the first instance of the following general mathematical principle. You will encounter this multiple times over the course of these notes, and you will also be asked to execute this principle yourself.

Principle B.9. To prove that an object $\star$ is unique, assume that you have another object $\star'$ that satisfies the same properties. Then show that $\star = \star'$.

Using this principle, we can prove a similar statement on the uniqueness of multiplicative and additive inverses.

Corollary B.10. *For each $x \in \mathbb{F}$, the element x' satisfying $x + x' = 0$ is unique. If $x \neq 0$, the element x' satisfying $x' \cdot x = 1$ is unique.*

Proof. Exercise. $\square$

These corollaries allow us to talk about *the* additive identity, *the* multiplicative identity, *the* additive inverse of an element, etc. Furthermore, we can make the following notational definition.

Definition B.11. Let $\mathbb{F}$ be a field, and let $x \in \mathbb{F}$. The unique additive inverse of x is denoted $-x$, and the unique multiplicative inverse (if $x \neq 0$) is denoted x^{-1} or $\frac{1}{x}$.

We continue by proving more familiar properties of real numbers that are true in the general setting of fields.

Proposition B.12. Let $\mathbb{F}$ be a field, and let $x, y \in \mathbb{F}$. Then

- (i) $0 \cdot x = 0$,
- (ii) $(-x) \cdot y = x \cdot (-y) = -(x \cdot y)$,
- (iii) $(-x) \cdot (-y) = x \cdot y$,
- (iv) and 0 has no multiplicative inverse.

Proof. Exercise. □

We can also now define the notions of subtraction and division in a field, more things that you've taken for granted!

Definition B.13. Let $\mathbb{F}$ be a field. For $x, y \in \mathbb{F}$, define

$$x - y := x + (-y).$$

Similarly, if $y \neq 0$, define

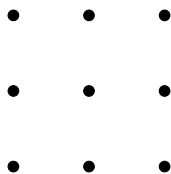
$$\frac{x}{y} := x \cdot \frac{1}{y}.$$

I'll repeat one last time that the point of this section is to abstractly re-develop the concept of a number system from the ground up. We formally forget everything we knew about the real numbers, algebraic manipulation, or even basic concepts like addition and multiplication. Thus, we begin with a raw collection of elements (a set) and we impose our own axioms of algebra on this set, leading to the completely abstract notion of a field. This abstract notion of a field generalizes familiar number systems like $\mathbb{R}$, $\mathbb{Q}$, and $\mathbb{C}$, and it also leads to some surprising new structures like the field with two elements $\mathbb{F}_2 = \{0, 1\}$.

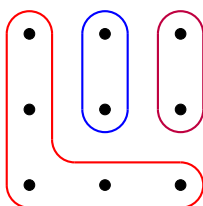
APPENDIX C. EQUIVALENCE RELATIONS

The notion of an *equivalence relation* is ubiquitous in math. In the context of real analysis, equivalence relations are crucial for rigorously defining integers, rational numbers, and real numbers.

C.1. The definition of an equivalence relation. Before giving the precise definition of an equivalence relation, let me describe the intuitive idea. Suppose we have a collection of objects, which I will represent below as a collection of dots.



This collection of dots represents a set of 9 elements. Suppose that for some reason, we wanted to group the objects together so that we could treat groups as one, new object. For example, we could group the objects together like this:

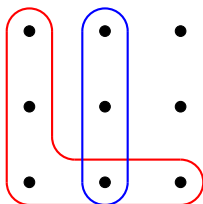


With this grouping, we now think of having 3 total objects, not 9. The red group is a single blob, the blue group is a single blob, and the purple group is a single blob. If you pick up one of the dots in the red blob, you pick up the entire blob. If you tell your friend *hey look at that dot in the middle of the top row*, all your friend will see is the entirety of the blue blob. If you ask your friend, *hey what's the difference between the top right dot and the dot in the middle of the third column*, they will tell you that there is no difference; they both exist as one in the purple blob. This is an equivalence relation! It's just a way to group collections of objects in a way that you treat the groups as one.

Now, there are some features of the above grouping that are important. I will highlight two.

- ◇ Every dot is in a group.
- ◇ No dot is in more than one group.

For example, consider the same collection above but with this attempt at grouping.



This does *not* describe an equivalence relation (which I still have yet to define). This grouping is not great, for a few reasons. First, two of the dots aren't part of a grouping at all; they are left out. We don't want to leave any object out of a grouping. Secondly, the middle dot in the bottom row is part of both the red and blue group. We don't want this to happen, because we want each group to be completely distinct so that we can treat them as separate objects.

With these pictures in mind, here is the official definition of an equivalence relation. After making some sense of the definition, we will consider a few linear-algebra-relevant examples.

Definition C.1. Let S be a set. An **equivalence relation on S** is a symbol $\sim$ that satisfies the following axioms.

- (1) (Reflexivity) For all $x \in S$, $x \sim x$.
- (2) (Symmetry) For all $x, y \in S$, we have $x \sim y$ if and only if $y \sim x$.
- (3) (Transitivity) For all $x, y, z \in S$, if $x \sim y$ and $y \sim z$, then $x \sim z$.

For any $x \in S$, let $[x] := \{y \in S \mid y \sim x\}$. The set $[x]$ is called the **equivalence class of x** .

In this definition, you should read the symbol “ $\sim$ ” as “is in the same group as”. So $x \sim x$, the reflexivity condition, means that x must be in the same group as itself. In terms of our above pictures, this is a trivial requirement. The symmetry condition means that if x is in the same group as y , then y must be in the same group as x . This is also a bit tautological in terms of the above groupings. The transitivity condition says that if x and y are in the same group, and if y and z are in the same group, then x and z have to be in the same group. This condition is violated by the figure above with overlapping blue and red groups!

C.2. An example involving finite fields. Our first example of an equivalence relation will be on the set of integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$. Define an equivalence relation (i.e., a grouping) $\sim$ as follows:

$$n \sim m \quad \text{if and only if } n - m \text{ is divisible by } 3.$$

To be clear, an integer M is **divisible by 3** if $M = 3k$ for some $k \in \mathbb{Z}$. Let’s verify that this is an equivalence relation, and then we’ll discuss how to think about what this gives us.

- (1) (Reflexivity) First we show that $n \sim n$ for all $n \in \mathbb{Z}$. Note that $n - n = 0 = 3 \cdot 0$ and so $n - n$ is divisible by 3. Thus, $n \sim n$.
- (2) (Symmetry) Suppose that $n \sim m$. This means that $n - m = 3k$ for some $k \in \mathbb{Z}$. It follows that $m - n = 3(-k)$, and so $m \sim n$. Thus, the symmetry property is satisfied.
- (3) (Transitivity) Suppose that $n \sim m$ and $m \sim \ell$. We wish to show that $n \sim \ell$. Since $n \sim m$ and $m \sim \ell$, we have $n - m = 3k_1$ and $m - \ell = 3k_2$ for some $k_1, k_2 \in \mathbb{Z}$. Thus,

$$n - \ell = (n - m) + (m - \ell) = 3k_1 + 3k_2 = 3(k_1 + k_2)$$

and so $n \sim \ell$, as desired.

Thus, $\sim$ as defined above is indeed an equivalence relation on $\mathbb{Z}$. But what is it telling us?

To understand this, let’s consider the *set of equivalence classes*. For example, consider the integer $0 \in \mathbb{Z}$. What is the equivalence class $[0]$? By the above definition, $[0]$ is the set of integers n such that $n \sim 0$. That is, the set of integers n such that $n - 0 = 3k$. This means that $[0]$ is the set of all multiples of 3:

$$[0] = \{\dots, -9, -6, -3, 0, 3, 6, 9, \dots\}.$$

To emphasize the point of an equivalence relation, we also have $[0] = [3] = [6]$ and so on. The notation $[a]$ just means “the group containing the element a .” What about $[1]$, the equivalence class containing 1? Well, this would be the set of integers n such that $n - 1 = 3k$, so that $n = 3k + 1$. This means that

$$[1] = \{\dots, -8, -5, -2, 1, 4, 7, 10, \dots\}.$$

Likewise, one can check that

$$[2] = \{\dots, -7, -4, -1, 2, 5, 8, 11, \dots\}.$$

You'll notice that with these three equivalence classes (groups), we have now partitioned all of the integers. That is, we have decomposed $\mathbb{Z} = [0] \cup [1] \cup [2]$. This is a grouping of integers according to their remainder after division by 3. This should feel familiar, if you've spent some time thinking about finite fields. In particular, the field $\mathbb{F}_3$ can be described precisely using the three equivalence classes of $\mathbb{Z}$ under $\sim$ as above!

C.3. An example involving subspaces of vector spaces. Next, here is an equivalence relation in the realm of vector spaces.¹⁰ This equivalence relation is used in defining *quotient vector spaces*, informally, a way to “divide” a vector space by a subspace.

Let V be any vector space, and let $W \subset V$ be a subspace. Define $\sim$ on V as follows:

$$v \sim w \quad \text{if and only if } v - w \in W.$$

Let's first verify that this is in fact an equivalence relation, and then we will discuss what it means.

- (1) (Reflexivity) For any $v \in V$, note that $v - v = 0 \in W$, since W is a subspace. Thus, $v \sim v$. This proves that $\sim$ is reflexive.
- (2) (Symmetry) Suppose that $v \sim w$. Then $v - w \in W$. Since W is a subspace, and hence is closed under scalar multiplication, we have that $(-1)(v - w) = w - v \in W$, and hence $w \sim v$. This proves that $\sim$ is symmetric.
- (3) (Transitivity) Suppose that $v \sim w$ and $w \sim u$. Then $v - w, w - u \in W$. Note that

$$v - u = (v - w) + (w - u) \in W$$

since W is closed under addition. Thus, $v \sim u$, and so $\sim$ is transitive.

Thus, $\sim$ is in fact an equivalence relation on V .

To see what this equivalence class gives us, let's consider the set of equivalence classes. First, consider $[0]$, the equivalence class containing the 0 vector. This is the set of all vectors equivalent to 0, which is the set of all vectors v such that $v - 0 \in W$. But this is precisely all of the vectors in W , and so $[0] = W$. That is, under this equivalence relation we think of the entire subspace W as being one object.

All of the other objects (equivalence classes) are parallel shifts of the subspace W . For example, let $v_0 \in V$ be a vector which is *not* in the subspace W . Then

$$[v_0] = \{ w \in V \mid w - v_0 \in W \} = \{ w \in V \mid w \in \{v_0\} + W \}.$$

In words, the equivalence class containing v_0 is given by adding v_0 to every single vector in W , that is, shifting the subspace W by v_0 . Note that $[v_0]$ is not a subspace of V , because it won't contain the 0 vector — instead, it is a parallel shift of W away from the origin by v_0 . Thus, the equivalence classes of V under $\sim$ foliate the vector space V by parallel copies of W .

¹⁰To understand this subsection, you should know what a subspace of a vector space is.

APPENDIX D. INDUCTION

Induction is a well-known and widely used proof technique. It is used to prove statement that come with the descriptor *for all* $n \in \mathbb{N}$, or something similar. That is, suppose you have a collection of mathematical statements $P(n)$, indexed by the natural numbers. As a silly example,

$$P(1) = \text{"The number 1 is not a color."}$$

$$P(2) = \text{"The number 2 is not a color."}$$

$$P(3) = \text{"The number 3 is not a color."}$$

$$\vdots$$

$$P(n) = \text{"The number } n \text{ is not a color."}$$

$$\vdots$$

We would like to prove the validity of the statements $P(n)$ for all $n \geq 1$. One possible approach is to prove them one by one. That is, first prove $P(1)$, then $P(2)$, then $P(3)$, and so on, until we have proven all statement. This, unfortunately, is not a very smart idea, because there are infinitely many statements. We would spend the rest of our lives proving the statements and would die before we ever finished. We could conscript all future humans to continue proving statements one by one after we perish, and this would still not be enough. Humanity would go extinct before we can prove infinitely many statements directly. This approach is not feasible.

Another approach is to write down a proof for the validity of $P(n)$ completely abstractly, without making an argument that depends on the specific value of the number n . This *is* a perfectly valid technique. You shouldn't blindly use induction, just because a statement contains the phrase "for all n ." Sometimes a direct and n -abstract argument will work.

However, sometimes a direct n -abstract argument can be difficult, and it can be easier to use induction (which I have yet to explain). The philosophy of induction is to treat the sequence of statements $P(1), P(2), P(3), \dots$ like a chain of standing dominoes. Proving the statement $P(n)$ corresponds to tipping over the n th domino. Instead of tipping over each domino individually (the first approach) or calling upon an earthquake to simultaneously tip over all dominoes at once (the second method), we can take a more refined approach by only tipping over the first domino, and then arguing that each falling domino will tip over the next. By starting a chain reaction — and proving that a chain reaction must take place — we can delicately finish the job.

To say this in a more mathematical formulation, a proof by induction proceeds as follows. You prove directly the first statement, say, $P(1)$. This corresponds to tipping the first domino. Then, you must prove that if a statement $P(k)$ is true, then the following statement $P(k + 1)$ must also be true. This proves that one falling domino will cause the next to fall. I'll summarize this in the following principle, and then we will work through some examples.

Principle D.1 (Principle of induction.). To prove a statement of the form “ $P(n)$ for all $n \geq 1$ ” using induction, you:

- ◊ Establish a *base case* by proving the statement $P(1)$ directly. This tips the first domino.
- ◊ Prove that validity of $P(k)$ implies validity of $P(k+1)$. That is, assume that $P(k)$ is true — this is the *inductive hypothesis* — and prove that $P(k+1)$ must also then be true. This shows that if one domino falls, then the next will fall too.

D.1. The size of power sets. As a first example, let’s consider a concept rooted in the basics of set theory.

Definition D.2. Let S be any set (possibly infinite). The **power set of S** , denoted $\mathcal{P}(S)$, is the set consisting of all subsets of S . That is,

$$\mathcal{P}(S) := \{ X \mid X \subset S \}.$$

Example D.3. Let $S = \{1, 2\}$. Then

$$\mathcal{P}(S) = \{ \emptyset, \{1\}, \{2\}, \{1, 2\} \}.$$

Note that the empty set is a subset of every set, and hence always an element of the power set of anything! Similarly, S is always a subset of S , and so $S \in \mathcal{P}(S)$.

Example D.4. Let $S = \{1, 2, 3\}$. Then

$$\mathcal{P}(S) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\} \}.$$

Example D.5. Nothing stops us from taking power sets of power sets. Let $S = \{1, 2\}$. Then

$$\begin{aligned} \mathcal{P}(\mathcal{P}(S)) = & \left\{ \emptyset, \{\emptyset\}, \{\{1\}\}, \{\{2\}\}, \{\{1, 2\}\}, \right. \\ & \{\emptyset, \{1\}\}, \{\emptyset, \{2\}\}, \{\emptyset, \{1, 2\}\}, \{\{1\}, \{2\}\}, \{\{1\}, \{1, 2\}\}, \{\{2\}, \{1, 2\}\}, \\ & \left. \{\emptyset, \{1\}, \{2\}\}, \{\emptyset, \{1\}, \{1, 2\}\}, \{\emptyset, \{2\}, \{1, 2\}\}, \{\{1\}, \{2\}, \{1, 2\}\}, \{\emptyset, \{1\}, \{2\}, \{1, 2\}\} \right\} \end{aligned}$$

Power sets of power sets and beyond get difficult to itemize explicitly because of the mess of nested parentheses, but I’ll point out one thing: the elements $\emptyset$ and $\{\emptyset\}$ are distinct elements of the double power set!

Now I’ll get to the point where we can use some induction. For any (finite) set S , let $|S|$ denote the number of elements in the set. For example, if $S = \{1, 2\}$ then $|S| = 2$, and if $S = \{1, 2, 3\}$ then $|S| = 3$. If you count the elements in the above examples, you’ll pick up on the following pattern:

- ◊ When $S = \{1, 2\}$, so that $|S| = 2$, we have $|\mathcal{P}(S)| = 4 = 2^2 = 2^{|S|}$.
- ◊ When $S = \{1, 2, 3\}$, so that $|S| = 3$, we have $|\mathcal{P}(S)| = 8 = 2^3 = 2^{|S|}$.
- ◊ When $S = \{1, 2\}$, so that $|S| = 2$, we have $|\mathcal{P}(S)| = 4$ and $|\mathcal{P}(\mathcal{P}(S))| = 16 = 2^4 = 2^{|\mathcal{P}(S)|}$.

This pattern is no accident. In general, if you have a set of size n , then the size of the power set will be 2^n .

Proposition D.6. Let S be a set with $|S| = n$. Then $|\mathcal{P}(S)| = 2^n$.

Proof. Before we write a purple-quality proof, I will point out that, although not explicit, this proposition does say implicitly that something is true for all n . In this case, we have to make out the implication that

the statement of the proposition implies “for all $n \geq 0$.” Next, because this is our first induction proof, I will intersperse my own commentary (in black print) with the professional-quality writing that would eventually make up my final proof (purple print). Keep in mind that if I were writing a proof for this problem on a test, or on homework, or in a research paper, I would *only* present the purple parts of this proof.

The first step in an induction proof is to directly proof a base case. Above, I hinted that this statement is true for all $n \geq 0$, and so we can take $n = 0$ as our base case.

We proceed by induction on n . We begin with the base case $n = 0$. Let S be a set with $|S| = 0$. The only such set is the empty set, so $S = \emptyset$. Note that

$$\mathcal{P}(S) = \{\emptyset\}$$

and so $|\mathcal{P}(S)| = 1 = 2^0$. This verifies the base case.

A few more comments before proceeding. Note that I began my proof by telling the reader *I am going to use induction*. Comments like this help the readability of your proofs, as they help the reader understanding your argument on a greater scale. Next, I’ll once again point out the subtlety in that the empty set $\emptyset$, has *no* elements. It is the set $\emptyset = \{\}$. The set $\{\emptyset\}$ is very different than $\emptyset$.

Next, we perform the inductive step. We assume that we know that for some $k \geq 0$ that the statement is already true. We will prove that this forces the statement to be true for $k + 1$. We are not *proving* the statement is true for k , and we are not even claiming that it is true. We are simply hypothetically assuming that the statement is true for k , and then proving that, if this were the case, it would also have to be true for $k + 1$.

Next, we perform the inductive step. Assume that for some $k \geq 0$, it is true that if S is a set with $|S| = k$, then $|\mathcal{P}(S)| = 2^k$. Let $\tilde{S}$ be a set with $|\tilde{S}| = k + 1$. We must prove that $|\mathcal{P}(\tilde{S})| = 2^{k+1}$.

This is the set up for our inductive step. Note that we will make significant use of the inductive assumption to prove validity for $k + 1$. This is the point of using induction. To do this, we have to figure out how to use our assumption of sets of size k . Our current set of interest, $\tilde{S}$, contains $k + 1$ elements. It seems like a reasonable idea to me to write $\tilde{S}$ as a union of a subset of size k and one other element. From there, we can understand the power set of $\tilde{S}$ in terms of the power set of the smaller subset and the extra element.

Write $\tilde{S} = S \cup \{x\}$, where $S \subset \tilde{S}$ is a subset with $|S| = k$. By the inductive hypothesis, we know that $|\mathcal{P}(S)| = 2^k$. Thus, we can write

$$\mathcal{P}(S) = \{X_1, X_2, \dots, X_{2^k}\}.$$

Note that

$$\mathcal{P}(\tilde{S}) = \left\{ \underbrace{X_1, X_2, \dots, X_{2^k}}_{2^k \text{ elements}}, \underbrace{X_1 \cup \{x\}, X_2 \cup \{x\}, \dots, X_{2^k} \cup \{x\}}_{2^k \text{ elements}} \right\}.$$

Thus, $|\mathcal{P}(\tilde{S})| = 2^k + 2^k = 2 \cdot 2^k = 2^{k+1}$, as desired. By the principle of induction, this completes the proof.

□

Alternate proof. Before giving you some more examples of proofs by induction, I want to point out that we can prove the above proposition about sizes of power sets directly, without using induction. Arguably, a direct proof is easier.

Let S be a set with $|S| = n$. We can enumerate the elements (in order) as follows:

$$S = \{x_1, \dots, x_n\}.$$

We can identify each element $X \in \mathcal{P}(S)$ with a string of length n of 0s and 1s, where the j th digit, 0 or 1, indicates whether $x_j \notin X$ or $x_j \in X_j$, respectively. For example, the element $X = \{x_1, x_3, x_n\} \in \mathcal{P}(S)$ would correspond to the string

$$1010 \cdots 01.$$

Note that there are 2^n such strings, as each digit has two possibilities (0 or 1). Thus, $|\mathcal{P}(S)| = 2^n$, as desired. $\square$

D.2. Some numerical inequalities. Proving the size of power sets is a fairly sophisticated example with which to introduce the concept of induction. For some simpler examples, we can consider some numerical inequalities; statements about integers that holds for all n larger than some initial value.

Proposition D.7. *For all $n \geq 4$, we have $n! \geq 2^n$.*

Proof. We proceed by induction on n . First we establish the base case $n = 4$. Note that $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24 \geq 16 = 2^4$, as desired.

Next, we perform the inductive step. Suppose that for some $k \geq 4$ we know that $k! \geq 2^k$. Note that

$$\begin{aligned} (k+1)! &= (k+1) \cdot k! \\ &\geq (k+1) \cdot 2^k \\ &\geq 2 \cdot 2^k \\ &= 2^{k+1}. \end{aligned}$$

In the second inequality we used the inductive hypothesis, and in the third inequality we used the fact that $k+1 \geq 2$, since $k \geq 4$. By the principle of induction this proves that $n! \geq 2^n$ for all $n \geq 4$. $\square$

Proposition D.8. *Prove that for all integers $n \geq 0$ and all real numbers $x \geq -1$, we have $(1+x)^n \geq 1+nx$.*

Proof. We proceed by induction on n . First, we establish the base case $n = 0$. Note that for any $x \geq -1$, we have $(1+x)^0 = 1 \geq 1 = 1+0x$ as desired.

Next we perform the inductive step. Suppose that for some $k \geq 1$, we know that $(1+x)^k \geq 1+kx$ for all $x \geq -1$. Then

$$\begin{aligned} (1+x)^{k+1} &= (1+x)^k(1+x) \\ &\geq (1+kx)(1+x) \\ &= 1+kx+x+kx^2 \\ &= 1+(k+1)x+kx^2 \\ &\geq 1+(k+1)x. \end{aligned}$$

The first inequality comes from the inductive hypothesis and crucially uses the fact that $x \geq -1$, and hence $1+x \geq 0$. Otherwise the inequality would flip directions. The last inequality comes from the fact that $kx^2 \geq 0$. $\square$

I'll close our discussion of induction with some observations about these two examples. First, note the overall structural similarity between the proofs of these inequalities with the power set example above. I began by telling the reader I would perform induction, and then I verified the base case. After that, I did the inductive step and very explicitly used the inductive hypothesis to make the relevant conclusion. All of these features are common in any induction proof.